Virtuoso® Layout Pro: T5 Interactive Routing

Course Version IC25.1

Lecture Manual **Revision 1.0**

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Virtuoso Layout Pro: T5 Interactive Routing

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following courses:

- Virtuoso® Layout Pro: T1 Environment and Basic Commands
- Virtuoso® Layout Pro: T2 Create and Edit Commands
- Virtuoso® Layout Pro: T3 Basic Commands
- Virtuoso® Layout Pro: T4 Advanced Commands

Or, you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux

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Course Objectives

In this course, you

- Analyze the basic concepts and settings in wire creation
- Create and modify the wires
- Use the assisted wires creation
- Use the smart auto via feature (Appendix)

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Course Agenda

- Basic Concepts and Settings in Wire Creation
  - Using Wire Editing Options with the Create Wire Command
  - Using Override Constraints in the Wire Assistant
- Creating and Modifying Wires
  - Connecting to Target Pins with the Create Wire Command
  - Using the Interactive Via Alignment Feature
- Assisted Wires Creation
  - Using Pin-to-Trunk Routing
  - Creating Assisted Wires
  - Using Auto Router on Selected Nets
- Appendix: Smart Auto Via
  - Using Auto Via Preview
  - Using Fast Interactive Via Editing – FIVE

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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/85091.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.

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Become Cadence Certified by Earning a Digital Badge

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  - Expand career opportunities
- Professional credibility
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- For more information, go to www.cadence.com/training or email es_digitalbadge@cadence.com.

How do I register to take the exam?

- Log in to our Learning Management System, click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- Credly validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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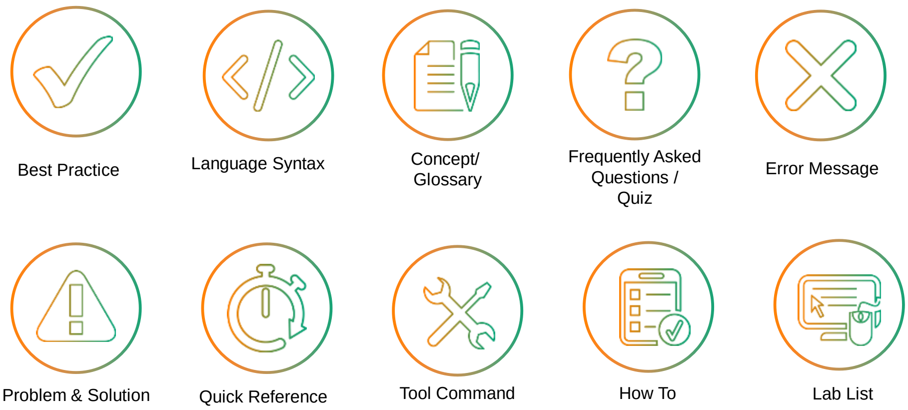
Icons Used in This Class



Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.

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Module 2

Basic Concepts and Settings in Wire Creation

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Module Objectives

In this module, you

- Use Wire Editing Options with the Create Wire command
  - Auto Merge Wires
  - Allow Loops
  - Adjust Edited Vias Parameters
  - Show Alignment Markers
- Use Override Constraints in the Wire Assistant
  - Use existing PRO to populate the Override Constraints section in the Wire Assistant
  - Create new PRO dynamically to populate the Override Constraints section in the Wire Assistant
  - Experiment switching between different constraint groups

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In this module, you will use the Wire Editing Options with the Create Wire command.

You will explore the Auto Merge Wires, Allow Loops, Adjust Edited Vias Parameters, and Show Alignment Markers options available in the Layout Editor Option form.

You will use the Override Constraints section in the Wire Assistant.

You will use the existing Process Rule Overrides, to populate the Override Constraints section in the Wire Assistant.

You will create new Process Rule Overrides dynamically, to populate the Override Constraints section in the Wire Assistant.

You will experiment switching between different constraint groups.

Terms and Definition

Process Rule Overrides (PRO)

Process Rule Overrides (PRO) can be used to override, on a given object, the generic foundry and design process rules with specific, more stringent, rules as defined in Constraint Groups (which are stored in designs or technology files and act as a collection of process constraints that are applicable to the objects in a design).

Wire Assistant (WA)

The Wire Assistant is a dockable assistant that provides a quick, easy, and single point of access to the commonly used options for all routing purposes: automatic routing, interactive routing, assisted routing, and post-route optimization. The Wire Assistant is available in Layout Suite XL and higher tiers.

Point to Point Routing (P2P Routing)

The Point-to-Point command is an interactive routing command that automatically generates routing between two digitized points in a given cellview. The two specific locations can be the two click points. Point-to-point routing is available in the Virtuoso® Layout Suite XL/EXL/MXL tiers.

Fast Interactive Via Editing (FIVE)

The Fast Interactive Via Editing (FIVE) is very handy to edit/modify the existing vias when the overlapping area increases/decreases to match the new topology and to edit/modify the existing vias after a techfile update.

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Here you see some useful Acronyms and Terms used in this course.

Take some time to go over these options.

Types of Wires: Geometric Wires and Symbolic Wires

You can create a wire using the *Path* or *Path Segment* (*pathSeg*).

The different types of wires you can create are:

- Geometric Wires
- Symbolic Wires

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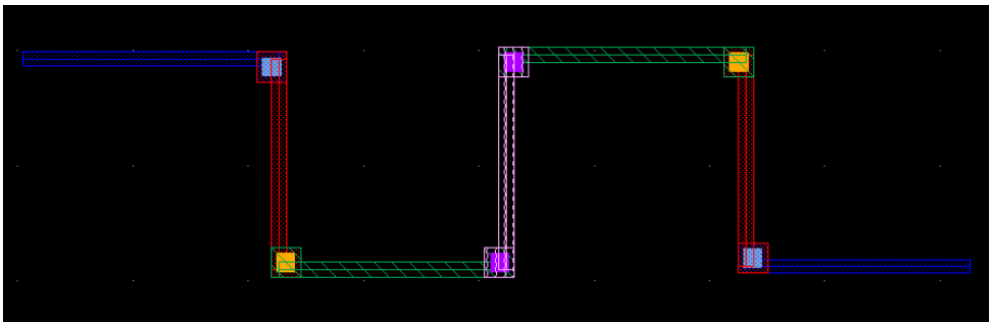
What Are Geometric Wires?

Geometric wires are created using paths, pathSegs, and vias that are not contained within a route.

- Paths cannot be placed in routes; therefore, paths are always geometric data.
- Geometric wires correspond to the DEF SPECIALNETS statement definition.



Geometric wires can be created in the Virtuoso Layout Suite XL/EXL/MXL tiers.

What Are Geometric Wires?

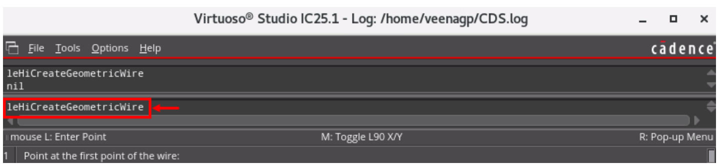
Geometric wires are created using paths, pathSegs, and vias that are not contained within a route. Paths cannot be placed in routes; therefore, paths are always geometric data. Geometric wires correspond to the DEF SPECIALNETS statement definition. Geometric wires can be created in Virtuoso Layout Suite XL/EXL/MXL tiers. Geometric wires can be used to create custom interconnect that auto signal routers do not modify when re-routing, ripping-up, and pushing wires. The autorouter does not create, delete, or re-route geometric data but it may connect to geometric data.

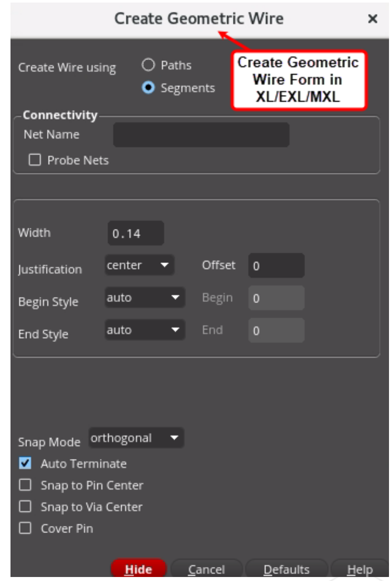
Creating Geometric Wires

To create geometric wires, enter in CIW:

`leHiCreateGeometricWire`

- In CIW > **`leHiCreateGeometricWire`**
- These type of wires are used for special routing (poGeometric wires, ground, clock, and so on).





Creating Geometric Wires

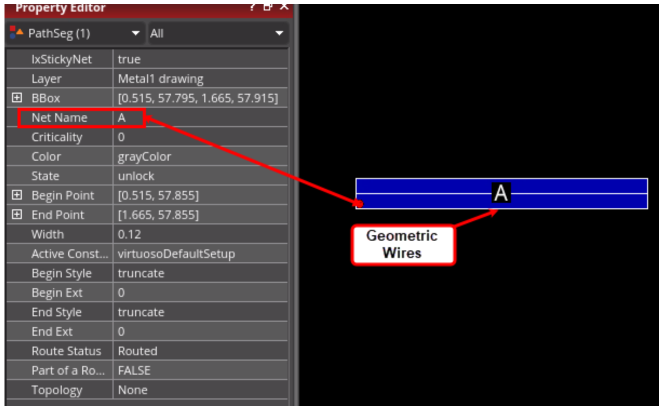
Geometric Wires can be created by using the *leHiCreateGeometricWire* command in Virtuoso Layout Suite XL/EXL/MXL tiers.

To create the geometric wires, enter in CIW: *leHiCreateGeometricWire* command to open the Create Geometric Wire form.

Geometric wires are usually used for special routing (power, ground, clock, and so on), which is not the default for creating wires in Virtuoso.

Results of Creating Geometric Wires

Geometric Wires are created using the **leHiCreateGeometricWire** command.



Geometric Wires are indicated as **Net Name**, in this example as, **A**.

Results of Creating Geometric Wires

You see the results of the Geometric Wires created here using the leHiCreateGeometricWire command.

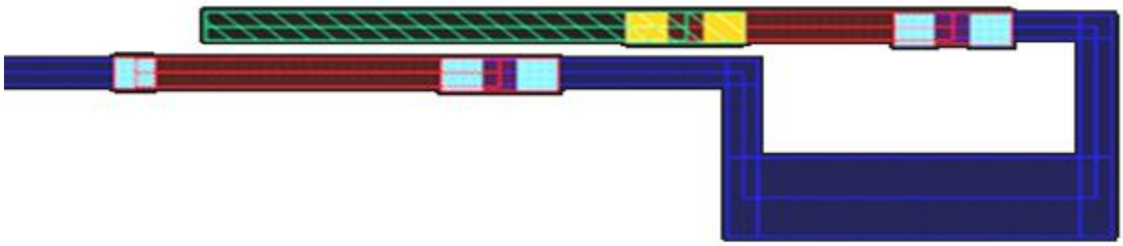
Geometric Wires are indicated with Net Name.

In this example, the Net Name created using the leHiCreateGeometricWire command is A.

What Are Symbolic Wires?

Symbolic wires are created using pathSegs and vias that are contained within routes.

- Routes are internal containers used by the system to group pathSegs and vias.
- Symbolic wires correspond to the DEF NETS statement definition.
- The *Create Wire* command is used only for single wire creation. For multiple wires or bus creation, the *Create Bus* command is used.



Symbolic wires can be created in the Virtuoso Layout Suite XL/EXL/MXL tiers.

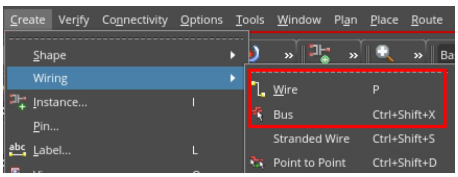
What Are Symbolic Wires?

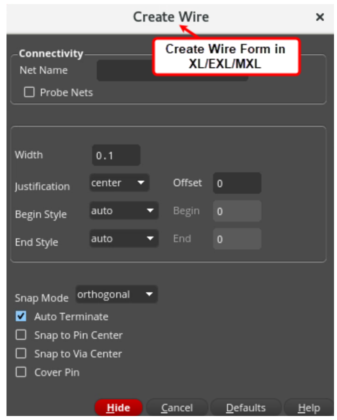
Symbolic wires are created using pathSegs and vias that are contained within routes. Routes are internal containers used by the system to group pathSegs and vias. Symbolic wires correspond to the DEF NETS statement definition. Symbolic wires can be created in Virtuoso Layout Suite XL/EXL/MXL tiers. They are usually attached to nets and can have default constraint groups assigned. The Virtuoso Space-Based Router creates only well-formed symbolic wires with matching endpoints.

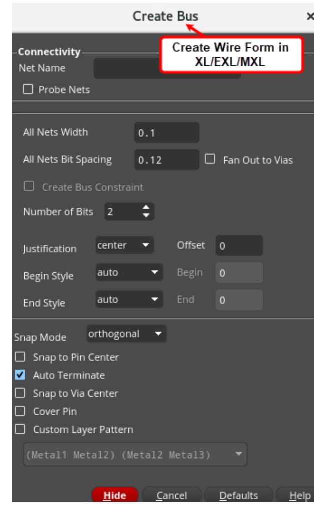
Creating Symbolic Wires

Symbolic wires can be created by using the **Create – Wiring – Wire** or **Create – Wiring – Bus** command in VLS-XL/EXL/MXL.

Symbolic wires are typically used for signal routing, which is the default for creating wires in Virtuoso.







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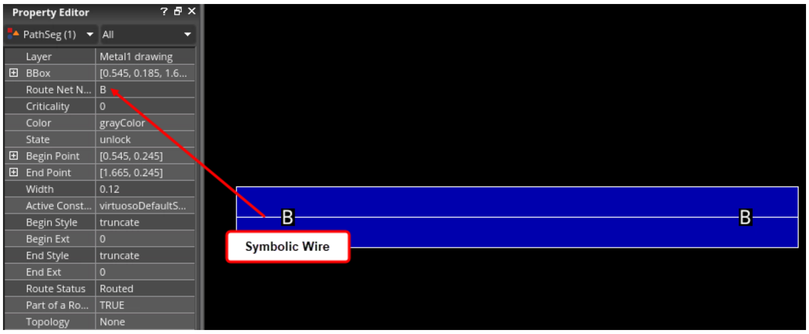
Creating Symbolic Wires

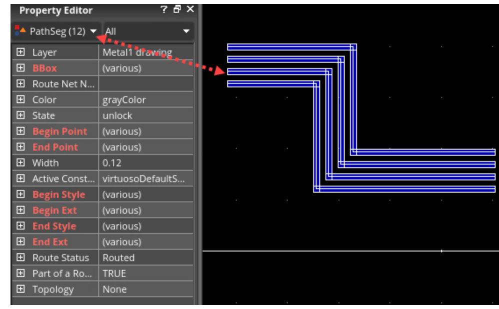
In Virtuoso, symbolic wires can be created by using the *Create – Wiring – Wire* or *Create – Wiring – Bus* command in all tiers of Layout Suite.

Symbolic wires are typically used for signal routing which is the default for creating wires in Virtuoso.

Results: Creating Symbolic Wires

Symbolic Wires are created using the **Create – Wiring – Wire** or **Create – Wiring – Bus** command.





Symbolic Wires are indicated as **Route Net Name**, in this example as **B**.

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Results of Creating Symbolic Wires

You see the results of the Symbolic Wires created here using the *Create Wire* and *Create Bus* commands.

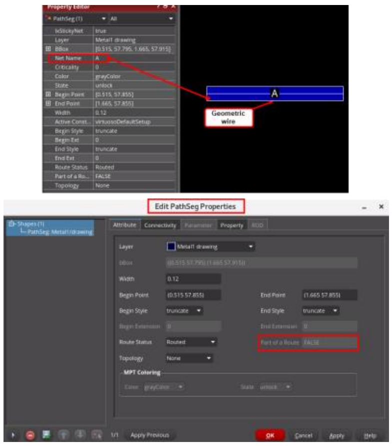
Symbolic Wires are indicated with *Route Net Name*.

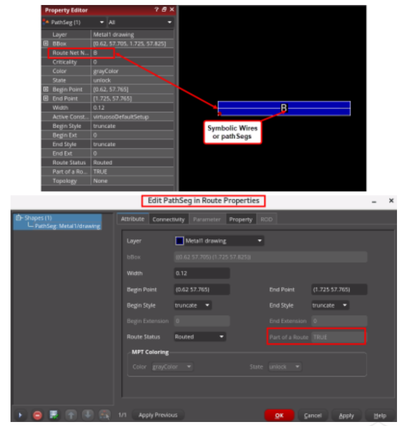
In this example, the *Route Net Name* created with *Create Wire* command is *B*.

Identifying If the Wire Is Geometric or Symbolic

The easiest and fastest way to identify if a wire is **geometric** is by using either the Property Editor assistant or the **Edit – Basic – Properties form**.

The easiest and fastest way to identify if a wire is **symbolic** is by using either the Property Editor assistant or the **Edit – Basic – Properties form**.





Differentiating Between Geometric and Symbolic Wires

The easiest and fastest way to identify if a wire is geometric or symbolic is by using either the *Property Editor* assistant or by using the *Edit – Basic – Properties* form.

As you see in this example, the geometric wire *A* and the symbolic wire *B* are identified in the *Property Editor* assistant and in the *Properties* form.

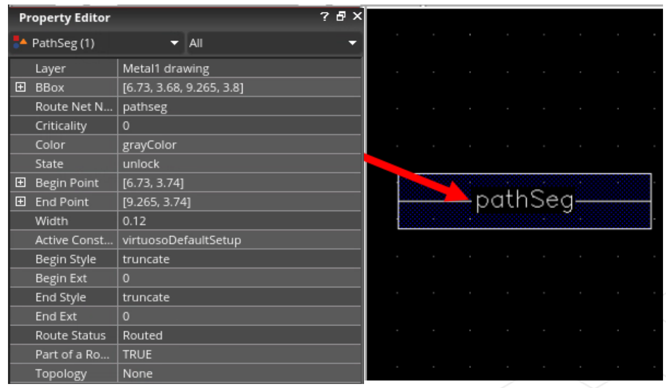
What Is a Wire Object?

A **wire object** is a shaped-based object represented by a specified width, a layer-purpose pair, and a two-point routing segment with a style associated with each of the points.

- Wires can either be orthogonal or diagonal and can be used to create on grid 45 degree wires.
- Custom end styles containing octagonal edges are used in diagonal pathSegs to put all vertices on the grid.

Wire objects can be defined as follows:

- Path Segment (pathSeg)
- Contiguous pathSegs
- T-junction



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What Is Wire Object?

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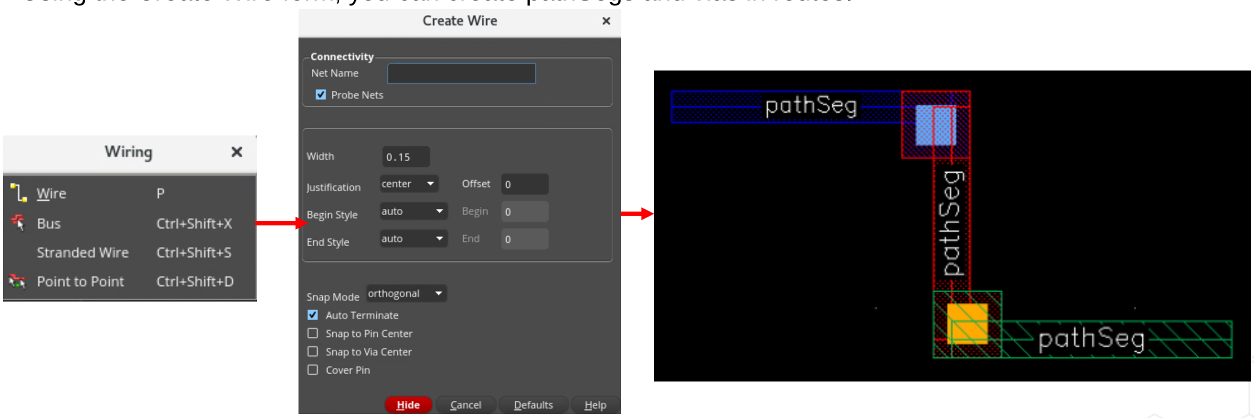
Wire Objects can be defined as follows:

- Path Segment (pathSeg)
- Contiguous pathSegs
- T-junction

Creating Path Segment

To create symbolic wires or pathsegs, in the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the *Create* toolbar or use the bindkey **P** to open the Create Wire form.

- Using the *Create Wire* form, you can create pathSegs and vias in routes.



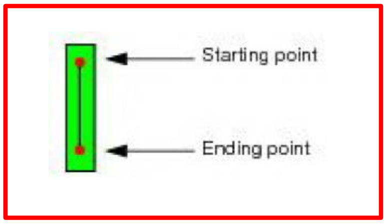
Creating Symbolic Wires or pathSegs

To create symbolic wires or pathsegs, in the design canvas, choose the **Create** – **Wiring** – **Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.

Using the *Create Wire* form, you can create pathSegs and vias in routes.

Wire Object: Path Segment (pathSeg)

A **pathSeg** is a shape used to realize physical routing that can either be orthogonal or diagonal. A pathSeg has a start and end point.

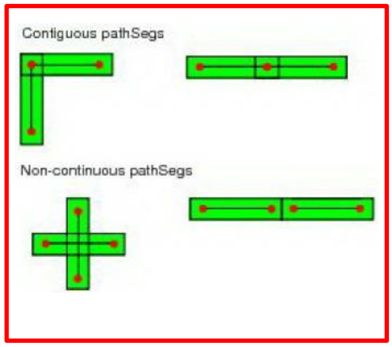


Wire Object: Path Segment (pathSeg)

A **pathSeg** is a shape used to realize physical routing that can either be orthogonal or diagonal. A pathSeg has a start and end point.

Wire Object: Contiguous pathSegs

Two **pathSegs** are contiguous when they are on the same layer and when they have an overlapping point (start or end).

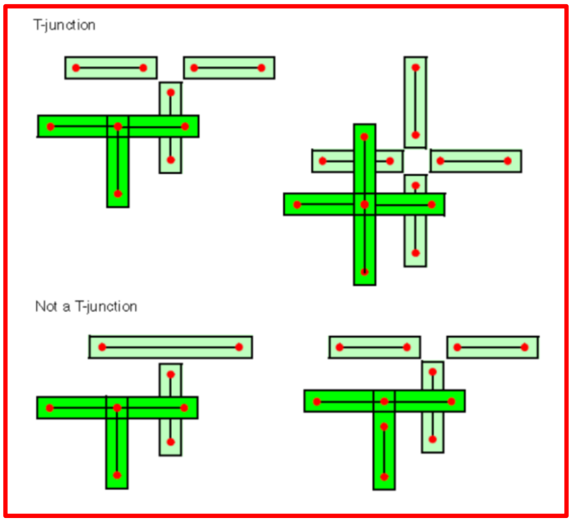


Wire Object: Contiguous pathSegs

Two **pathSegs** are contiguous when they are on the same layer and when they have an overlapping point (start or end).

Wire Object: T-junction

A **T-junction** is a point where there are more than two contiguous pathSegs.



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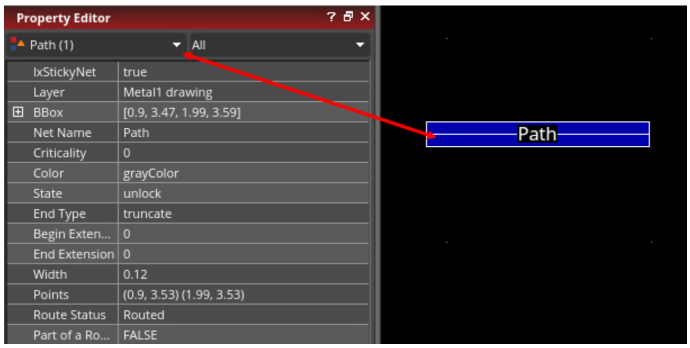
Wire Object: T-junction

A **T-junction** is a point where there are more than two contiguous pathSegs.

What Is a Path?

A **path** object is a shaped-based object represented by a point array with a specified width, style, and layer-purpose pair.

- Paths support anyAngle mode.
- When used to create non-orthogonal routing, paths can lead to off-grid vertices depending on the width and angle being used.



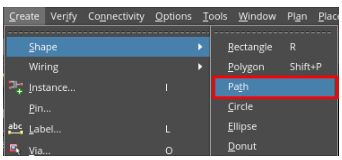
What Is Path?

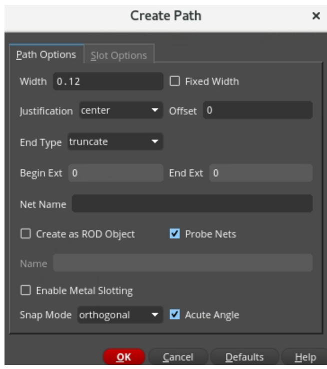
A **path** object is a shaped-based object represented by a point array with a specified width, style, and layer-purpose pair. Paths support anyAngle mode. When used to create non-orthogonal routing, paths can lead to off-grid vertices depending on the width and angle being used.

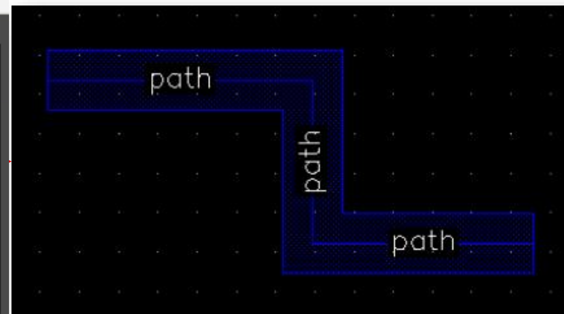
Creating Path

To create a path in the design canvas, choose the **Create – Shape – Path** menu command to invoke the Create Path form.

- Using the *Create Path* form, you can specify how a path is created in your design.
- Using paths for routing creates geometric data.







Creating Path

To create path, in the design canvas, choose the *Create – Shape – Path* menu command to invoke the Create Path form.

Using the *Create Path* form, you can specify how a path is created in your design.

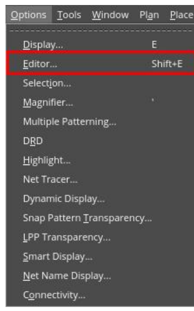
Using paths for routing creates geometric data.

Invoking the Layout Editor Options Form

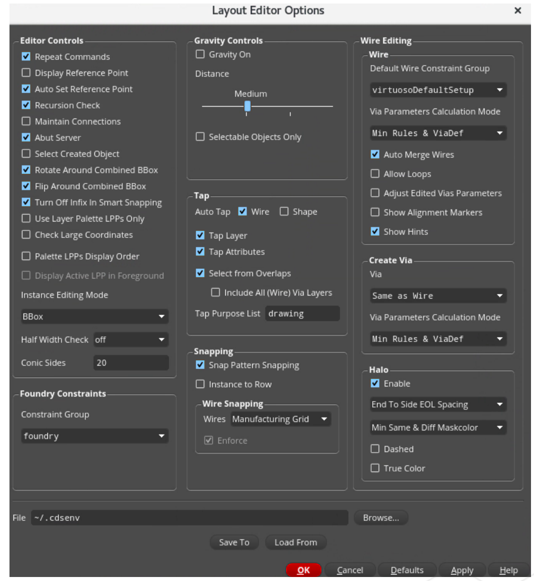
In the layout canvas, choose the **Options – Editor** menu command or press the bindkey **Shift+E** to invoke the Layout Editor Options form.

- Using the *Layout Editor Options* form, you can set the fields appropriately for your layout editing tasks.

Choose Options – Editor



Layout Editor Options Form



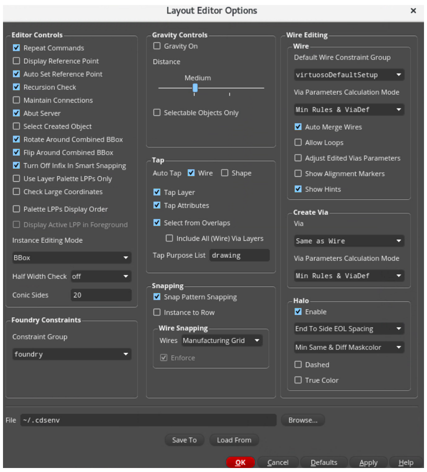
Invoking the Layout Editor Options Form

In the layout canvas, choose the *Options – Editor* menu command or press the bindkey *Shift+E* to invoke the Layout Editor Options form.

Using this form, you can set the fields as required for your layout editing tasks.

Layout Editor Options Form: Editor Controls

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Repeat Commands	Sets all commands to automatically repeat, if you first select the command and then the object to edit. Does not affect commands if you first select an object, then the command.
Display Reference Point	Displays a bold asterisk (*) on the current reference point in your layout.
Auto Set Reference Point	Sets a reference point automatically whenever you enter a new point.
Recursion Check	Turns recursion checking on. Recursion checking prevents the creation of recursive hierarchy (where a cell has an instance of itself at some level) by commands such as Create Instance, Make Cell, and Find/Replace.
Maintain Connections	Turns on an automatic re-connection mode for pins, paths, and pathsegs that have been separated during any editing activity that moves pins.
Abut Server	Turns automatic abutment and automatic unabutment on and off.
Select Created Object	Ensures that the object you create remains selected after you complete the create, copy and paste commands. If you create multiple objects, all of them remain selected after being created. The Repeat commands should be disabled for this option to work for copy-and-paste commands.
Rotate Around Combined BBox	Rotates multiple selected objects around the specified reference point of the combined bounding box of the objects.
Flip Around Combined BBox	Flips multiple selected objects around the specified reference point of the combined bounding box of the objects.
Turn Off Infix in Smart Snapping	Turns off infix when commands, such as Create Measurement and Quick Align, that use smart snapping are active.
Use Layer Palette LPPs Only	Ensures only valid layers displayed in the Layers assistant are populated in the layer field of the Property Editor, Move, Copy, Label, and Find/Replace forms.
Check Large Coordinates	Provides control for checking for large coordinates during the copy operation.
Palette LPPs Display Order	Enables you to change the display order of layer-purpose pairs in the canvas.
Display Active LPP in Foreground	Displays the shapes on the active layer in the foreground, on top of the shapes drawn on all other layers.

Layout Editor Options Form: Editor Controls

We will see the *Editor Controls* section in the *Layout Editor Options* form.

Repeat Commands option sets all commands to automatically repeat, if you first select the command and then the object to edit. Does not affect the commands if you first select an object, then the command.

Auto Set Reference Point option automatically sets a reference point whenever you enter a new point.

Recursion Check option turns recursion checking on. Recursion checking prevents the creation of recursive hierarchy, where a cell has an instance of itself at some level, by commands such as *Create Instance*, *Make Cell*, and *Find/Replace*.

Abut Server option turns automatic abutment and automatic unabutment on and off respectively.

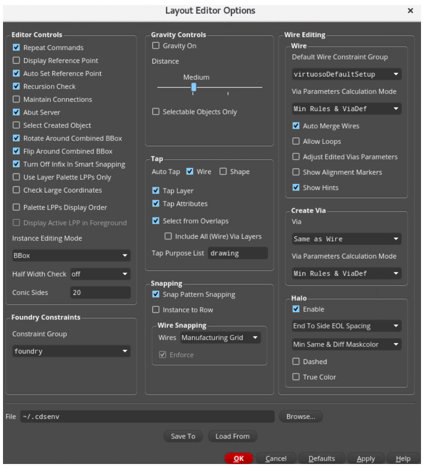
Rotate Around Combined Center option enables you to rotate multiple selected objects around the center of the selected bBox.

Flip Around Combined Center option enables you to flip multiple selected objects around the center of the selected bBox.

Turn Off Infix In Smart Snapping option if selected, turns off infix, when commands such as *Create Measurement* and *Quick Align*, that use smart snapping are active.

Layout Editor Options Form: Editor Controls (continued)

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Instance Editing Mode	Specifies whether the Flip, Rotate, and Create Instance (when creating an array of instances) commands use the instance bounding box (BBox), the place and route boundary (P&R Boundary), or the Snap Boundary as the instance border when operating on instances or mosaics. The default is Bbox.
Half Width Check	Sets whether or not you can create or edit a path that has a segment with a length less than half the path width. You can either disable the checking (off) or enable it (notify or enforce).
Conic Sides	Sets the number of segments to use when you convert conics to polygons. The maximum number of sides is 4000.

Foundry Constraints GROUP - Specifies the name of the constraint group that contains the foundry constraints.

Layout Editor Options Form: Editor Controls

We will see the *Editor Controls* section in the *Layout Editor Options* form.

Repeat Commands option sets all commands to automatically repeat, if you first select the command and then the object to edit. Does not affect the commands if you first select an object, then the command.

Auto Set Reference Point option automatically sets a reference point whenever you enter a new point.

Recursion Check option turns recursion checking on. Recursion checking prevents the creation of recursive hierarchy, where a cell has an instance of itself at some level, by commands such as *Create Instance*, *Make Cell*, and *Find/Replace*.

Abut Server option turns automatic abutment and automatic unabutment on and off respectively.

Rotate Around Combined Center option enables you to rotate multiple selected objects around the center of the selected bBox.

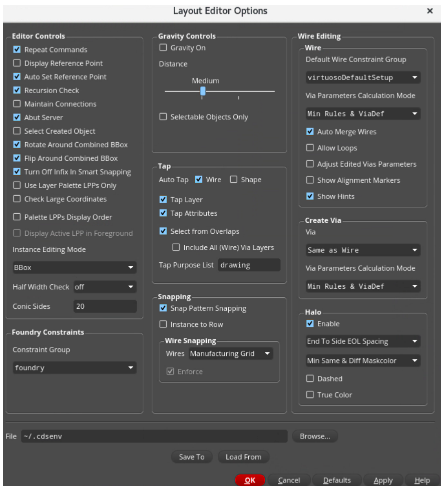
Flip Around Combined Center option enables you to flip multiple selected objects around the center of the selected bBox.

Turn Off Infix In Smart Snapping option if selected, turns off infix, when commands such as *Create Measurement* and *Quick Align*, that use smart snapping are active.

Foundry Constraints GROUP - Specifies the name of the constraint group that contains the foundry constraints.

Layout Editor Options Form: Gravity Controls

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Gravity On	Sets the drawing pointer to automatically snap to objects in the cellview. The pointer snaps to all the visible objects at any hierarchy level.
Distance slider	Sets the snap distance between the pointer and an object. You can choose from Small, Medium, Large, or Very Large using the slider.
Selectable Objects Only	Sets the drawing pointer to snap only to the selected objects in the cellview.

Layout Editor Options Form: Gravity Controls

We will see the *Gravity Controls* section in *the Layout Editor Options* form.

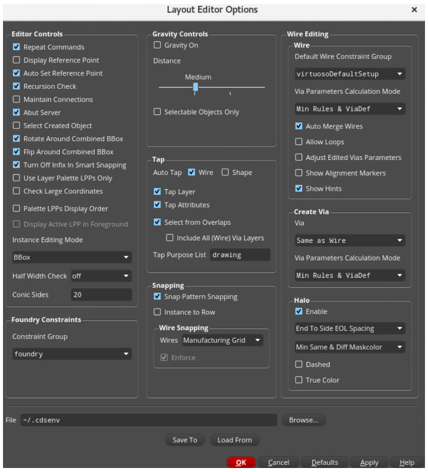
Gravity On option sets the drawing pointer to automatically snap to objects in the cellview. The pointer snaps to all the visible objects at any hierarchy level.

Distance slider sets the snap distance between the pointer and an object. You can choose from *Small*, *Medium*, *Large*, or *Very Large* using the slider.

Selectable Objects Only option sets the drawing pointer to snap only to the selected objects in the cellview.

Layout Editor Options Form: Tap

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Auto Tap	Sets the layer of the wire being created to be the same as that of an existing clicked wire automatically. Attributes of objects are also tapped. In addition to object attributes, an existing net name can be tapped.
Tap Layer	Sets the tapped layer as the layer for the wire or shape being created and updates the Layer Selection Window. Layer option is enabled only when one of Wire or Shape from Auto Tap option is on.
Tap Attributes	Captures attributes from the wire or shape you click. When tapping a wire or path, you can tap the LPP, extension type, and width of the wire. The Attributes option is enabled only when one of Wire or Shape from Auto Tap option is on.
Select from Overlaps	Prompts you to choose a layer or net name from the Choose Object to Tap form when clicking on a coordinate with objects on overlapping layers. The option is enabled only when one of Wire or Shape from Auto Tap option is on.
Include All (Wire) Via Layers	Prompts you with the Choose Object to Tap form when tapping a via in Layout XL and higher tiers. For more information, see Tapping Vias
Tap Purpose List	Prioritizes the purposes considered when changing the current layer. For example, if you have a metal1 pin layer, but you want to create a wire on metal1 drawing layer, you must have the purpose list set to drawing.

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Layout Editor Options Form: Tap

We will see the *Tap* section in the *Layout Editor Options* form.

Auto Tap option automatically sets the layer of the wire being created to be the same as that of an existing clicked wire.

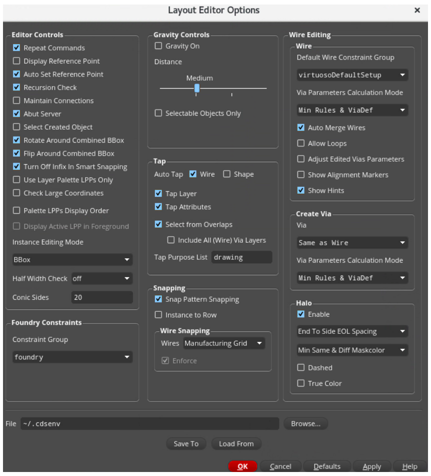
Tap Layer option sets the tapped layer as the layer for the wire or shape being created and updates the Layer Selection Window.

Tap Attributes option captures the attributes from the wire or shape you click.

Select from Overlaps option prompts you to choose a layer or net name from the *Choose Object to* form when clicking on a coordinate with objects on overlapping layers.

Layout Editor Options Form: Snapping

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Snap Pattern Snapping	Enables snap pattern snapping.
Instance to Row	Snaps instances to rows during the commands Edit - Move, Edit - Copy, Edit - Stretch, and Create - Instance
Layer Manufacturing Grid	(Virtuoso Photonics Option) Determines in Layout EXL or a higher tier whether objects are snapped to coarse manufacturing grids instead of the base manufacturing grid. This feature is supported for the Create Instance, Create Rectangle, Create Polygon, Create Path, Create Pin, Create Via, Move, Copy, and Stretch commands.
Wires	Lets you select the snapping mode. The available options are: • Manufacturing Grid snaps the wires to the Manufacturing Grid. This is the default option. • Routing Grid snaps the segments and vias other than the ones directly connecting to the off-grid pins to the routing grid. • Track Pattern snaps the segments and vias other than the ones directly connecting to the off-grid pins to track. The track patterns are saved on the design and so each design can have its own set of non-uniform track patterns. • Snap Pattern enables snapping of wires to the nearest snap pattern track.
Enforce	Snaps the wire to the centerline of the track pattern. Also, if this option is selected, the wire inherits the width of the active width spacing pattern and the direction in which the wire is created is the same as the direction of the track pattern.

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Layout Editor Options Form: Snapping

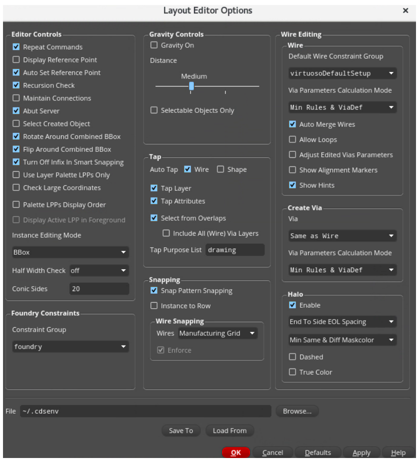
We will see the *Snapping* section in the *Layout Editor Options* form.

Instance to Row option snaps the instances to rows during the *Edit – Move*, *Edit – Copy*, *Edit – Stretch*, and *Create – Instance* commands.

In the *Wire Snapping* field using the *Wires* option you can select the snapping mode as *Manufacturing Grid*, *Routing Grid*, *Off Grid*, and *Track Pattern*.

Layout Editor Options Form: Wire Editing – Wire

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Default Wire Constraint Group	Allows the selection of one of the available constraint groups, which may be defined at the technology or the design level.
Via Parameters Calculation Mode	Sets the calculation mode for via parameters to Min Rule and ViaDef, Minimum Rules and ViaDef Defaults.
Auto Merge Wires	Merges the connected geometric paths automatically. Also, allows you to choose whether or not to merge the collinear geometric pathSegs while creating or stretching a geometric wire.
Allow Loops	Permits creation of loops when creating or editing wires. The option is off by default.
Adjust Edited Vias Parameters	Enables the re-calculation of via alignment of the vias if you stretch the wires connected to them. This check box is not selected by default.
Show Alignment Markers	Displays an alignment arrow and an alignment marker. The arrow points to the closest object on the active flightline. The marker snaps the cursor to the target object edges and center as you move the cursor.
Show Hints	Controls the display of the Interactive Routing Bindkey hint box in Layout EXL and higher tiers when you start the Create Wire, Create Bus, or Create Stranded Wire command. In Layout XL, the hint box is displayed only for the Create Stranded Wire command.

Layout Editor Options Form: Wire Editing – Wire

Wire Editing section comprises the *Wire* and *Create Via* sub-sections in the *Layout Editor Options* form.

We will see the *Wire* sub-section now.

Default Wire Constraint Group cyclic field allows the selection of one of the available constraint groups which may be defined at the technology or the design level.

Via Parameters Calculation Mode sets the calculation mode for via parameters to one of the following: namely, *Min Rules & ViaDef*, *Minimum Rules*, and *ViaDef Defaults*.

Auto Merge Wires option automatically merges the connected paths and collinear pathsegs while creating or stretching a wire.

Allow Loops option permits the creation of the loops when creating or editing wires.

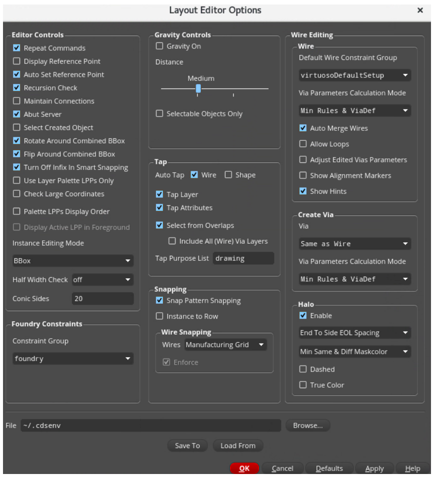
Adjust Edited Vias Parameters option enables the re-calculation of via alignment of the vias if you stretch the wires connected to them.

Show Alignment Markers option dynamically displays an alignment arrow and an alignment marker.

Show Hints option controls the display of the Interactive Routing Bindkey hint box when you start the *Create Stranded Wire* command.

Layout Editor Options Form: Wire Editing – Create Via

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Via	Specifies the constraint group from which vias will be available when adding vias using Create ? Via or while editing vias using the Edit Via Properties form, the Property Editor assistant, or the Create Via form.
Via Parameters Calculation Mode	Seeds the Create Via form the first time when a specified viaDef is used. Then onwards, the Create Via form is populated from Parameter Caching. The cyclic field values for this option include Min Rules & ViaDef, Minimum Rules and ViaDef Defaults.

Layout Editor Options Form: Wire Editing – Create Via

We will see the *Create Via* sub-section now.

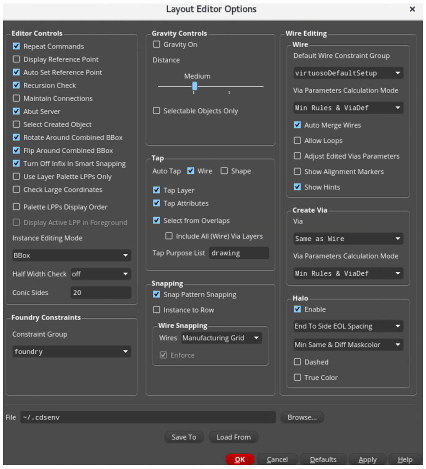
You can use the *Via* cyclic field to specify the constraint group from which vias will be available when adding vias or while editing vias.

Via Parameters Calculation Mode sets the calculation mode for via parameters to one of the following: namely, *Min Rules & ViaDef*, *Minimum Rules*, and *ViaDef Defaults*.

Using *Halo*, you can select a halo option.

Layout Editor Options Form: Wire Editing – Halo

You can use the Editor Controls, Gravity Controls, Tap, Snapping, and Wire Editing sections in the Layout Editor Options form to specify the required options for your layout editing tasks.



Field	Description
Enable	Enables or disables a halo display in the layout canvas. When the option is unselected, the halo is not shown when the wire is created.
EOL Spacing list	Lets you select the spacing for the halo.
Maskcolor list	Lets you select a scope for creating a halo from the drop-down list.
Dashed	Creates a dashed halo around a wire, bus, or stranded wire while creating or editing an object that violates a spacing constraint. By default, this option is deselected.
True Color	Creates a halo in the color of that layer for each layer. By default, this option is deselected.

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Layout Editor Options Form: Wire Editing – Create Via

We will see the *Create Via* sub-section now.

You can use the *Via* cyclic field to specify the constraint group from which vias will be available when adding vias or while editing vias.

Via Parameters Calculation Mode sets the calculation mode for via parameters to one of the following: namely, *Min Rules & ViaDef*, *Minimum Rules*, and *ViaDef Defaults*.

Using *Halo*, you can select a halo option.

Setting Up the Interactive Routing Environment

To set up the Interactive Routing Environment, you can configure the **Wire Editing** options to control the way the **Wire Editor** behaves as well as how it generates vias.

The **Wire Editing** section in the **Layout Editor Options** form comprises the **Wire** and **Create Via** sub-sections.

1. In the layout canvas, display the Layout Editor Options form by choosing the **Options – Editor** menu command or by pressing the bindkey **Shift+E**.
2. In the Layout Editor Options form, in the **Wire Editing** section, in the **Wire** sub-section, in the **Default Wire Constraint Group** cyclic field, select one of the available constraint groups that may be defined at the technology or the design level.
3. In the Layout Editor Options form, in the **Wire Editing** section, in the **Create Via** sub-section, and in the **Via** cyclic field, select the constraint group from which vias will be available when adding vias using **Create – Via** or while editing vias using the **Edit Via Properties** form, the **Property Editor** assistant, or the **Create Via** form.
4. Click **OK** in the Layout Editor Options form.

Result: During Interactive Routing, the options set in the **Wire** and **Create Via** sub-sections are used.

Setting up the Interactive Routing Environment

To set up the Interactive Routing Environment, you can configure the *Wire Editing* options to control the way the *Wire Editor* behaves as well as how it generates vias.

The *Wire Editing* section in the *Layout Editor Options* form comprises the *Wire* and *Create Via* sub-sections.

In the layout canvas, display the Layout Editor Options form by choosing the *Options – Editor* menu command or by pressing the bindkey *Shift+E*.

In the Layout Editor Options form, in the *Wire Editing* section, in the *Wire* sub-section, in the *Default Wire Constraint Group* cyclic field, you can select one of the available constraint groups which may be defined at the technology or the design level.

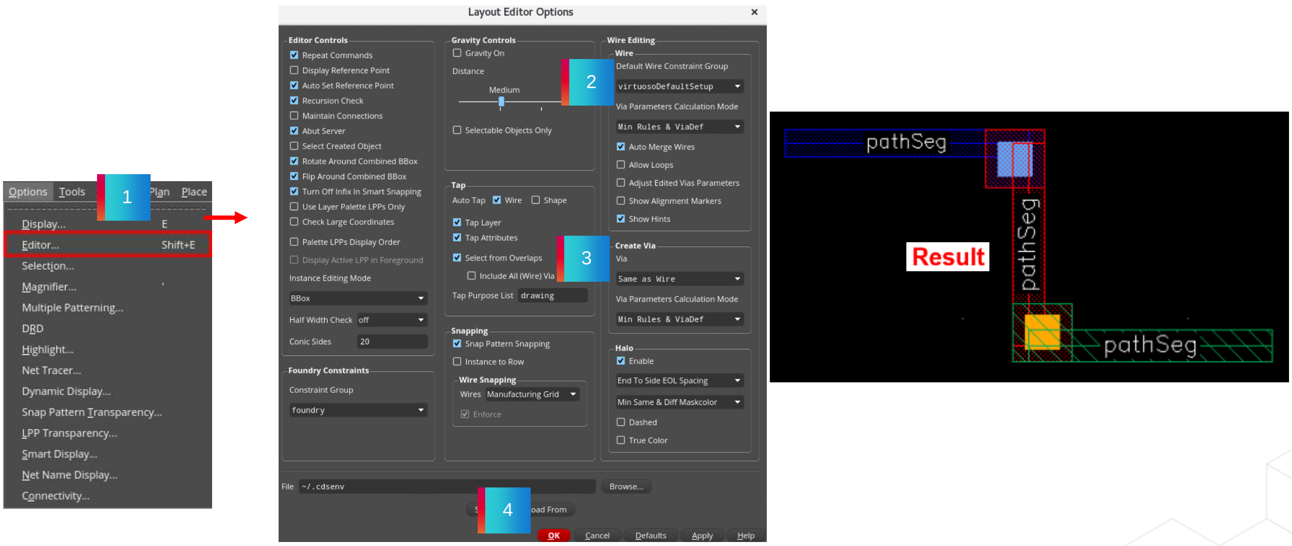
In the *Create Via* sub-section, you can use the *Via* cyclic field to specify the constraint group from which vias will be available when adding vias or while editing vias in the *Wire Editing* section in the *Layout Editor Options* form.

Then click *OK* in the form.

The options set in the *Wire* and *Create Via* sub-sections are used during the Interactive Routing.

Result: Setting Up the Interactive Routing Environment

Result: During Interactive Routing, the options set in the **Wire** and **Create Via** sub-sections are used.



Wire Editing

This section comprises the *Wire* and *Create Via* sub-sections.

Wire

Default Wire Constraint Group cyclic field allows the selection of one of the available constraint groups which may be defined at the technology or the design level. The constraint group defines the appropriate layers and vias for the application being used. Only the layers, vias, and any rules specified in the selected constraint group are used when creating wires. Selection of a constraint group in the *Wire* cyclic field specifies the application default constraint group for wire creation and editing commands. The default active constraint group is virtuosoDefaultSetup. See Constraint Groups in the *Virtuoso Space-based Router User Guide* for more information about defining constraint groups. Environment variable: *wireConstraintGroup*

Create Via

Via specifies the constraint group from which vias will be available when adding vias using *Create – Via* or while editing vias using the *Edit Via Properties* form, the *Property Editor* assistant, or the *Create Via* form. The via spacing values in the constraint group selected here populate the *Create Via* form, as well as reflect on the *Edit Via Properties* form and *Property Editor* assistant. For more information about defining constraint groups, see Constraint Groups in the *Virtuoso Space-based Router User Guide*. Environment variable: *viaConstraintGroup*

Invoke the *Layout Editor Options* form and specify the required constraint groups in the *Default Wire Constraint Group* cyclic field and in the *Via* cyclic field and *OK* the form.

The options set in the *Wire* and *Create Via* sub-sections of the *Wire Editing* section in the *Layout Editor Options* are used during the Interactive Routing.

Enabling Auto Merging of Wires

In the Layout Editor Options form, with the **Auto Merge Wires** option enabled, it automatically merges the connected paths and collinear pathSegs while creating or stretching a wire.

The checkbox is selected by default. ☑ Auto Merge Wires

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Enabling Auto Merging of Wires

In the Layout Editor Options form, with the *Auto Merge Wires* option enabled, it automatically merges the connected paths and collinear pathSegs while creating or stretching a wire.

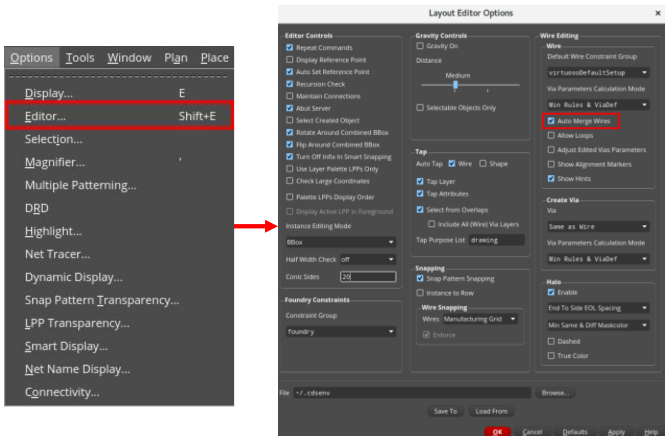
The checkbox is selected by default.

Auto Merge Wires Option ON

Turn Auto Merge Wires Option ON → Invoke Create Wire Command → Create a Wire

☑ Auto Merge Wires

In the Layout Editor Options form, in the **Wire Editing: Wire** section, enable the **Auto Merge Wires** option.



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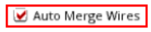
Auto Merge Wires Option ON

We will see the *Auto Merge Wires* feature now.

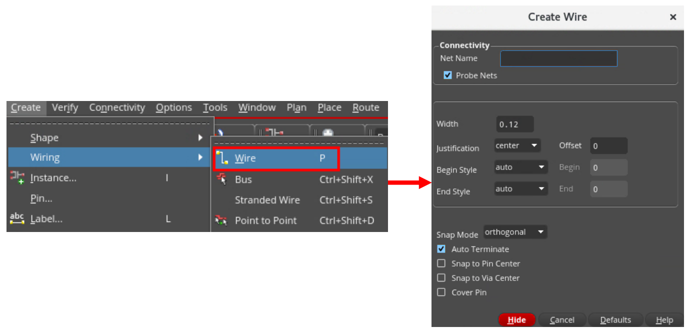
In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, enable the *Auto Merge Wires* option which is the default option.

Invoke Create Wire Command

Turn Auto Merge Wires Option ON → Invoke Create Wire Command → Create a Wire



In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** to invoke the Create Wire command.



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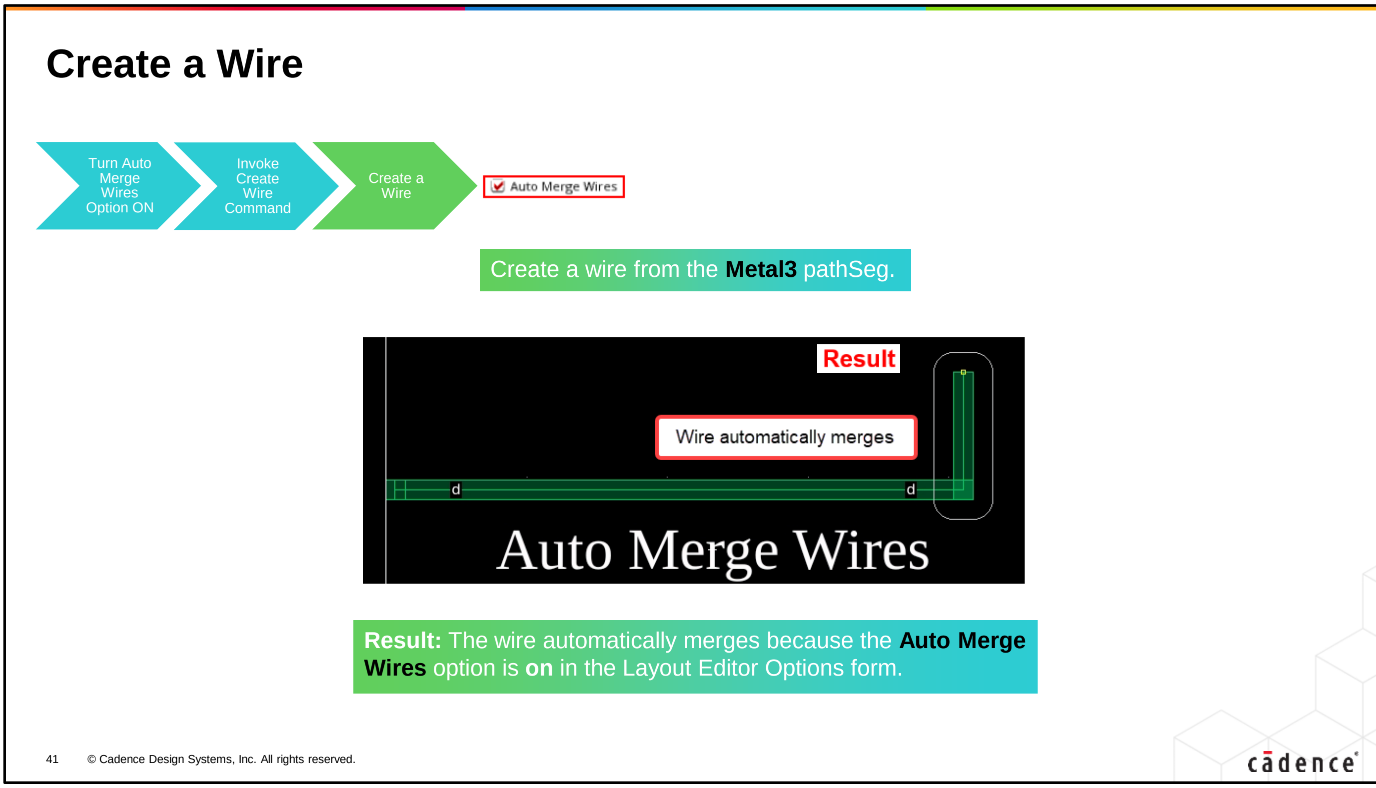
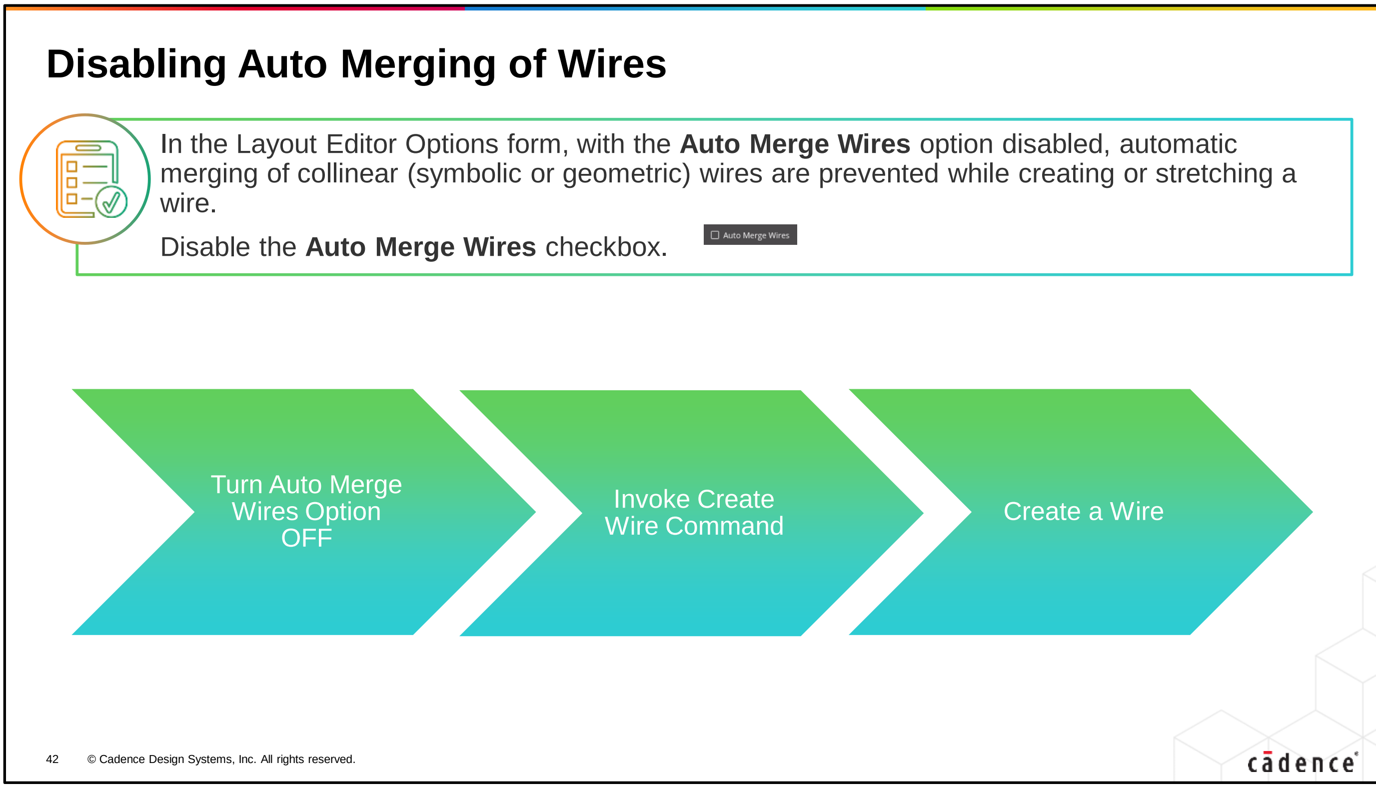
Invoke Create Wire Command

Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.



Create a Wire

Now, when you create a wire from the *Metal3* pathSeg, you can observe that the wire automatically merges because the *Auto Merge Wires* option is *on* in the Layout Editor Options form.

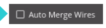


Disabling Auto Merging of Wires

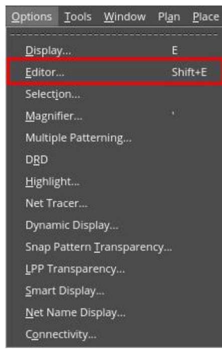
In the Layout Editor Options form, with the *Auto Merge Wires* option disabled, automatic merging of collinear wires are prevented while creating or stretching a wire.

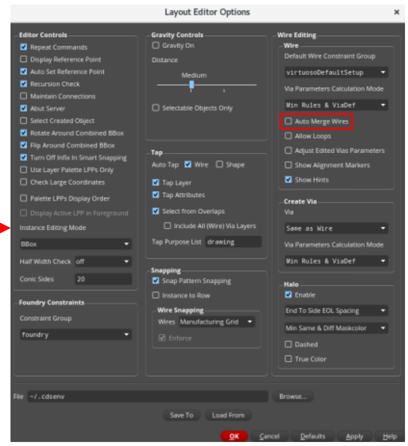
You can disable the *Auto Merge Wires* checkbox to get this feature.

Auto Merge Wires Option OFF



In the Layout Editor Options form, in the **Wire Editing: Wire** section, disable the **Auto Merge Wires** option.





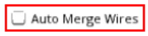
Auto Merge Wires Option OFF

We will see the *Auto Merge Wires* feature now.

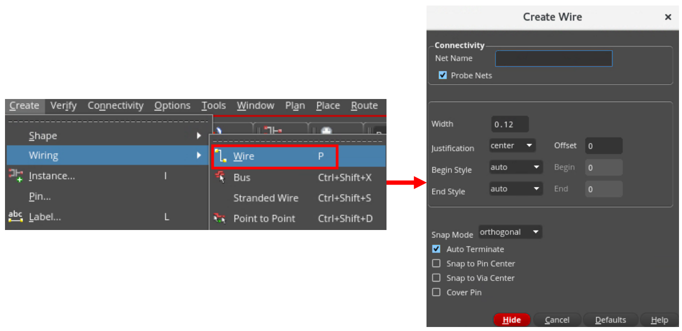
In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, disable the *Auto Merge Wires* option.

Invoke Create Wire Command

Turn Auto Merge Wires Option OFF → Invoke Create Wire Command → Create a Wire



In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** to invoke the Create Wire command.

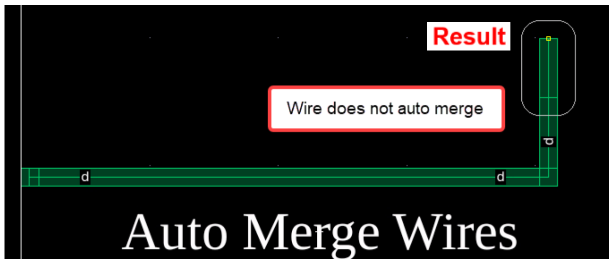


Invoke Create Wire Command

Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.

Create a Wire

Create a wire from the **Metal3** pathSeg.



Result: The wire does not auto-merge because the **Auto Merge Wires** option is **off** in the Layout Editor Options form.

Create a Wire

Now, create a wire from the *Metal3* pathSeg and you can observe that the wire does not auto merge because the *Auto Merge Wires* option is *off* in the Layout Editor Options form.

Removing Loops

The “loops” can be automatically removed by the **Create Wire** command with the **Allow Loops** option disabled in the Layout Editor Options form.

The checkbox is deselected by default. ☐ Allow Loops

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, disable the **Allow Loops** option.
2. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** to invoke the Create Wire command.
3. Create a wire that loops.

Result: The “loops” are removed automatically while creating the wire because the **Allow Loops** option is **off** in the Layout Editor Options form.

Removing Loops

The “loops” can be automatically removed by the *Create Wire* command with the *Allow Loops* option disabled in the Layout Editor Options form.

The checkbox is deselected by default.

We will see the *Allow Loops* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, disable the *Allow Loops* option which is the default option.

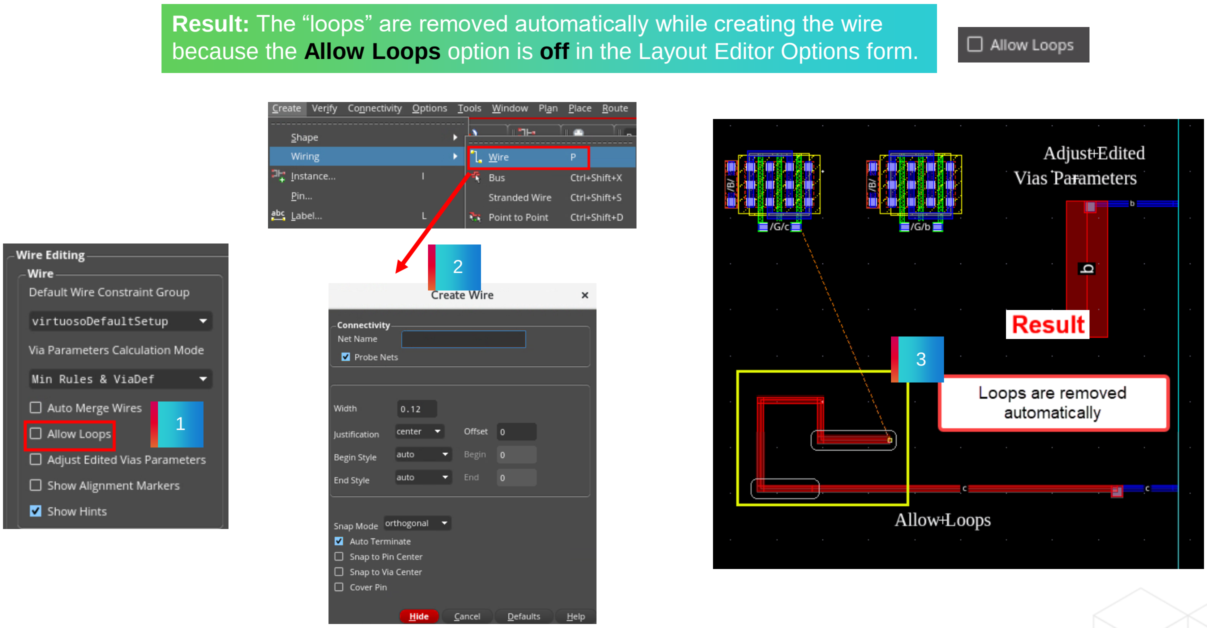
Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.

Now, create a wire which loops.

You can observe that the “loops” are removed automatically while creating the wire because the *Allow Loops* option is *off* in the Layout Editor Options form.

Result: Removing Loops

Result: The "loops" are removed automatically while creating the wire because the **Allow Loops** option is **off** in the Layout Editor Options form.



Result: Removing Loops

Invoke the Layout Editor Options form and disable the *Allow Loops* option.

When you create a wire which loops, you can observe that the "loops" are removed automatically while creating the wire because the *Allow Loops* option is *off* in the Layout Editor Options form.

Allowing Loops

The “loops” can be automatically allowed by the **Create Wire** command with the **Allow Loops** option enabled in the Layout Editor Options form.

Enable the **Allow Loops** option.

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, enable the **Allow Loops** option.
2. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** to invoke the Create Wire command.
3. Create a wire that loops.

Result: The “loops” stay unchanged and are allowed while creating the wire because the **Allow Loops** option is **on** in the Layout Editor Options form.

Allowing Loops

The “loops” can be automatically allowed by the *Create Wire* command with the *Allow Loops* option enabled in the Layout Editor Options form.

You can enable the *Allow Loops* checkbox to get this feature.

We will see the *Allow Loops* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, enable the *Allow Loops* option.

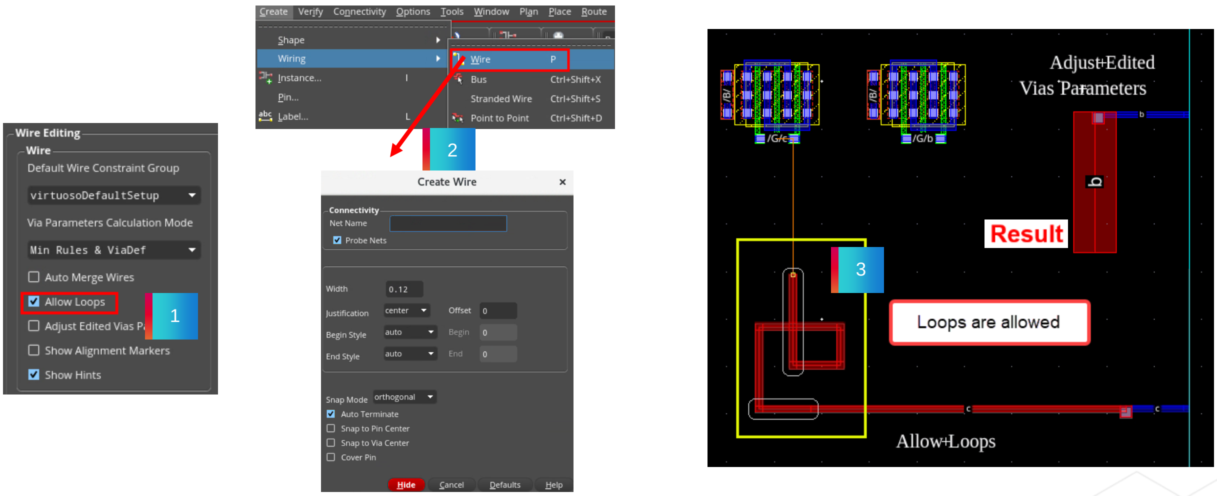
Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.

Now, create a wire which loops.

You can observe that the “loops” stay unchanged and are allowed when creating the wire because the *Allow Loops* option is *on* in the Layout Editor Options form.

Result: Allowing Loops

Result: The “loops” stay unchanged and are allowed while creating the wire because the **Allow Loops** option is **on** in the Layout Editor Options form.



Result: Allowing Loops

Invoke the Layout Editor Options form and enable the *Allow Loops* option.

When you create a wire which loops, you can observe that the “loops” stay unchanged and are allowed when creating the wire because the *Allow Loops* option is *on* in the Layout Editor Options form.

Disabling Adjust Edited Vias Parameters Option

In the Layout Editor Options form, with the **Adjust Edited Vias Parameters** option disabled, when the size of a wire element (pathSeg) is changed, the via stays unchanged and the via calculation is not triggered.

The checkbox is deselected by default. ☐ Adjust Edited Vias Parameters

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, disable the **Adjust Edited Vias Parameters** option.
2. In the design canvas, select the **Metal1** pathSeg in the Adjust Edited Vias Parameters section.
3. In the Property Editor assistant, change the **Metal1** pathSeg width from **0.12** to **0.7**.

Result: The via remains unchanged (single cut) even if there is enough place to use an array of vias because the **Adjust Edited Vias Parameters** option is **off** in the Layout Editor Options form.

Disabling Adjust Edited Vias Parameters Option

In the Layout Editor Options form, with the *Adjust Edited Vias Parameters* option disabled, when the size of a wire element is changed, the via stays unchanged and the via calculation is not triggered.

The checkbox is deselected by default.

We will see the *Adjust Edited Vias Parameters* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, disable the *Adjust Edited Vias Parameters* option which is the default option.

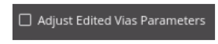
In the design canvas, select the *Metal1* pathSeg in the Adjust Edited Vias Parameters section.

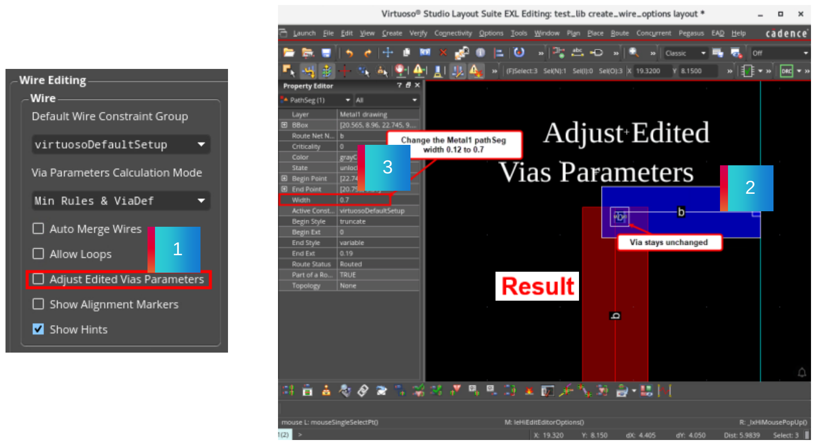
In the Property Editor assistant, change the *Metal1* pathSeg width from *0.12* to *0.7*.

You can observe that the via remains unchanged as a single cut even if there is enough place to use an array of vias because the *Adjust Edited Vias Parameters* option is *off* in the Layout Editor Options form.

Result: Disabling Adjust Edited Vias Parameters Option

Result: The via stays unchanged, and the via calculation is not triggered because the **Adjust Edited Vias Parameters** option is **off** in the Layout Editor Options form.





Result: Disabling Adjust Edited Vias Parameters Option

Invoke the Layout Editor Options form and disable the *Adjust Edited Vias Parameters* option.

When you change the *Metal1* pathSeg width, you can observe that, the via stays unchanged and the via calculation is not triggered because the *Adjust Edited Vias Parameters* option is *off* in the Layout Editor Options form.

Enabling Adjust Edited Vias Parameters Option

In the Layout Editor Options form, with the **Adjust Edited Vias Parameters** option enabled, when the size of a wire element (pathSeg) is changed, the via calculation is triggered.

Enable the **Adjust Edited Vias Parameters** option.

Adjust Edited Vias Parameters

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, enable the **Adjust Edited Vias Parameters** option.
2. In the design canvas, select the **Metal1** pathSeg in the Adjust Edited Vias Parameters section.
3. In the Property Editor assistant, change the **Metal1** pathSeg width from **0.12** to **0.7**.

Result: The via calculation is triggered, and a larger via (an array of 2x2 cuts) is used because the **Adjust Edited Vias Parameters** option is **on** in the Layout Editor Options form.

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Enabling Adjust Edited Vias Parameters Option

In the Layout Editor Options form, with the *Adjust Edited Vias Parameters* option enabled, when the size of a wire element is changed, the via calculation is triggered.

You can enable the *Adjust Edited Vias Parameters* checkbox to get this feature.

We will see the *Adjust Edited Vias Parameters* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, enable the *Adjust Edited Vias Parameters* option.

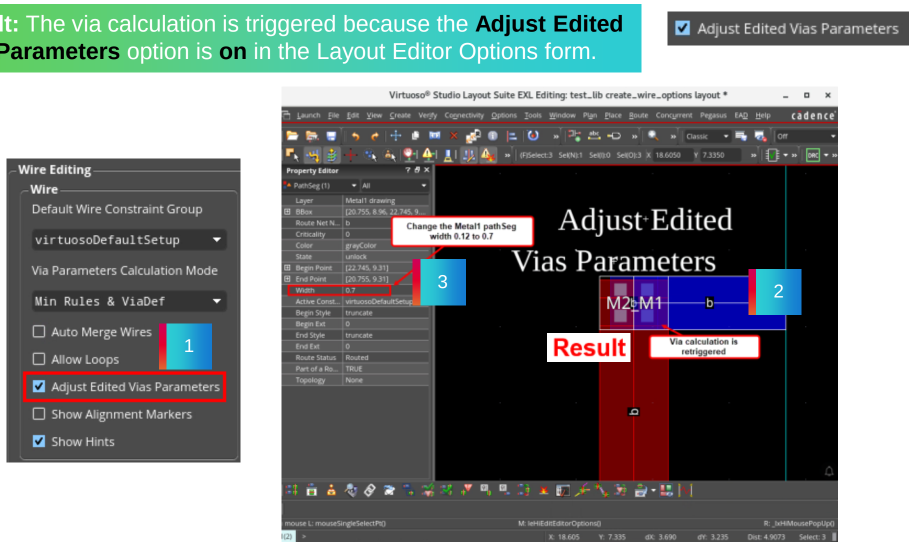
In the design canvas, select the *Metal1* pathSeg in the Adjust Edited Vias Parameters section.

In the Property Editor assistant, change the *Metal1* pathSeg width from *0.12* to *0.7*.

You can observe that the via calculation is triggered, and a larger via of an array of 2x2 cuts is used because the *Adjust Edited Vias Parameters* option is *on* in the Layout Editor Options form.

Result: Enabling Adjust Edited Vias Parameters Option

Result: The via calculation is triggered because the **Adjust Edited Vias Parameters** option is **on** in the Layout Editor Options form.



Result: Enabling Adjust Edited Vias Parameters Option

Invoke the Layout Editor Options form and enable the *Adjust Edited Vias Parameters* option.

When you change the *Metal1* pathSeg width, you can observe that, the via calculation is triggered because the *Adjust Edited Vias Parameters* option is *on* in the Layout Editor Options form.

Disabling Show Alignment Markers Option

In the Layout Editor Options form, with the **Show Alignment Markers** option disabled, when the wire is aligned to the target pin, the marker is not visible.

The checkbox is deselected by default. ☐ Show Alignment Markers

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, disable the **Show Alignment Markers** option.
2. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** to invoke the Create Wire command.
3. Create a wire from the **Metal1** pathSeg and move the mouse pointer upwards.

Result: No marker is visible because the **Show Alignment Markers** option is **off** in the Layout Editor Options form.

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Disabling Show Alignment Markers Option

In the Layout Editor Options form, with the *Show Alignment Markers* option disabled, when the wire is aligned to the target pin, the marker is not visible.

The checkbox is deselected by default.

We will see the *Show Alignment Markers* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, disable the *Show Alignment Markers* option which is the default option.

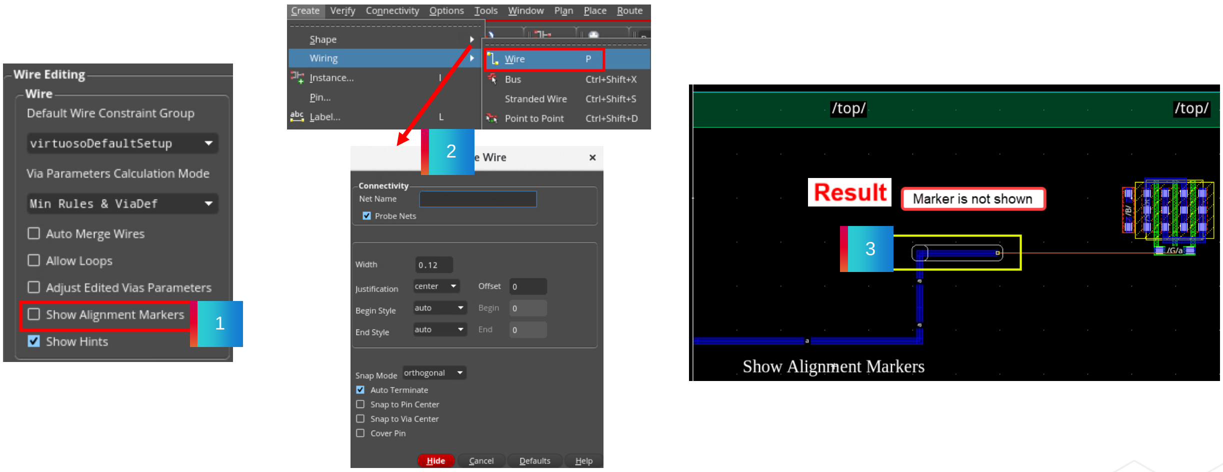
Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.

Create a wire from the *Metal1* pathSeg and move the mouse pointer upwards.

You can observe that no marker is visible when creating the wire because the *Show Alignment Markers* option is *off* in the Layout Editor Options form.

Result: Disabling Show Alignment Markers Option

Result: Marker is not visible because the **Show Alignment Markers** option is **off** in the Layout Editor Options form.



Result: Disabling Show Alignment Markers Option

Invoke the Layout Editor Options form and disable the *Show Alignment Markers* option.

When you create a wire and move the mouse pointer upwards, you can observe that marker is not visible because the *Show Alignment Markers* option is *off* in the Layout Editor Options form.

Enabling Show Alignment Markers Option

In the Layout Editor Options form, with the **Show Alignment Markers** option enabled, when the wire is aligned to the target pin, the marker is visible.

Enable the Show Alignment Markers option. 

1. In the Layout Editor Options form, in the **Wire Editing: Wire** section, enable the **Show Alignment Markers** option.
2. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire command.
3. Create a wire from the **Metal1** pathSeg and move the mouse pointer upwards.

Result: Marker is visible because the **Show Alignment Markers** option is **on** in the Layout Editor Options form.

Enabling Show Alignment Markers Option

In the Layout Editor Options form, with the *Show Alignment Markers* option enabled, when the wire is aligned to the target pin, the marker is visible.

You can enable the *Show Alignment Markers* checkbox to get this feature.

We will see the *Show Alignment Markers* feature now.

In the Layout Editor Options form, in the *Wire Editing* section, from the *Wire* section, enable the *Show Alignment Markers* option.

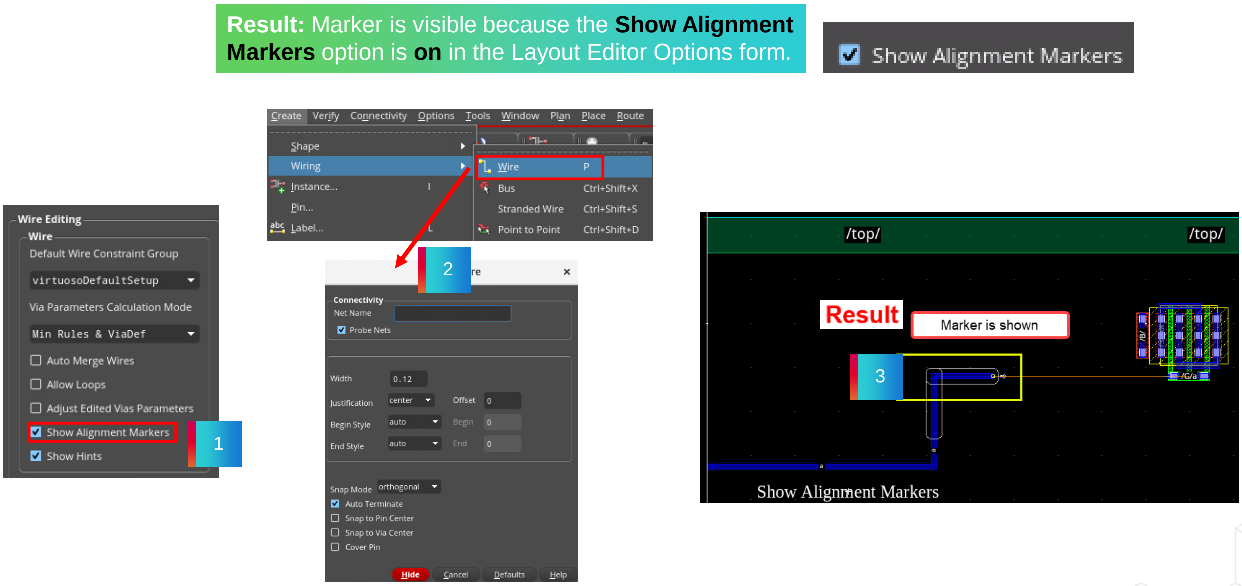
Then in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire command.

Create a wire from the *Metal1* pathSeg and move the mouse pointer upwards.

You can observe that marker is visible when creating the wire because the *Show Alignment Markers* option is *on* in the Layout Editor Options form.

Result: Enabling Show Alignment Markers Option

Result: Marker is visible because the **Show Alignment Markers** option is **on** in the Layout Editor Options form.



Result: Enabling Show Alignment Markers Option

Invoke the Layout Editor Options form and enable the *Show Alignment Markers* option.

When you create a wire and move the mouse pointer upwards, you can observe that marker is visible because the *Show Alignment Markers* option is *on* in the Layout Editor Options form.

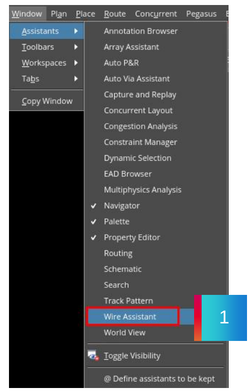
Invoking the Wire Assistant (WA)

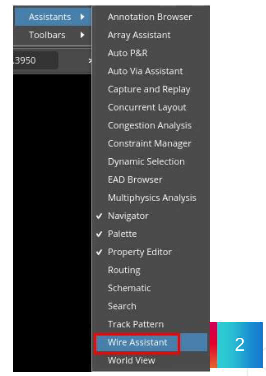
The **Wire Assistant** is a dockable assistant that provides a quick, easy, and single point of access to the commonly used options for all routing purposes: automatic routing, interactive routing, assisted routing, and post-route optimization.

To access the Wire Assistant in the layout canvas:

1. Select **Window – Assistants – Wire Assistant**.
2. Or **right**-click in the main menu or toolbar area and select **Assistants – Wire Assistant**.

Result: The **Wire Assistant** is displayed in the design canvas.





Invoking the Wire Assistant (WA)

The *Wire Assistant* is a dockable assistant that provides a quick, easy, and single point of access to the commonly used options for all routing purposes: automatic routing, interactive routing, assisted routing, and post-route optimization.

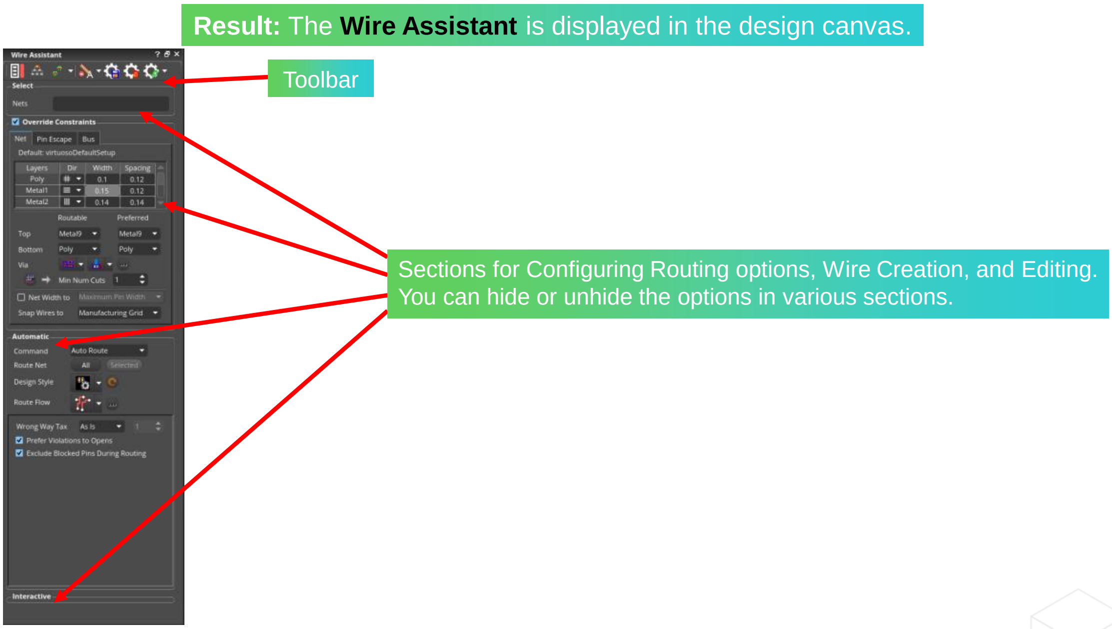
To access the Wire Assistant, in the layout canvas:

you can select *Window – Assistants – Wire Assistant* menu command.

Or you can *right*-click in the main menu or toolbar area and select *Assistants – Wire Assistant* menu command.

You can observe that the *Wire Assistant* is displayed in the design canvas.

Result: Invoking the Wire Assistant (WA)



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Result: Invoking the Wire Assistant (WA)

Once launched, the *Wire Assistant* is added as a docked assistant pane within the current session window.

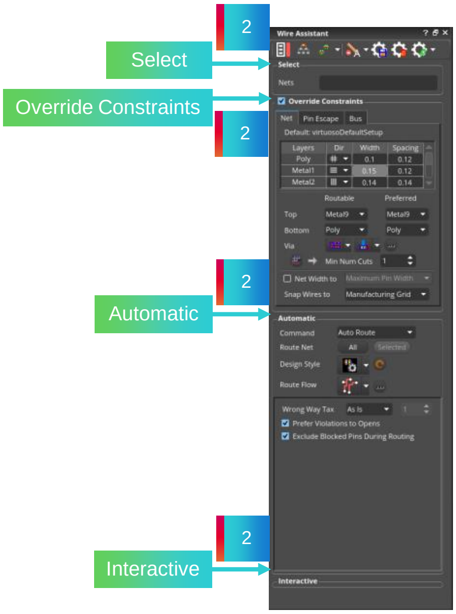
By default, the *Wire Assistant* is positioned at the lower right area of the session window.

The *Wire Assistant* contains two components, namely: the toolbar and the different sections, which are used for configuring the routing settings and configuring the settings while creating and editing a wire or a bus.

You can hide or unhide the options in various sections.

You can also configure the fields and options to appear in each section as required.

The Wire Assistant User Interface



The **Wire Assistant** contains two components:

1. Wire Assistant Toolbar
2. Wire Assistant Sections: Select, Override Constraints, Automatic, and Interactive

The Wire Assistant User Interface

The *Wire Assistant User Interface* comprises of two components, namely: *Wire Assistant Toolbar* and *Wire Assistant Sections*: which are *Select*, *Override Constraints*, *Automatic*, and *Interactive* which can be used for Configuring Routing options, Wire Creation, and Editing.

The Wire Assistant User Interface (continued)

- The Wire Assistant provides single-point access to some of the constraint values to be used by the interactive routing for creating and editing wires.
- You can override values for constraints such as minimum width, minimum spacing, minimum number of cuts, valid layers, and valid vias.

The **Wire Assistant** is available in the Virtuoso Layout Suite XL/EXL/MXL. tiers.

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The Wire Assistant User Interface

The *Wire Assistant* provides single-point access to some of the constraint values to be used by the interactive routing for creating and editing wires.

You can override values for constraints such as minimum width, minimum spacing, minimum number of cuts, valid layers, and valid vias.

The *Wire Assistant* is available in Virtuoso Layout Suite XL, EXL and MXL tiers.

How to Populate the Override Constraints Section in the Wire Assistant

Using the constraint look-up feature, the Wire Assistant provides the values to be used and a way to create transient constraints to override some of the values in the fields.

- You can override values for constraints such as minimum width, minimum spacing, minimum number of cuts, valid layers, and valid vias.

1 Using Existing Pro

2 Creating New Pro

This page does not contain notes.

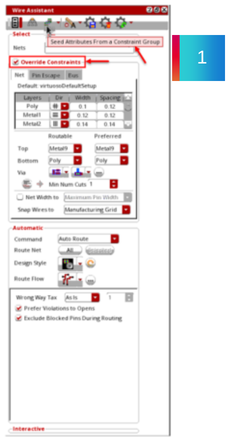
Using Existing PRO to Populate the Override Constraints Section in the Wire Assistant

The design rules included in the Constraint Group can be overridden using the option:

Seed Attributes From a Constraint Group

1 Click on the **Seed Attributes From a Constraint Group** drop-down list on the **Wire Assistant** toolbar.

2 Select the **dsn:M1_M3_200n (Net)** constraint group to populate the **Override Constraints** section with the new overridden values.



Using Existing PRO to Populate the Override Constraints Section in the Wire Assistant

You can override the design rules included in the Constraint Group using the Seed Attributes From a Constraint Group option in the Wire Assistant.

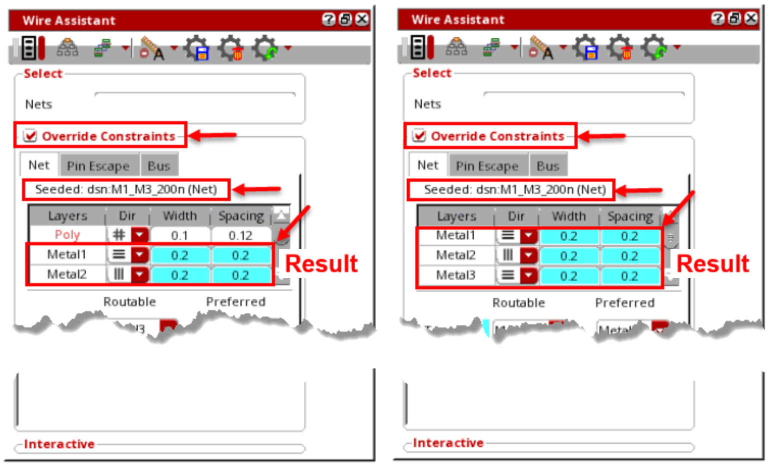
Click on the Seed Attributes From a Constraint Group drop-down list on the Wire Assistant toolbar.

Select the dsn:M1_M3_200n (Net) constraint group to populate the Override Constraints section with the new overridden values.

Result: Using Existing PRO to Populate the Override Constraints Section in the Wire Assistant

Result

The **Override Constraints** section populates with the new set of rules (PRO) of **dsn:M1_M3_200n (Net)** constraint group.



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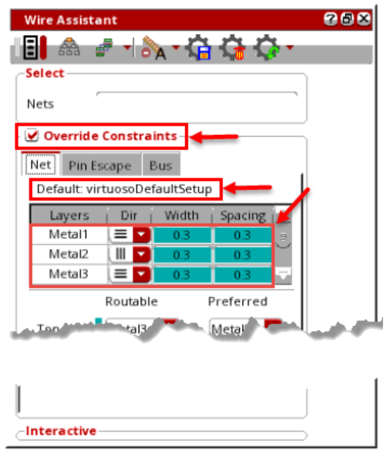
Result: Using Existing PRO to Populate the Override Constraints Section in the Wire Assistant

Once you have selected the *dsn:M1_M3_200n (Net)* constraint group from the *Seed Attributes From a Constraint Group* drop-down list on the wire assistant toolbar, the *Override Constraints* section populates with the new set of rules of this constraint group as seen in this example.

Creating New PRO Dynamically to Populate the Override Constraints Section in the Wire Assistant

Modify the rules in the **Override Constraints** section as in this example:
Width and **Spacing** values of **Metal1**, **Metal2**, and **Metal3** to **0.3**



Creating New PRO Dynamically to Populate the Override Constraints Section in the Wire Assistant

Modify the rules in the *Override Constraints* section, say for example:

Width and *Spacing* values of *Metal1*, *Metal2*, and *Metal3* to say, *0.3*.

You can *Route With WA Overrides* with dynamically created Process Rule Overrides.

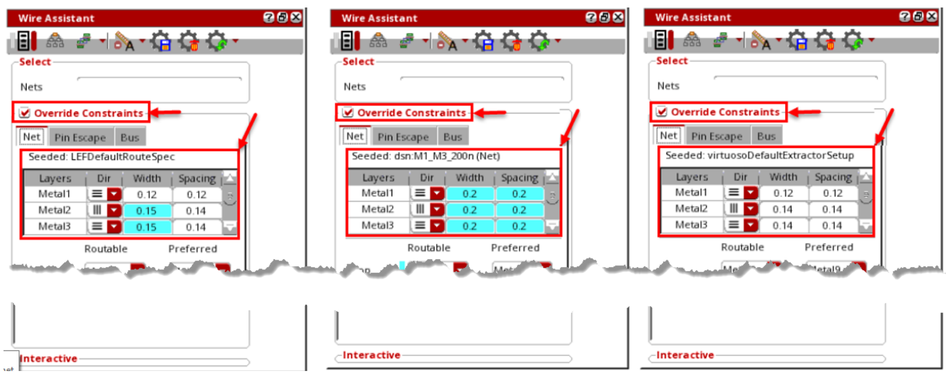
Switching Between Different Constraint Groups

Experiment switching between these three different sets of constraint groups.

LEFDefaultRouteSpec

dsn:M1_M3_200n (Net)

virtuosoDefaultExtractorSetup



Every set of rules brings in different values for the **Width** and **Spacing** values of the layers in the **Override Constraints** section.

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Switching Between Different Constraint Groups

Experiment switching between these three different sets of constraint groups, namely:

LEFDefaultRouteSpec, *dsn:M1_M3_200n (Net)* and *virtuosoDefaultExtractorSetup*.

Every set of rules bring in different values for the *Width* and *Spacing* values of the layers in the *Override Constraints* section.

FAQ: Different Types of Wires

What are the two different types of wires you can create?

The different types of wires you can create are **Geometric Wires** and **Symbolic Wires**.

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This page does not contain notes.

FAQ: Path Segment

What Is **Path Segment** (**pathSeg**)?

A **pathSeg** is a specialized shaped-based object represented by a specified width, a layer-purpose pair, and a two-point routing segment with a style associated with each of the points.

This page does not contain notes.

FAQ: Path

What Is **Path**?

A **path** object is a shaped-based object represented by a point array with a specified width, style, and layer-purpose pair.

This page does not contain notes.

FAQ: Layout Editor Options Form

How do you invoke the **Layout Editor Options** form?

In the layout canvas, choose the **Options – Editor** menu command or press the bindkey **Shift+E** to invoke the Layout Editor Options form.

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FAQ: Overriding the Design Rules in the Constraint Group

Which option in the **Wire Assistant** can you use to override the design rules included in the Constraint Group?

You can use the **Seed Attributes From a Constraint Group** option in the **Wire Assistant** to override the design rules included in the Constraint Group.

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Module Summary

In this module, you learned how to

- Use Wire Editing Options with the Create Wire command
  - Auto Merge Wires
  - Allow Loops
  - Adjust Edited Vias Parameters
  - Show Alignment Markers
- Use Override Constraints in the Wire Assistant
  - Use existing PRO to populate the Override Constraints section in the Wire Assistant
  - Create a new PRO dynamically to populate the Override Constraints section in the Wire Assistant
  - Experiment switching between different constraint groups

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Labs

Lab 2-1 Using Wire Editing Options with the Create Wire Command

Lab 2-2 Using Override Constraints in the Wire Assistant

Now, you will do lab exercises for Module 2.

There are 2 labs in this module.

In the first lab, Using Wire Editing Options with the Create Wire command, you explore the Auto Merge Wires, Allow Loops, Adjust Edited Vias Parameters, and Show Alignment Markers options available in the Layout Editor Options form.

In the second lab, Using Override Constraints in the Wire Assistant, you Use Existing Process Rule Overrides to Populate the Override Constraints Section in the Wire Assistant, Create New Process Rule Overrides Dynamically to Populate the Override Constraints Section in the Wire Assistant, and experiment Switching Between Different Constraint Groups.

Quiz: Type of Wire Used for Signal Routing

Which type of wire can you typically use for signal routing which is the default for creating wires in Virtuoso?

Select all that apply.

A. Geometric
B. Symbolic
C. Signal
D. Trunk
E. None of the above

Answer: B

You can use the **Symbolic** wire for signal routing which is the default for creating wires in Virtuoso.

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Quiz: The Wire Assistant in VLS

The **Wire Assistant** is available in Virtuoso Layout Suite XL/EXL/MXL tiers.

A. True

B. False

Answer: B

The **Wire Assistant** is available in Virtuoso Layout Suite XL/EXL/MXL tiers.

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Quiz: Creation of Loops

Which option in the **Layout Editor Options** form permits the creation of loops when creating or editing wires? ________________

Enter the answer in the blank field.

Answer: **Allow Loops**

Allow Loops option in the **Layout Editor Options** form permits creation of loops when creating or editing wires.

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Quiz: Auto Merging Wires

While creating or stretching a wire, the **Auto Merge Wires** option in the **Layout Editor Options** form automatically:

Select all that apply.

A. Merges connected paths

B. Merges collinear pathSegs

C. Merges paths on adjacent nets

D. Fixes any loops in a wire

E. All of the above

Answer: A and B

While creating or stretching a wire, the **Auto Merge Wires** option in the **Layout Editor Options** form automatically merges connected paths and merges collinear pathSegs.

This page does not contain notes.

Quiz: The Assistant Used for Routing

Which dockable assistant provides a quick, easy, and single point of access to the commonly used options for all routing purposes: automatic routing, interactive routing, assisted routing, and post-route optimization?

Select all that apply.

A. Annotation Browser Assistant
B. Navigator Assistant
C. Property Editor Assistant
D. Wire Assistant
E. None of the above

Answer: D

The **Wire Assistant** provides a quick, easy, and single point of access to the commonly used options for all routing purposes.

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Module 3

Creating and Modifying Wires

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In this module, we will discuss Creating and Modifying Wires.

Module Objectives

In this module, you

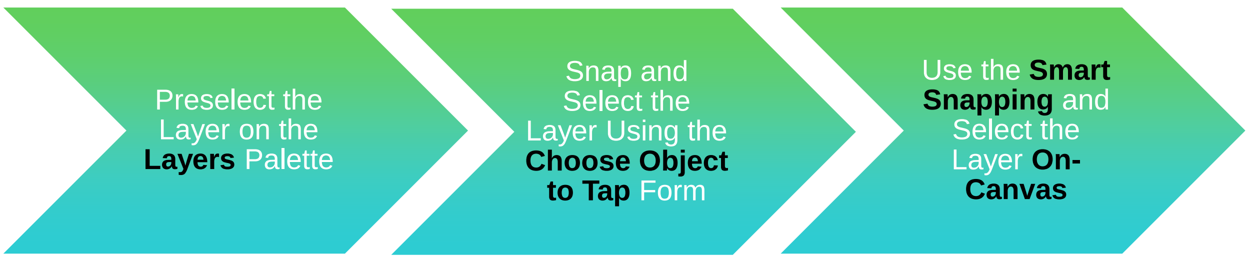
- Connect to target pins with the Create Wire command
- Use the Interactive Via Alignment feature

In this module, you will explore how to connect to target pins with the Create Wire command.

You will explore how to create wire using the Interactive Via Alignment feature.

Selecting the Layer to Create Wire

When creating wires, you can use the different methods available to select the layer. The options available are:



The **Create Wire** command works only with routing layers.

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Selecting the Layer to Create Wire

When creating wires, you can use the different methods available to select the layer.

The options available are:

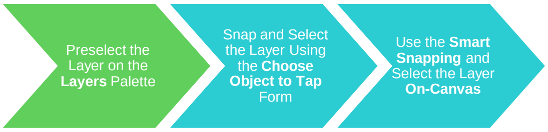
Preselect the Layer on the *Layers* Palette.

Snap and Select the Layer Using the *Choose Object to Tap* Form.

Use the *Smart Snapping* and Select the Layer *On-Canvas*.

The *Create Wire* command works only with routing layers.

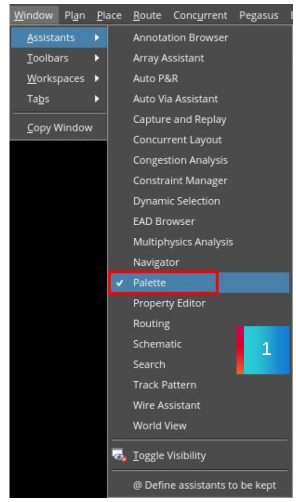
Preselect the Layer on the Layers Palette

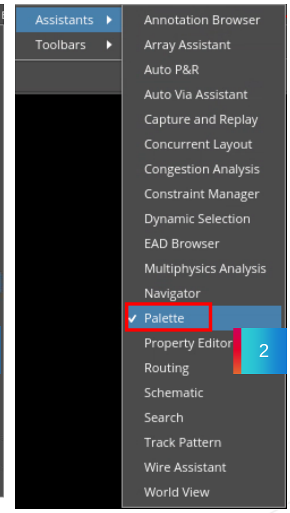


To access the **Layers** palette in the layout canvas:

1. Select **Window – Assistants – Palette**.
2. Or **right**-click in the main menu or toolbar area and select **Assistants – Palette**.

Result: The **Layers** palette is added as a docked assistant pane in the layout window.





Preselect the Layer on the Layers Palette

To access the *Layers* Palette, in the layout canvas:

Select *Window – Assistants – Palette* menu command.

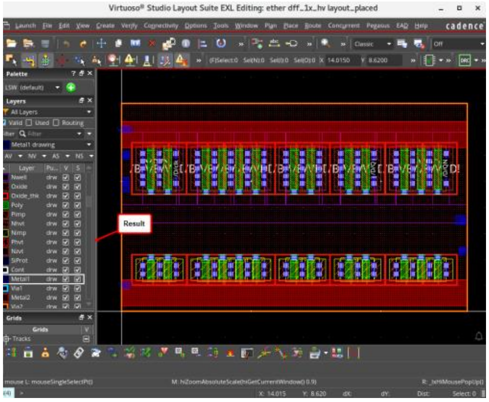
Or *right*-click in the main menu or toolbar area and select the *Assistants – Palette* menu command.

You can notice that the *Layers* Palette is added as a docked assistant pane in the layout window.

Result: Preselect the Layer on the Layers Palette

Result: Once launched, the **Layers** palette is added as a docked assistant pane within the current session window.

By default, the **Layers** palette is positioned at the left area of the session window.



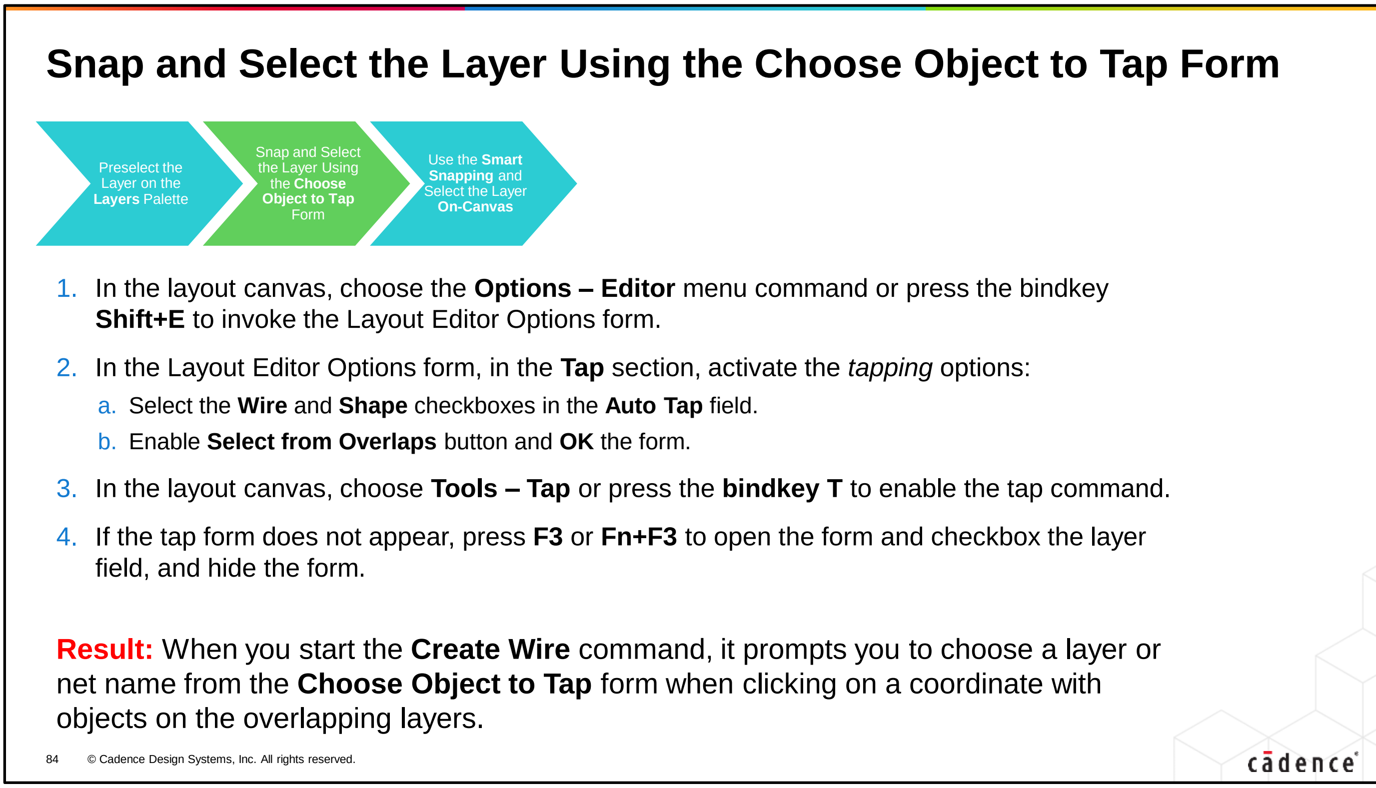
If the current active layer is available, it will be used to start the new wire without any further request.

Result: Preselect the Layer on the Layers Palette

You can observe that, once launched, the *Layers* Palette is added as a docked assistant pane within the current session window.

By default, the *Layers* Palette is positioned at the left area of the session window.

If the current active layer is available, it will be used to start the new wire, without any further request.



Snap and Select the Layer Using the Choose Object to Tap Form

In the layout canvas, choose the *Options – Editor* menu command or press the bindkey *Shift+E* to invoke the Layout Editor Options form.

In this form, in the *Tap* section, activate the following *tapping* options:

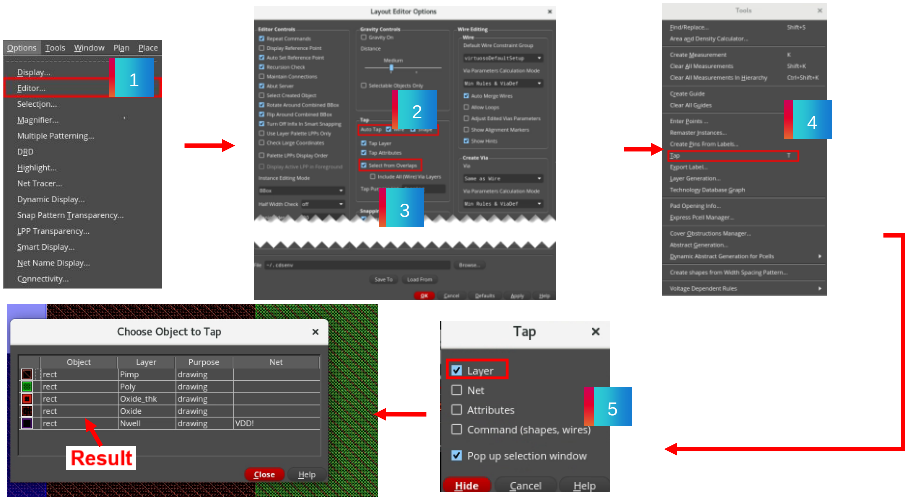
Select the *Wire* and *Shape* checkboxes in the *Auto Tap* field.

Enable *Select from Overlaps* button and *OK* the form.

You can observe that, when you start the *Create Wire* command, it prompts you to choose a layer or net name from the *Choose Object to Tap* form when clicking on a coordinate with objects on the overlapping layers.

Result: Snap and Select the Layer Using the Choose Object to Tap Form

Result: When you start the **Create Wire** command, it prompts you to choose a layer or net name from the **Choose Object to Tap** form when clicking on a coordinate with objects on the overlapping layers.



You can choose from the list of all the layers under the cursor in the **Choose Object to Tap** form.

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Result: Snap and Select the Layer Using the Choose Object to Tap Form

As you see in this example, in the *Layout Editor Options* form, in the *Tap* section, after setting the required options when starting the *Create Wire* command when clicking on a coordinate with objects on the overlapping layers, the *Choose Object to Tap* form pops up from which you can choose from the list of all the layers under the cursor in this form.

Use the Smart Snapping and Select the Layer On-Canvas

Preselect the Layer on the **Layers** Palette → Snap and Select the Layer Using the **Choose Object to Tap** Form → Use the **Smart Snapping** and Select the **Layer On-Canvas**

1. In the layout canvas, choose the **Options – Editor** menu command or press the bindkey **Shift+E** to invoke the Layout Editor Options form.
2. In the Layout Editor Options form, in the **Tap** section, activate the *tapping* options:
   a. Select the **Wire** and **Shape** checkboxes in the **Auto Tap** field and **OK** the form.
3. Enable **Smart Snapping** in the **Create Single Wire** context-sensitive menu.

Result: Start the **Create Wire** command and move the mouse around the starting point; the different available layers will be shown directly on the canvas.

Use the Smart Snapping and Select the Layer On-Canvas

In the layout canvas, choose the *Options – Editor* menu command or press the bindkey *Shift+E* to invoke the Layout Editor Options form.

In this form, in the *Tap* section, activate the following *tapping* options:

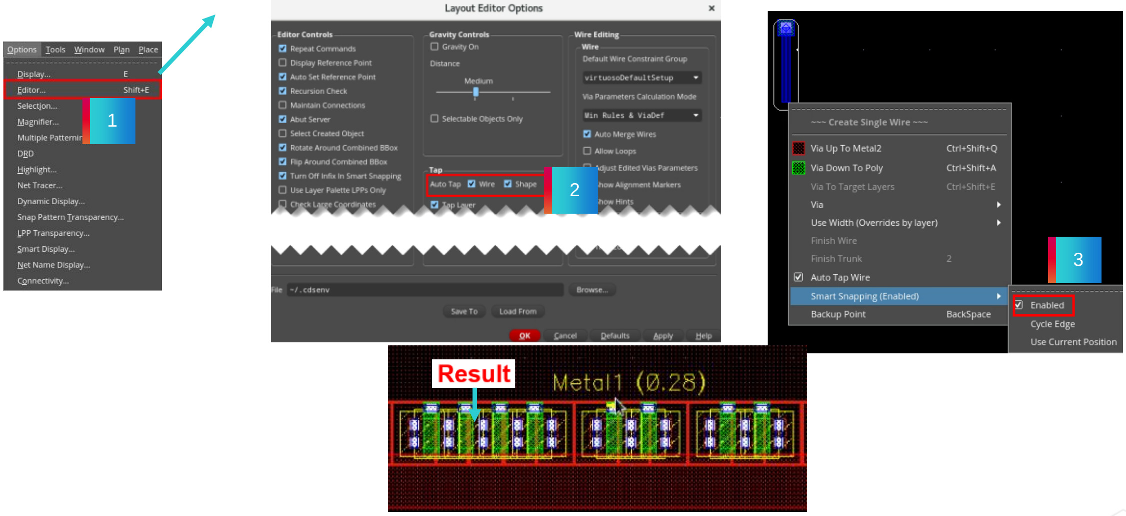
Select the *Wire* and *Shape* checkboxes in the *Auto Tap* field and *OK* the form.

Enable *Smart Snapping* in the *Create Single Wire* Context-Sensitive Menu.

You can observe that, when you start the *Create Wire* command and move the mouse around the starting point, the different available layers will be shown directly on the canvas.

Result: Use the Smart Snapping and Select the Layer on Canvas

Result: Start the **Create Wire** command and move the mouse around the starting point; the different available layers will be shown directly on the canvas.



Result: Use the Smart Snapping and Select the Layer On-Canvas

As you see in this example, in the *Layout Editor Options* form, in the *Tap* section, after setting the required options and with the *Smart Snapping* option enabled in the *Create Single Wire* context-sensitive menu, when you start the *Create Wire* command and move the mouse around the starting point, the different available layers will be shown directly on the canvas.

Invoking the Create Wire Form

In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.

- Using the *Create Wire* form, you can create pathSegs and vias in routes.

Create Wire Form in XL/EXL/MXL Tiers

Connectivity

Net Name lets you enter a net name for the current wire. The created wire is added to the specified net. A new net is created if the one specified does not already exist. To specify net names for creating a wire, enter a list of net names separated by commas or spaces.

Width displays the width of the current wire.

Justification controls the direction in which to offset the pathSeg from the digitized points of the path or pathSeg.

Offset creates a wire at a specified distance from the digitized points with respect to the centerline of the pathSeg, in the direction specified by *Justification*. The default is 0.

Begin Style controls how the beginning wire segment ends are created.

End Style controls how the ending wire segment ends are created.

Snap Mode controls how the cursor snaps when you create the wire.

Auto Terminate finishes the *Create Wire* command as soon as a point is digitized on a flightline target object (pin or existing wire or via). When creating a bus and *Auto Terminate* is on, all wires in the bus snap to its target pin. This option is selected by default.

Snap to Pin Center snaps wires to the center of rectangular pins or of pins that are created as polygons and have rectangular shapes. If this checkbox is selected, wires snap to the starting and ending pins. The *Create – Shape – Geometric Wire* and *Create – Wiring – Wire* commands snap wires to the centerline of paths and pathSegs, and the origin of vias when tapping those objects regardless of whether *Snap to Pin Center* is on or off. The *Snap to Pin Center* checkbox is off by default.

Snap to Via Center snaps wires to the center of vias. If this checkbox is selected and you click a via, the starting wire starts from the clicked point and two small segments are added to connect to the via origin. With this checkbox selected, tapping a via and a wire gives preference to the via. This checkbox is off by default.

Cover Pin adjusts the wire segment to automatically cover the source and target pins. Otherwise, the wire segment is not adjusted, and the wire terminates at a point you had chosen. By default, this option is deselected.

Displaying the Create Single Wire Context-Sensitive Menu

When creating wires, you can use the options available in the **Create Single Wire** context-sensitive menu which enhances the routing functionality.

This context-sensitive menu can be accessed when the ***Create Geometric Wire*** or ***Create Wire*** (XL/EXL/MXL) command is active.

Displaying the Create Single Wire Context-Sensitive Menu

When creating wires, you can use the options available in the *Create Single Wire* context-sensitive menu which enhances the routing functionality.

This context-sensitive menu can be accessed when the *Create Geometric Wire* or *Create Wire* command is active in all the tiers of Virtuoso Layout Suite.

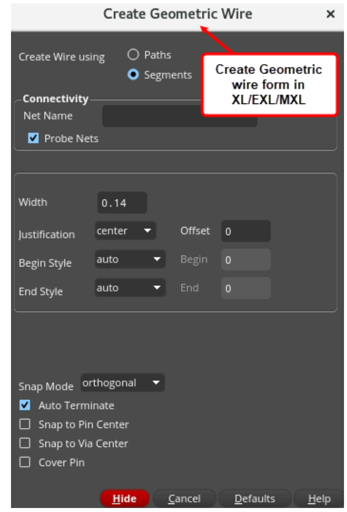
Enabling Create Geometric Wire Command Active

Enabling **Create Geometric Wire Command** active

Enabling **Create Wire Command** active

1. To create geometric wires, enter the **CIW>leHiCreateGeometricWire** command to open the Create Geometric Wire form.
2. **Right**-click in the design canvas when the *Create Geometric Wire* (XL/EXL/MXL) command is active.

Result: The **Create Single Wire** context-sensitive menu is displayed in the layout canvas.



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When the Create Geometric Wire Command is active

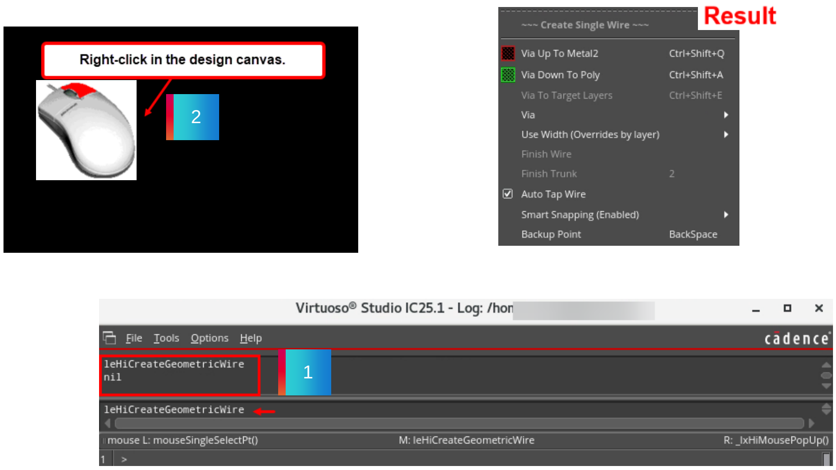
To create geometric wires, in the CIW window, enter the *leHiCreateGeometricWire* SKILL® command to open the Create Geometric Wire form.

Right-click in the design canvas when the *Create Geometric Wire* command is active in all the tiers of Virtuoso Layout Suite.

The *Create Single Wire* context-sensitive menu is displayed in the layout canvas.

Result: When the Create Geometric Wire Command Is Active

Create Single Wire context-sensitive menu in XL/EXL/MXL.



Result: When the Create Geometric Wire Command is active

When creating geometric wires, you can **right**-click in the design canvas to display the *Create Single Wire* context-sensitive menu.

You can use the options available in this context-sensitive menu to enhance your routing features when creating geometric wires.

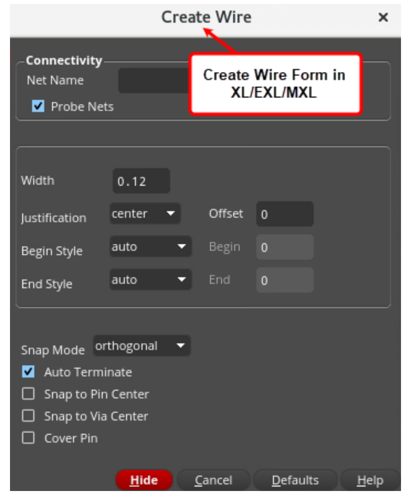
Enabling Create Wire Command Active

Enabling **Create Geometric Wire Command** active

Enabling **Create Wire Command** active

1. To create symbolic wires, in the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.
2. **Right**-click in the design canvas when the *Create Wire* (XL/EXL/MXL) command is active.

Result: The **Create Single Wire** context-sensitive menu is displayed in the layout canvas.

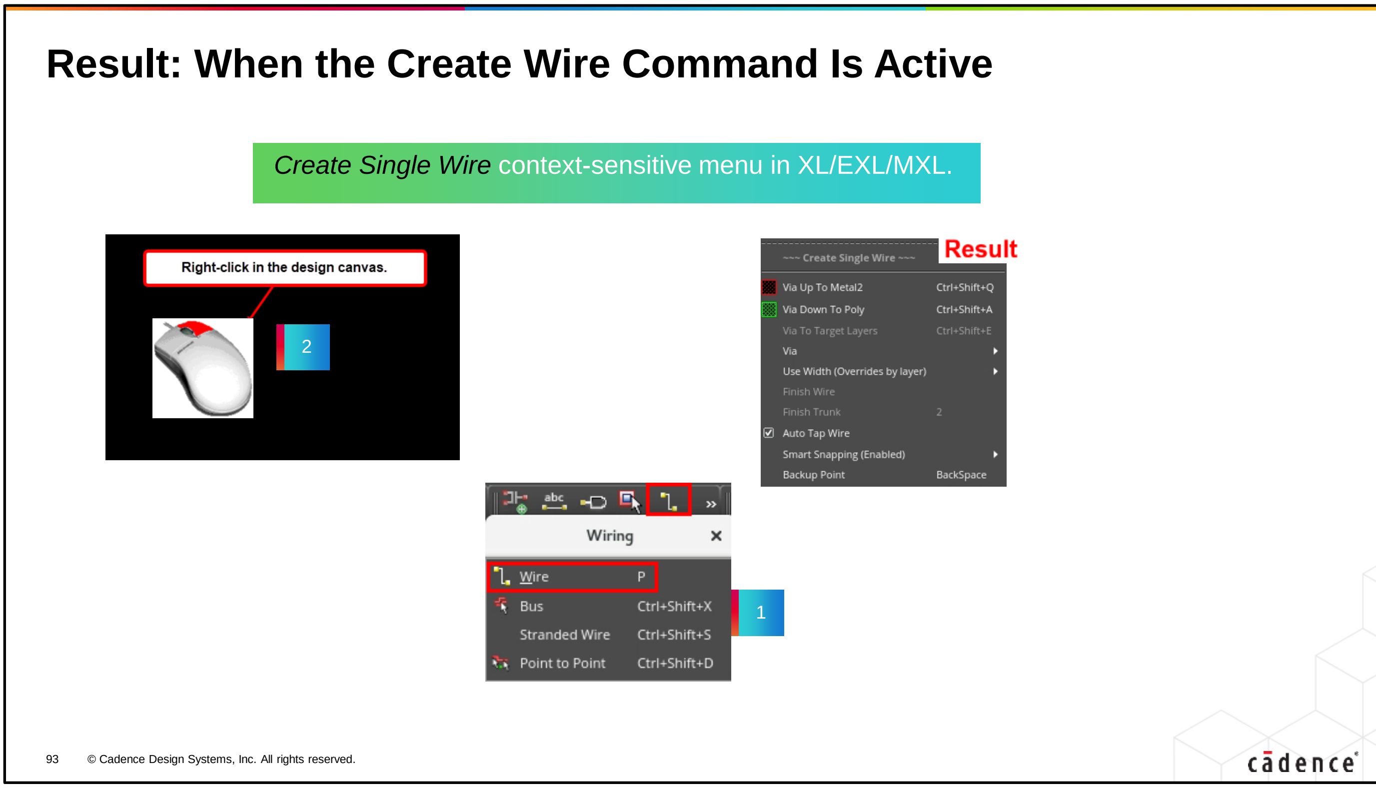


When the Create Wire Command is active

To create symbolic wires, in the design canvas, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire form.

Right-click in the design canvas when the *Create Wire* command is active in the XL/EXL/MXL tiers of Virtuoso Layout Suite.

The *Create Single Wire* context-sensitive menu is displayed in the layout canvas.

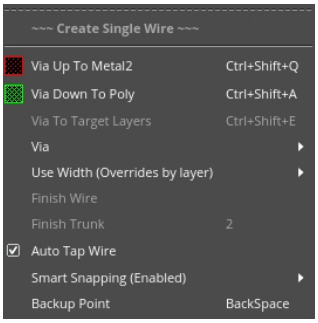


Result: When the Create Wire Command is active

When creating symbolic wires, you can *right*-click in the design canvas to display the *Create Single Wire* context-sensitive menu.

You can use the options available in this context-sensitive menu to enhance your routing features when creating symbolic wires.

Create Single Wire Context-Sensitive Menu Options



Via Up To	Changes the layer to the next layer up.
Via Down To	Changes the layer to the next layer down.
Via To Target Layers	Adds via stack to the layer of the target object.
Via	Lets you select the via options.
Use Width	Use this option to specify the width.
Finish Wire	Automatically routes the active flightline following the layer, via, and width specifications defined in the application default constraint group or any overrides defined in either the F3 Options form or the Wire Assistant. The Finish Wire command is available only for symbolic wires.

The Create Single Wire Context-Sensitive Menu Options

When creating wires, you can use the options available in the *Create Single Wire* context-sensitive menu which enhances the routing functionality.

Via Up To option changes the layer to the next layer up.

Via Down To option changes the layer to the next layer down.

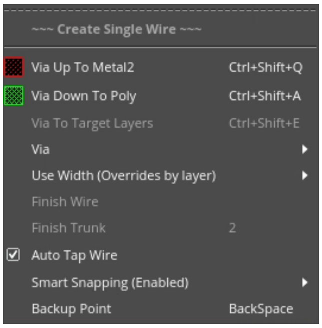
Via To Target Layers option adds via stack to the layer of the target object.

Via option lets you to select the via options.

Use the *Use Width* option to specify the width.

Finish Wire option automatically routes the active flightline following the layer, via, and width specifications defined in the application default constraint group or any overrides defined in either the F3 Options form or the Wire Assistant. The *Finish Wire* command is available only for symbolic wires.

Create Single Wire Context-Sensitive Menu Options (continued)



Finish Trunk	Automatically converts the pathSeg that is being created using the Create Wire or Create Bus command into a trunk. The default bindkey is 2.
Auto Tap Wire	Automatically sets the layer of the wire being created to be the same as that of an existing clicked wire. This option is on by default.
Smart Snapping	This command is available on the Create Single Wire context-sensitive menu in XL,EXL, MXL.
Backup Point	Removes the last digitized point. If a via had been added to the last digitized point, the via is also removed.

The Create Single Wire Context-Sensitive Menu Options (continued)

Finish Trunk option automatically converts the pathSeg that is being created using the Create Wire or Create Bus command into a trunk. The default bindkey is 2.

Auto Tap Wire option automatically sets the layer of the wire being created to be the same as that of an existing clicked wire. This option is on by default.

Smart Snapping option is available on the Create Single Wire context-sensitive menu in XL and GXL.

Constraint-Aware Editing option enforces constraints for nets grouped in a bus constraint and for setting shielding wires. This option is on by default and is available only in XL,EXL,MXL.

Backup Point option removes the last digitized point. If via had been added to the last digitized point, the via is also removed.

Connecting to Target Pins with the Create Wire Command

Use the **Create Wire** command to connect to the target pins.

1. In the design canvas, select the net **D** in the Navigator assistant.
2. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.
3. Start routing from either the **NM0** or the **PM0** device toward pin **D**.

Result: As you near pin **D**, you see the possible connecting targets appear. As you move your mouse pointer around the pin, the targets will cycle.

You can use the **Ctrl+Space** key combination to cycle the targets.

Connecting to Target Pins with the Create Wire Command

You can use the Create Wire command to connect to the target pins.

In the design canvas, select the net *D* in the Navigator assistant.

Then, choose the *Create – Wiring – Wire* menu command or use the *Create Wire* icon in the Create toolbar or use the bindkey *P* to invoke the Create Wire form.

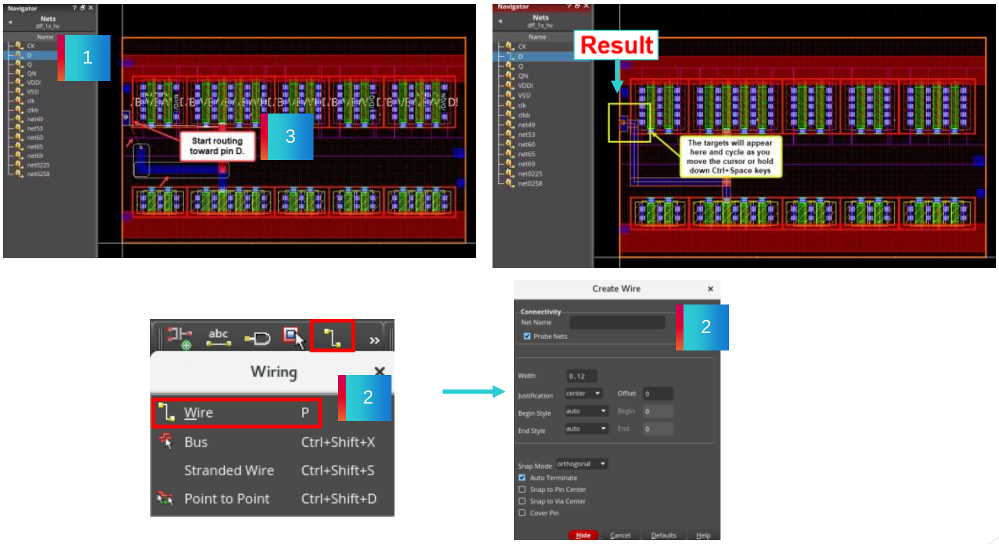
When you start routing from either the *NMOS* or the *PMOS* transistor toward pin *D*, you can see that, as you near pin *D*, you see the possible connecting targets appear.

As you move your mouse pointer around the pin, the targets will cycle.

You can use the *Ctrl+Space* key combination to cycle the targets.

Result: Connecting to Target Pins with the Create Wire Command

You can use the connecting targets to connect to the pin.



Result: Connecting to Target Pins with the Create Wire Command

As you see in this example, after selecting the net *D* in the Navigator assistant, when you start routing from one of the transistor device, when you near pin *D*, you see the possible connecting targets appear which are cycling when you move your mouse pointer around the pin or use the *Ctrl+Space* key.

You can use the connecting targets to connect to the pin.

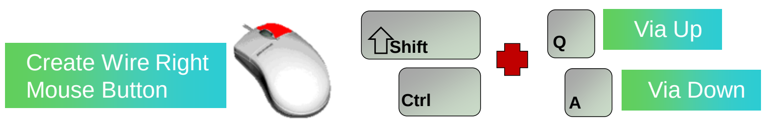
Creating Wire Using the Via Up, Via Down, and Finish Wire Options

When creating wires, the next upper/lower metal can be quickly reached by using the **Via Up** / **Via Down** menu entries on the ***Create Single Wire*** Context-Sensitive Menu or by using the bindkeys.

To complete creating the wire automatically, use the **Finish Wire** command on the ***Create Single Wire*** Context-Sensitive Menu.

1. In the design canvas, create a wire from pin **D**.
2. When changing layers, **right**-click and select **Via Up** on the *Create Single Wire* context-sensitive menu. **Middle**-click to rotate the via if needed.
3. To finish the connection, **right**-click and select **Via Down**. **Middle**-click to rotate the via if required.
4. **Right**-click and select **Finish Wire** to end the wiring.

Result: The vias created have the center of the layer enclosures matching the pathSeg centerline coinciding endpoints.



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Creating Wire Using the Via Up, Via Down, and Finish Wire Options

When creating wires, the next upper and lower metal can be quickly reached by using the *Via Up* and *Via Down* menu entries on the *Create Single Wire* Context-Sensitive Menu or by using the bindkeys.

To complete creating the wire automatically, use the *Finish Wire* command on the *Create Single Wire* Context-Sensitive Menu.

In the design canvas, create a wire from the pin *D*.

When changing layers, *right*-click and select *Via Up* on the *Create Single Wire* context-sensitive menu. Middle-click to rotate the via if needed.

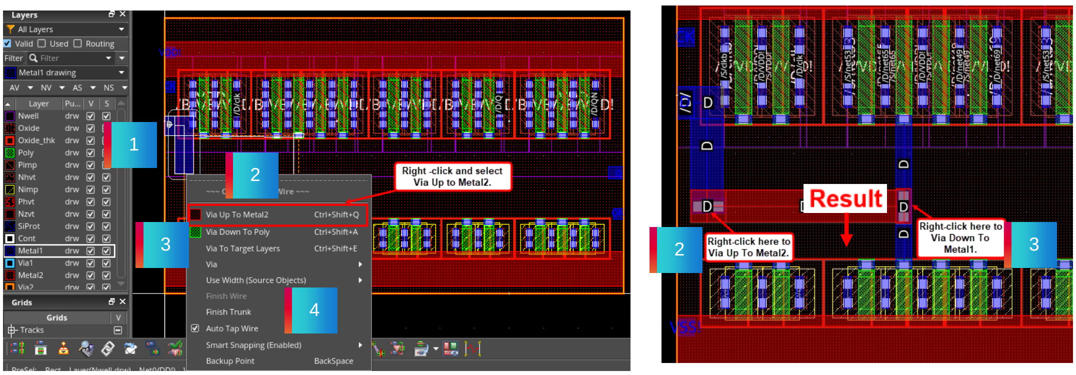
To finish the connection, *right*-click and select *Via Down*. *Middle*-click to rotate the via if required.

Right-click and select *Finish Wire* to end the wiring.

You can see that, the vias created have the center of the layer enclosures matching the pathSeg centerline coinciding endpoints.

Result: Creating Wire Using the Via Up, Via Down, and Finish Wire Options

You have used the **Via Up**, **Via Down**, and **Finish Wire** options on the *Create Single Wire* context-sensitive menu for your wiring connections.



The **Finish Wire** command is available only for symbolic wires.

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Result: Creating Wire Using the Via Up, Via Down, and Finish Wire Options

As you see in this example, you have used the *Via Up*, *Via Down*, and *Finish Wire* options available on the *Create Single Wire* context-sensitive menu for your wiring connections.

The *Finish Wire* command is available only for symbolic wires.

Creating Wire Using the Select Via Option

When creating wires, use the **Select Via** form to change to other layers, rotate the via, and control the via alignment which can be invoked from the ***Create Single Wire*** Context-Sensitive Menu or by using the bindkey.

1. In the design canvas, create a wire from pin **D**.
2. When changing layers, **right**-click and choose the **Via – Select Via** menu entry from the *Create Single Wire* context-sensitive menu or press the bindkey **Space** to invoke the Select Via form.
3. Choose the following options in the form: choose **Metal2** in the *Layer Transitions* section, select **Automatic** in the *Via Alignment* section, and select **Rotate Via(s) Cut Pattern** if required.
4. Complete the wiring connection.

Result: Alignment and rotation of the vias coincide with the direction of the routing.

Create Wire Right Mouse Button  Space Select Via

Creating Wire Using the Select Via Option

When creating wires, use the *Select Via* form to change to other layers, rotate the via, and control the via alignment which can be invoked from the *Create Single Wire* Context-Sensitive Menu or by using the bindkey.

In the design canvas, start the wiring connection from the pin *D*.

When changing layers, **right**-click and choose the *Via – Select Via* menu entry from the *Create Single Wire* context-sensitive menu or press the bindkey *Space* to invoke the Select Via form.

Choose the required options for the wiring in this form: say for example,

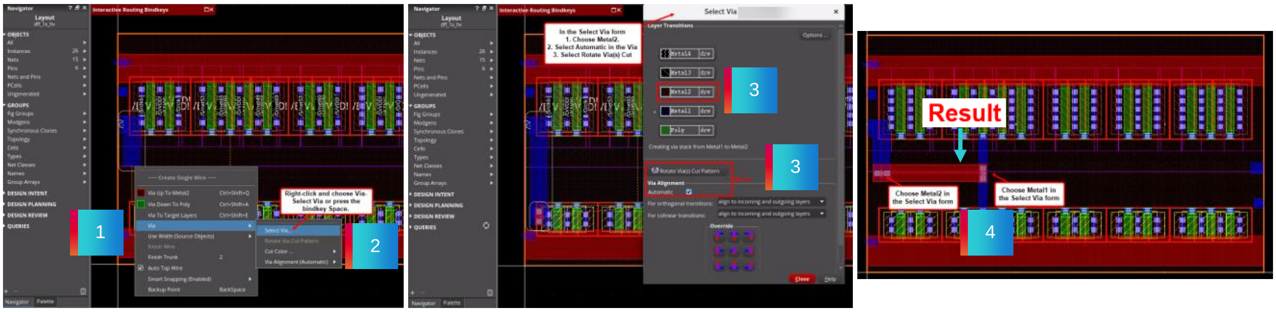
Choose *Metal2* in the *Layer Transitions* section, select *Automatic* in the *Via Alignment* section, and select *Rotate Via(s) Cut Pattern* if required.

Now complete the wiring connection.

You can see that, the alignment and rotation of the vias coincide with the direction of the routing.

Result: Creating Wire Using the Select Via Option

The **Select Via** form will present a list showing the complete metal stack.



The orientation and position of the vias can be automatically decided by the tool or it can be defined by the user.

Alignment and rotation of the vias coincide with the direction of the routing.

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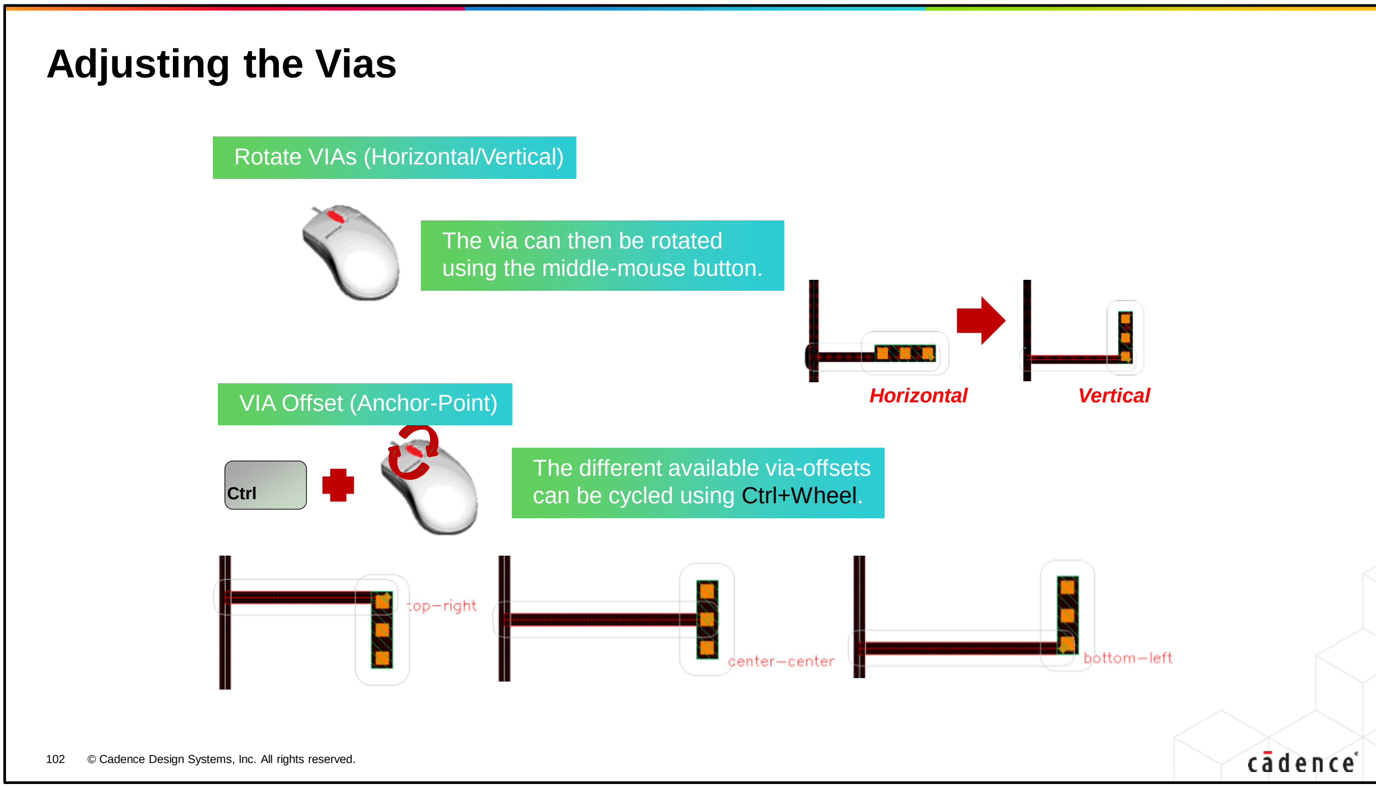
Result: Creating Wire Using the Select Via Option

As you see in this example, the *Select Via* form will present a list showing the complete metal stack.

The orientation and position of the vias can be automatically decided by the tool or it can be defined by the user.

Once you complete the wiring connection, you can see that, the alignment and rotation of the vias coincide with the direction of the routing.



Adjusting the Vias

After placing the vias, they can be rotated either horizontally or vertically.

As you see in this example, after placing the via horizontally, the via can then be rotated vertically using the middle mouse button.

The different available via offsets can be cycled using the *Ctrl+Wheel* mouse key options.

Creating Wire by Specifying the Min Num Cuts Value in the Wire Assistant

By specifying the via cuts as minimum number of cuts (**Min Num Cuts**) in the **Wire Assistant**, you can override the **minNumCut** rule value for creating and editing wires.

1. In the design canvas, set the value of **Min Num Cuts** to **4** in the Wire Assistant.
2. In the design canvas, create a wire from pin **D**.
3. Complete the wiring connection.

Result: This setting has forced the **Create Wire** command to use at least **4** cuts in all the vias that have been created in the routing.

Creating Wire By Specifying the Min Num Cuts Value in the Wire Assistant

By specifying the via cuts as minimum number of cuts in the *Wire Assistant*, you can override the *minNumCut* rule value for creating and editing wires.

In the design canvas, set the value of *Min Num Cuts*, to say for example *4*, in the Wire Assistant.

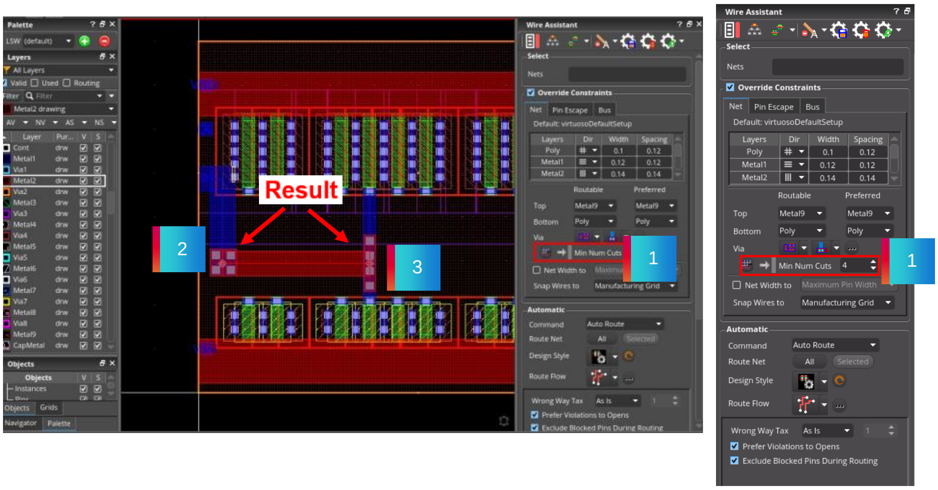
Then, create a wire from the pin *D*.

Complete the wiring connection using the *Via Up*, *Via Down*, and *Select Via* menu commands.

You can observe that, this setting has forced the *Create Wire* command to use at least *4* cuts in all the vias that has created in the routing.

Result: Creating Wire By Specifying the Min Num Cuts Value in the Wire Assistant

This setting has forced the **Create Wire** command to use at least **4** cuts in all the vias that have been created in the routing.



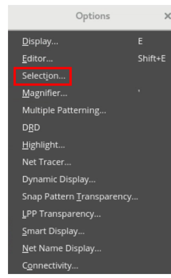
Result: Creating Wire By Specifying the Min Num Cuts Value in the Wire Assistant

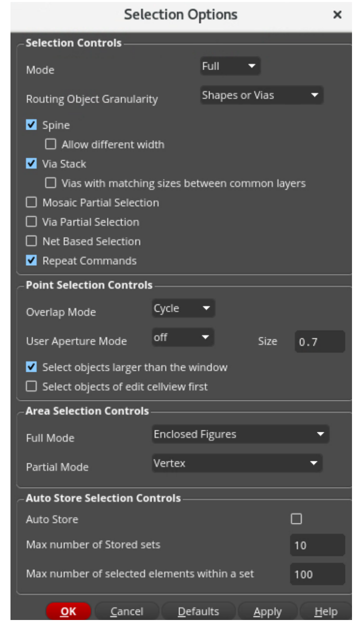
After specifying the *Min Num Cuts* value in the *Wire Assistant*, when you do the wiring, you can notice that, this setting has forced the *Create Wire* command to use at least the minimum number of cuts set in the *Wire Assistant* in all the vias that has created in the routing.

Invoking the Selection Options Form

In the design canvas, choose the **Options – Selection** menu command to invoke the Selection Options form.

Use the **Selection Options** form to control how selection is performed in the layout window.



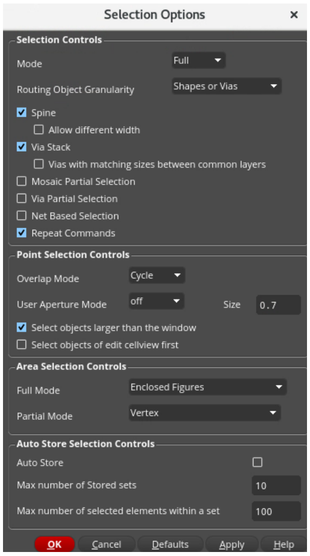


Invoking the Selection Options Form

In the design canvas, choose the *Options – Selection* menu command to invoke the Selection Options form.

You can use this form to control how the selection is performed in the layout window.

Selection Options Form: Controls Sections



The Selection Options form includes *Selection Controls*, *Point Selection Controls*, *Area Selection Controls*, and *Auto Store Selection Controls* sections.

Selection Controls	Use this section to choose between full and partial section modes.
Point Selection Controls	Use this section to control the behavior of point-and-click functionality.
Area Selection Controls	Use this section to control the behavior of area selection functionality.
Auto Store Selection Controls	Use this section to specify automatic selection expansion for routing objects and to control the size and number of selected sets that can be saved in the system for later reuse.

Selection Options Form – Controls Sections

The *Selection Options* form includes: *Selection Controls*, *Point Selection Controls*, *Area Selection Controls*, and *Auto Store Selection Controls* sections.

You can use the *Selection Controls* section to choose between full and partial section modes.

You can use the *Point Selection Controls* section to control the behavior of point-and-click functionality.

You can use the *Area Selection Controls* section to control the behavior of area selection functionality.

You can use the *Auto Store Selection Controls* section to specify the automatic selection expansion for routing objects and to control the size and number of selected sets that can be saved in the system for later reuse.

Changing the Width of an Existing Path Segment
Full Mode/Full Select

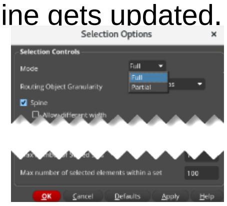
The *Mode* in the *Selection Controls* section of the **Selection Options form** affects the way the pathSegs are selected and modified.

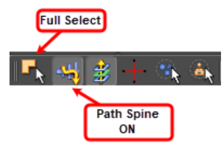
Full Mode: It fully selects objects inside the area selection box.

***Full Select* / Path Spine ON (default):** The complete spine is selected and modified.

1. In the **Selection Options** form, in the *Selection Controls* section, select *Full Mode* or enable *Full Select* from the **Options** toolbar.
2. Choose **Window – Assistants – Property Editor** to display the Property Editor assistant in the layout window.
3. Select a spine and change its width in the Property Editor assistant.

Result: The width of the entire spine gets updated.





Changing the Width of an Existing Path Segment Full Mode/Select

The *Mode* in the *Selection Controls* section of the *Selection Options* form affects the way the pathSegs are selected and modified.

Full Mode fully selects objects inside the area selection box.

With *Full Select* with Path Spine ON which is the default option, the complete spine is selected and modified.

In the *Selection Options* form, in the *Selection Controls* section, select *Full Mode* or enable *Full Select* from the *Options* toolbar.

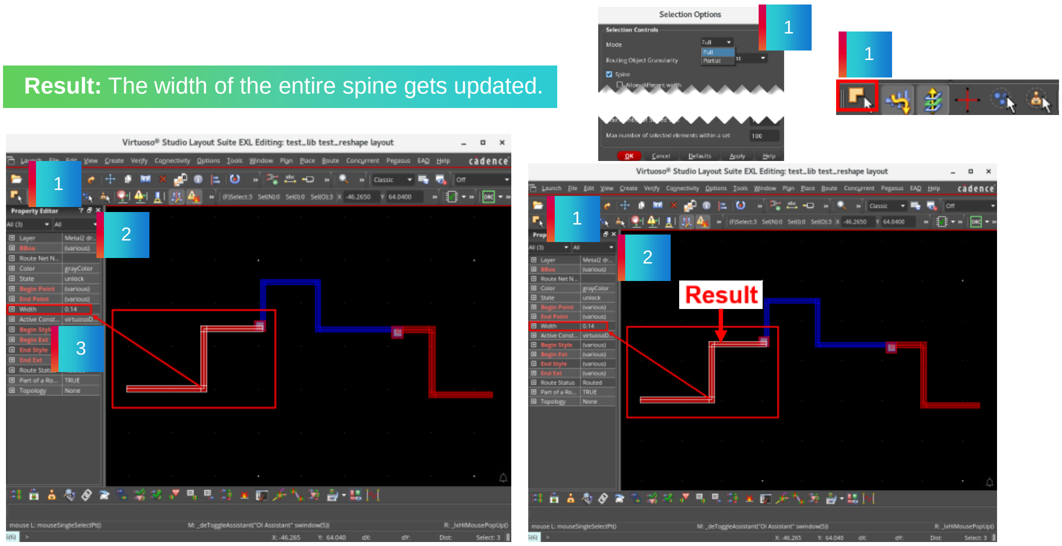
Then, choose *Window – Assistants – Property Editor* to display the Property Editor assistant in the layout window.

Now, select a spine and change its width in the Property Editor assistant.

You can see that the width of the entire spine gets updated.

Result: Changing the Width of an Existing Path Segment
Full Mode/Full Select

Result: The width of the entire spine gets updated.



Result: Changing the Width of an Existing Path Segment Full Mode/Select

After selecting either *Full Mode* in the *Selection Options* form or enabling *Full Select* in the *Options* toolbar, when you change the width of the selected spine, you can observe that the width of the entire spine gets updated.

Changing the Width of an Existing Path Segment
Partial Mode/Partial Select

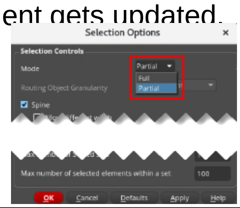
The *Mode* in the *Selection Controls* section of the **Selection Options** form affects the way the pathSegs are selected and modified.

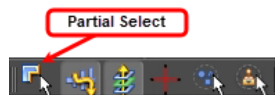
Partial Mode: It selects objects fully if you completely enclose them within the area selection box, and partially selects objects if you partially enclose them within the area selection box.

***Partial Select* (Spine not relevant):** Only the selected pathSeg is modified.

1. In the **Selection Options** form, in the *Selection Controls* section, select *Partial Mode* or enable *Partial Select* from the **Options** toolbar.
2. Choose **Window – Assistants – Property Editor** to display the Property Editor assistant in the layout window.
3. Select a spine and change its width in the Property Editor assistant.

Result: Only a single path segment gets updated.





Changing the Width of an Existing Path Segment Partial Mode/Select

The *Mode* in the *Selection Controls* section of the *Selection Options* form affects the way the pathSegs are selected and modified.

Partial Mode selects objects fully if you completely enclose them within the area selection box, and partially selects objects if you partially enclose them within the area selection box.

With *Partial Select* where Spine is not relevant, only the selected pathSeg is modified.

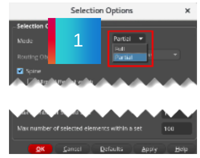
In the *Selection Options* form, in the *Selection Controls* section, select *Partial Mode* or enable *Partial Select* from the *Options* toolbar.

Then, choose *Window – Assistants – Property Editor* to display the Property Editor assistant in the layout window.

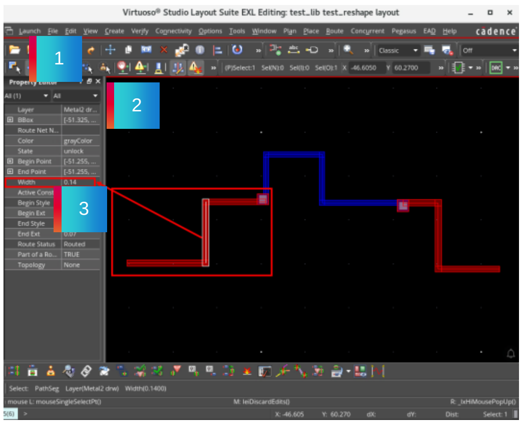
Now, select a spine and change its width in the Property Editor assistant.

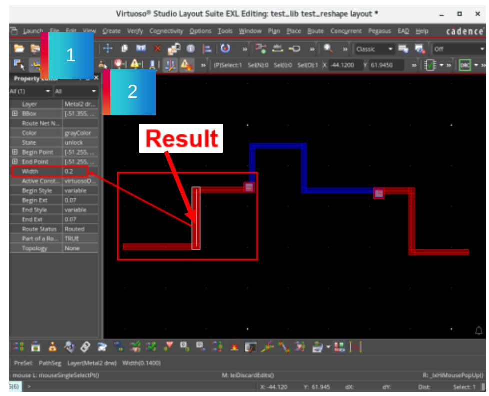
You can see that only a single path segment gets updated.

Result: Changing the Width of an Existing Path Segment
Partial Mode/Partial Select



Result: Only a single path segment gets updated.





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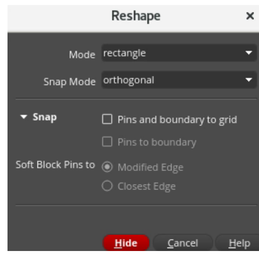
Result: Changing the Width of an Existing Path Segment Partial Mode/Select

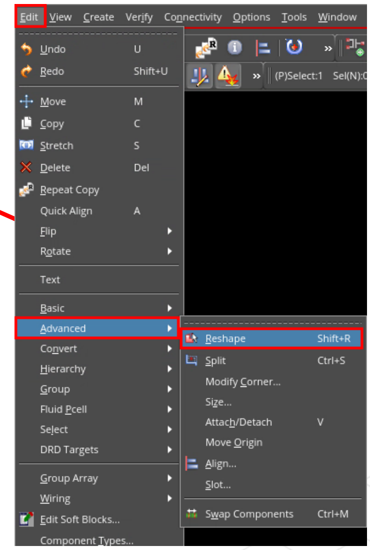
After selecting either *Partial Mode* in the *Selection Options* form or enabling *Partial Select* from the *Options* toolbar, when you change the width of the selected spine, you can observe that only a single path segment gets updated.

Invoking the Reshape Form

In the design canvas, choose the **Edit – Advanced – Reshape** menu command or click the **Reshape** icon in the Edit toolbar, or press the bindkey **Shift+R** to invoke the Reshape form.

Use the *Reshape* form to change the shape of a selected object.





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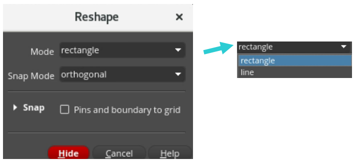
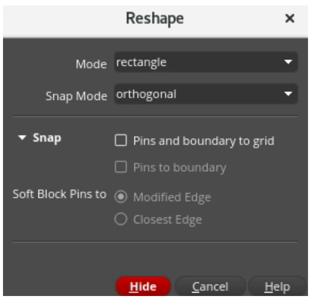
Invoking the Reshape Form

In the design canvas, choose the *Edit – Advanced – Reshape* menu command or click the *Reshape* icon in the Edit toolbar or press the *Shift+R* bindkey to invoke the Reshape form.

You can use this form to change the shape of a selected object.

The Reshape Form: Options in VLS-XL/EXL/MXL





Options	Description
Mode	Sets the geometry to use for reshaping a selected object. *rectangle*: lets you add or remove a rectangle shape. *line*: lets you add a polygon to a shape or reshape a path. Only *line* can be used to reshape a path.
Snap Mode	Controls the shape of line segments; applies only when *Reshape Type* is set to *line*.
Snap: Pins and boundary to grid	Snaps the stretched object to a grid depending on the block type.
Pins to boundary	Automatically snaps pins to the place and route boundary.
Soft Block Pins to	Automatically snaps soft block pins either to the *Modified Edge* or to the *Closest Edge* after the reshape is complete.

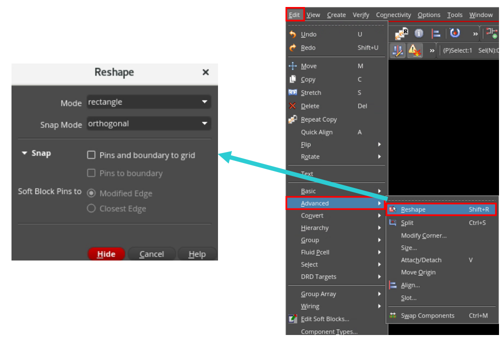
This page does not contain notes.

Reshaping the Wire

Use the **Reshape** command to change the shape of a selected object.

1. In the design canvas, select the object to be reshaped and choose the **Edit – Advanced – Reshape** menu command, or click the **Reshape** icon from the Edit toolbar or press the bindkey **Shift+R** to invoke the Reshape form.
2. Point at the first corner of the reshape rectangle.
3. Point at the opposite corner of the reshape rectangle.

Result: The selected shape is reshaped.



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Reshaping the Wire

You can use the *Reshape* command to change the shape of a selected object.

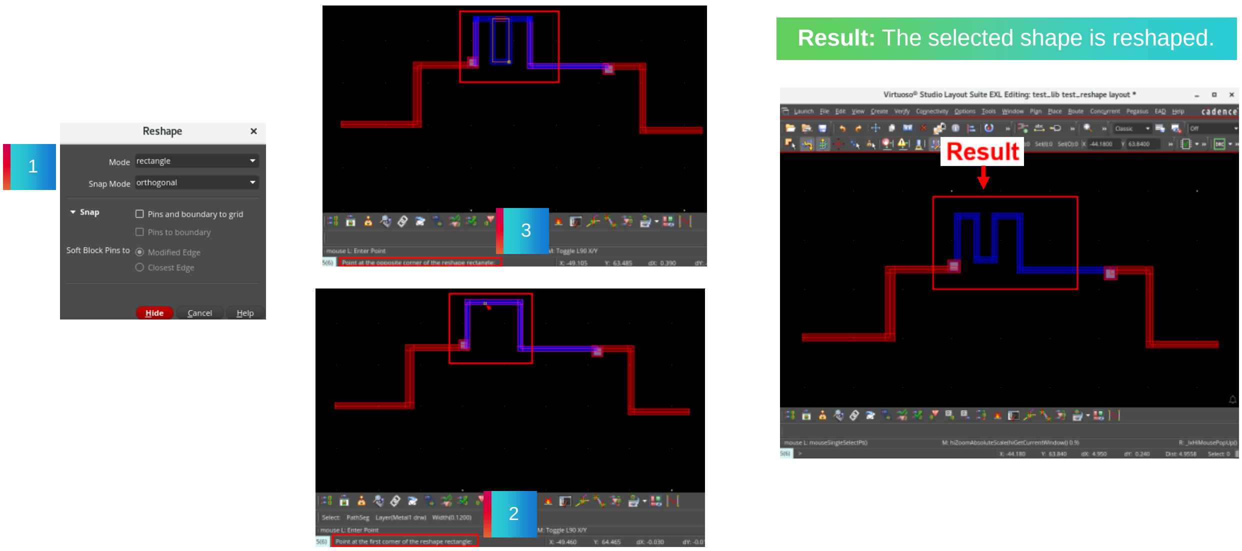
In the design canvas, select the object to be reshaped and choose the *Edit – Advanced – Reshape* menu command or click the *Reshape* icon from the Edit toolbar or press the *Shift+R* bindkey to invoke the Reshape form.

In the layout window, point at the first corner of the reshape rectangle.

Then, point at the opposite corner of the reshape rectangle to reshape the selected object.

You can observe that the selected shape is reshaped.

Result: Reshaping the Wire



Result: Reshaping the Wire

As you see in this example, when you apply the *Reshape* command on the selected object, the selected shape is reshaped.

Note: Once the new “part” is entered, the user can cycle through the different possible constellations using the MMB.

FAQ: Selecting the Layer to Create Wire

When creating wires, what are the different methods available to select the layer?

When creating wires, the different methods available to select the layer are:

- Preselect the layer on the **Layers** Palette.
- Snap and select the layer using the **Choose Object to Tap** form.
- Use Smart Snapping and select the layer **On-Canvas**.

This page does not contain notes.

FAQ: Create Wire Form

How do you invoke the **Create Wire** form?

In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon from the Create toolbar or use the bindkey **P** to invoke the Create Wire form.

This page does not contain notes.

FAQ: Create Geometric Wire Form

Which SKILL command can you use to open the **Create Geometric Wire** form?

To create geometric wires, enter the **CIW>leHiCreateGeometricWire** command to open the Create Geometric Wire form.

This page does not contain notes.

FAQ: The Finish Wire Command

When creating which types of wires, which **Finish Wire** command is available?

The **Finish Wire** command is available only for symbolic wires.

This page does not contain notes.

FAQ: Creating Wires

Which form can you use to change to other layers, rotate the via, and control the via alignment when creating wires?

You can use the **Select Via** form to change to other layers, rotate the via, and control the via alignment when creating wires.

This page does not contain notes.

Module Summary

In this module, you learned how to

- Connect to target pins with the Create Wire command
- Use the Interactive Via Alignment feature

This page does not contain notes.

Labs

Lab 3-1 Connecting to Target Pins with the Create Wire Command

Lab 3-2 Using the Interactive Via Alignment Feature

Now, you will do lab exercises for Module 3.

There are 2 labs in this module.

In the first lab, you will explore how to connect to target pins with the Create Wire command.

In the second lab, you will explore how to create wire using the Interactive Via Alignment feature.

Quiz: Selecting the Layer to Create Wire

When using the **Create Wire** command, if the current active layer is available, it will be used to start the new wire without any further requests.

A. True

B. False

Answer: A

When using the **Create Wire** command, if the current active layer is available, it will be used to start the new wire, without any further requests.

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This page does not contain notes.

Quiz: Create Single Wire Context-Sensitive Menu

Which commands offer the **Create Single Wire** context-sensitive menu options when creating wires?

Select all that apply.

A. Create Bus

B. Create Path

C. Create Geometric Wire

D. Create Wire

E. All of the above

Answer: C and D

This context-sensitive menu can be accessed when the **Create Geometric Wire** or **Create Wire** command is active.

This page does not contain notes.

Quiz: Create Wire Command

Which key combination can you use to cycle the targets when creating wires?

Select all that apply.

A. Ctrl+Space

B. Shift+Space

C. Ctrl+Tab

D. Shift+Tab

E. None of the above

Answer: A

You can use the **Ctrl+Space** key combination to cycle the targets.

This page does not contain notes.

Quiz: Controlling the Number of Vias When Creating Wires

Which assistant can you use to control the number of vias to override the **Min Num Cuts** value when creating wires?

Select all that apply.

A. Annotation Browser Assistant

B. Constraint Manager Assistant

C. Wire Assistant

D. Search Assistant

E. None of the above

Answer: C

You can use the **Wire Assistant** to control the number of vias to override the **Min Num Cuts** value when creating wires.

This page does not contain notes.

Quiz: Changing the Shape of a Selected Object

Which command can you use to change the shape of a selected object? -------

Enter the answer in the blank field.

Answer: **Reshape**

You can use the **Reshape** command to change the shape of a selected object.

This page does not contain notes.

Module 4

Assisted Wires Creation

In this module, we will discuss Assisted Wires Creation.

Module Objectives

In this module, you

- Use Pin-to-Trunk Routing
  - Start Pin-to-Trunk Router with Route With Default Lookup / Route With WA Overrides
  - Start Pin-to-Trunk Router with Wire Assistant
- Create Assisted Wires
  - Use Point-to-Point Routing
  - Use Bus Routing
  - Use Stranded Wiring
  - Use Finish Bus Command
- Launch Auto Router on the Selected Nets
  - Route with Default Lookup with default Constraint Group (CG)
  - Route with WA Overrides with seeded Constraint Group (CG)
  - Route with WA Overrides with dynamically created PRO

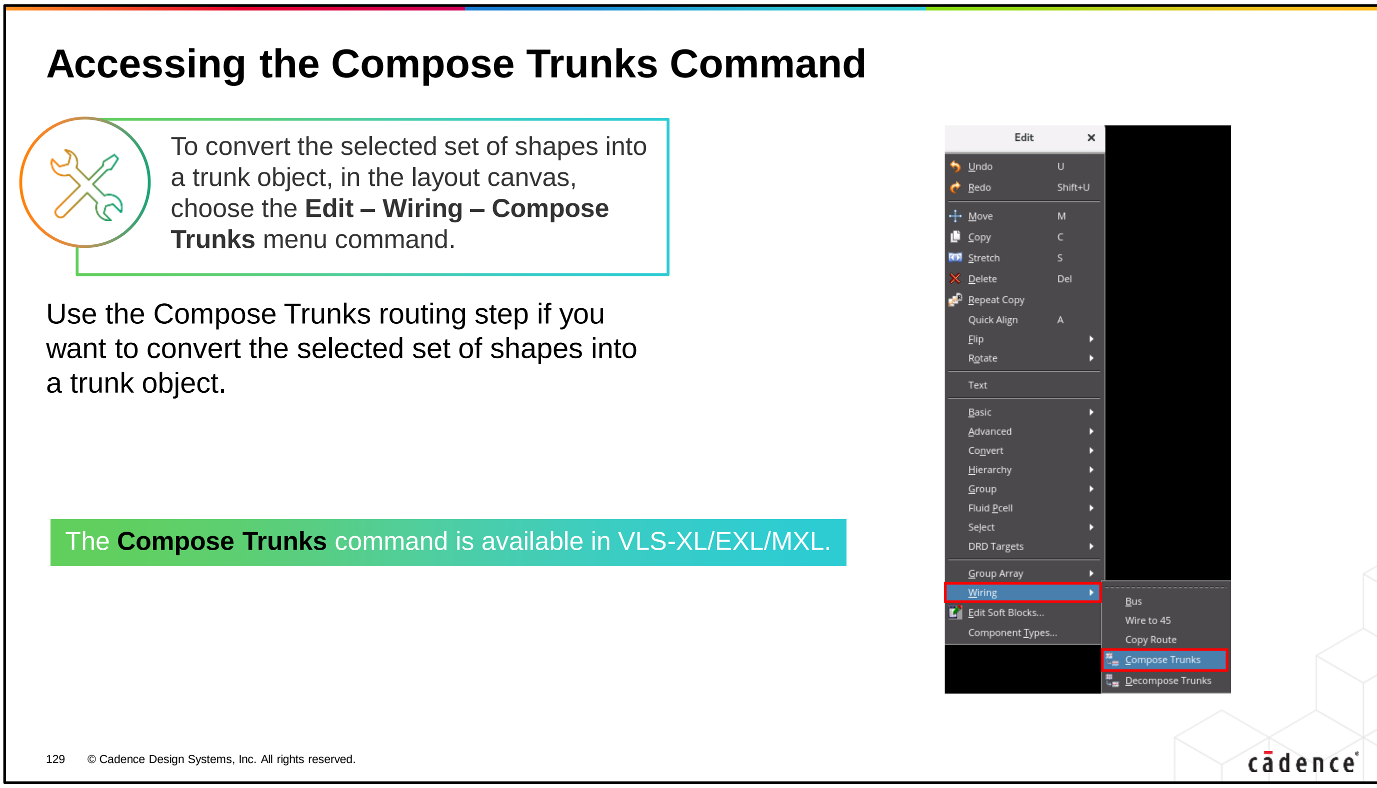
Module Objectives

In this module, you will use the Pin-to-Trunk Router with Route With Default Lookup and with Route With Wire Assistant Overrides.

You will use the Pin-to-Trunk Router with the Wire Assistant.

You will create the assisted wires using the Point to Point Routing, Bus Routing, Stranded Wiring, and Finish Bus command.

You will launch the auto router on the Selected Nets With Default Lookup with default Constraint Group, with Wire Assistant Overrides with seeded Constraint Group, and with dynamically created Process Rule Overrides.



Accessing the Compose Trunks Command

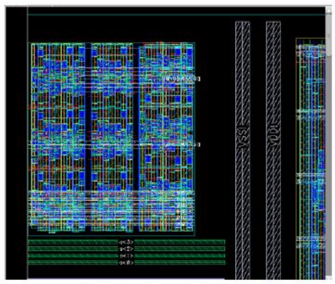
To convert the selected set of shapes into a trunk object, in the layout canvas, choose the *Edit – Wiring – Compose Trunks* menu command.

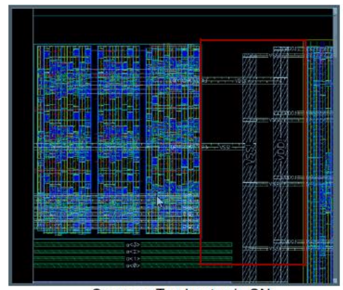
You can use the *Compose Trunks* routing step if you want to convert the selected set of shapes into a trunk object.

The *Compose Trunks* command is available in Virtuoso® Layout Suite XL/EXL/MXL tiers.

Result: Accessing the Compose Trunks Command

- Only pathSegs, vias, and rectangles are converted into trunk objects. Other shapes, such as paths and polygons, are not converted.
- These trunk objects can then be used to perform *Pin-to-Trunk* routing.
- The *Pin-to-Trunk* router is started by default when you run the route command from the *Net* context-sensitive menu after selecting the trunk net in the *Navigator* assistant.
- The *Pin-to-Trunk* routing step is performed after composing trunks.


Compose Trunks step is OFF


Compose Trunks step in ON

A message in the CIW summarizes the objects that are converted to form a trunk.

Result – Accessing the Compose Trunks Command

Only pathSegs, vias, and rectangles are converted into trunk objects. Other shapes, such as paths and polygons, are not converted.

These trunk objects can then be used to perform Pin-to-Trunk routing.

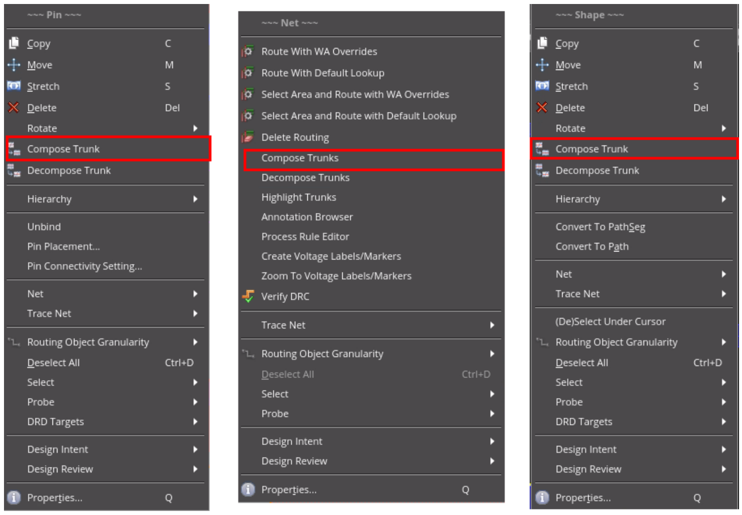
The Pin-to-Trunk router is started by default when you run the route command from the *Net* context-sensitive menu after selecting the trunk net in the *Navigator* assistant.

The Pin-to-Trunk routing step is performed after composing the trunks.

A message in the CIW window summarizes the objects that are converted to form a trunk.

Accessing the Compose Trunk Command from the Context-Sensitive Menus

You can access the **Compose Trunk** command from the **Pin**, **Net**, and **Shape** context-sensitive menus that display if you **right**-click in the design display area after selecting one or more objects.



Accessing the Compose Trunk Command From the Context-Sensitive Menus

You can access the *Compose Trunk* command from the *Pin*, *Net*, and *Shape* context-sensitive menus that display if you *right*-click in the design display area after selecting one or more objects.

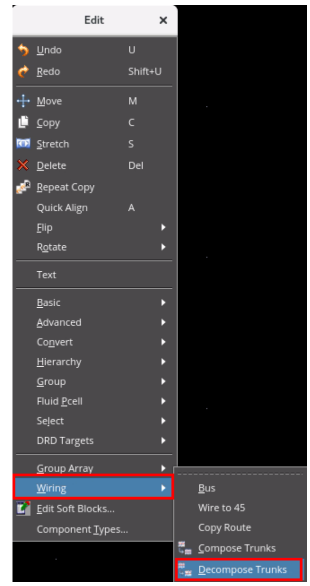
Accessing the Decompose Trunks Command

To convert the selected set of trunk objects to the original shapes, in the layout canvas, choose the **Edit – Wiring – Decompose Trunks** menu command.

- Use the *Decompose Trunks* command to convert the selected set of trunks, comprising selected set of shapes, to the original shapes from which it was formed.
- If the selected shapes are not part of a trunk and you run the *Decompose Trunks* command, the number of converted objects reported in the CIW is 0.

The **Decompose Trunks** command is available in VLS-XL/EXL/MXL.

A message in the CIW summarizes the objects that are converted to original shapes.



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Accessing the Decompose Trunks Command

To convert the selected set of trunk objects to the original shapes, in the layout canvas, choose the *Edit – Wiring – Decompose Trunks* menu command.

You can use this command to convert the selected set of trunks, comprising the selected set of shapes, to the original shapes from which it was formed.

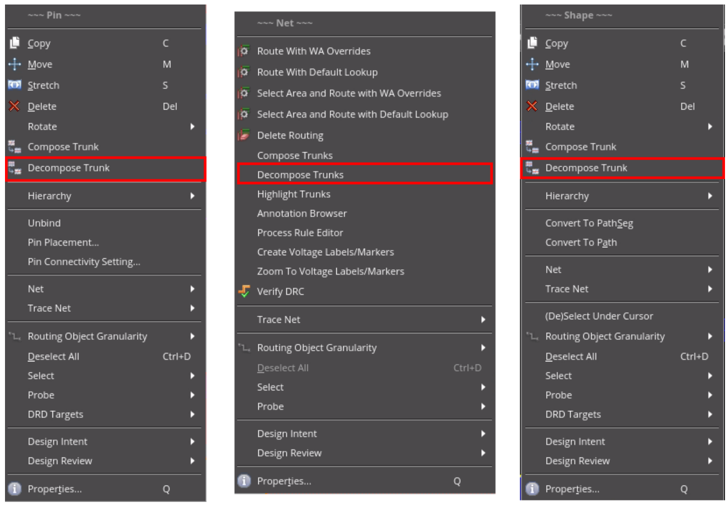
If the selected shapes are not part of a trunk and you run the *Decompose Trunks* command, the number of converted objects reported in the CIW window is 0.

This command is available in Virtuoso Layout Suite XL/EXL/MXL tiers.

A message in the CIW window summarizes the objects that are converted to original shapes.

Accessing the Decompose Trunk Command from the Context-Sensitive Menus

You can access the **Decompose Trunk** command from the **Pin**, **Net**, and **Shape** context-sensitive menus that display if you **right**-click in the design display area after selecting one or more objects.



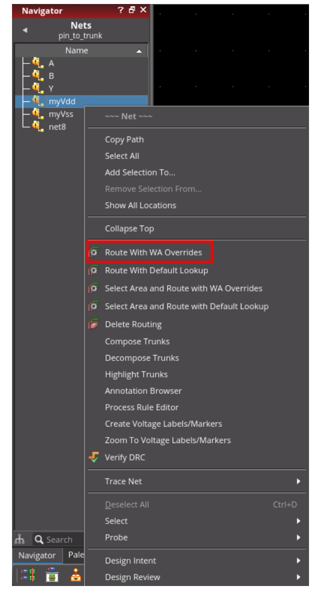
Accessing the Decompose Trunk Command from the Context-Sensitive Menus

You can access the *Decompose Trunk* command from the *Pin*, *Net*, and *Shape* context-sensitive menus that display if you *right*-click in the design display area after selecting one or more objects.

Routing With WA Overrides

In the design canvas, select a net in the Navigator assistant > **right**-click and choose **Route With WA Overrides** from the context-sensitive menu.

Use the *Route With WA Overrides* command to route the selected nets with the Wire Assistant Overrides taking precedence over the default constraint settings as defined by the Wire Editor Default Constraint Group or any Net Specific Constraints.



Routing With WA Overrides

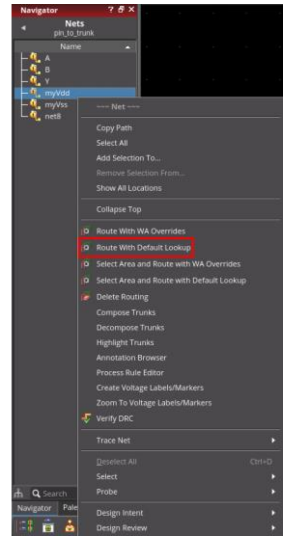
In the design canvas, select a net in the Navigator assistant and *right*-click and choose *Route With WA Overrides* command from the context-sensitive menu.

You can use this command to route the selected nets with the Wire Assistant Overrides, taking precedence over the default constraint settings as defined by the Wire Editor Default Constraint Group or any Net Specific Constraints.

Routing With Default Lookup

In the design canvas, select a net in the Navigator assistant > **right**-click and choose **Route With Default Lookup** from the context-sensitive menu.

Use the *Route With Default Lookup* command to route the selected net by using the default constraint lookup for routing.



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Routing With Default Lookup

In the design canvas, select a net in the Navigator assistant and *right*-click and choose the *Route With Default Lookup* command from the context-sensitive menu.

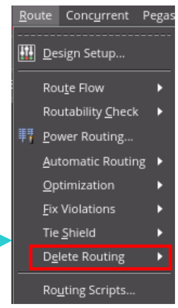
You can use this command to route the selected net by using the default constraint lookup for routing.

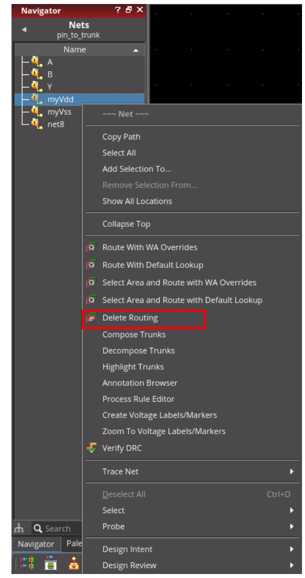
Deleting Routing on a Net

To delete the routing information > **right**-click the net in the Navigator and choose **Delete Routing** from the context-sensitive menu.

Use the *Delete Routing* command to delete the routing information and shielding wires on a net.

You can access the **Delete Routing** command from the **Route** menu in the layout window as well.





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Deleting Routing on a Net

To delete the routing information, *right*-click the net in the Navigator assistant and choose the *Delete Routing* command from the context-sensitive menu.

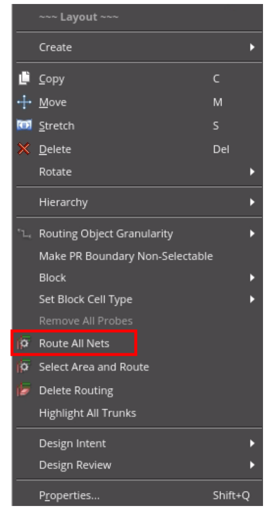
You can use this command to delete the routing information and shielding wires on a net.

You can also access this command from the *Route* menu in the layout window.

Routing All Nets on the Cell Level

To route all the nets in a cellview > **right**-click the cell and choose **Route All Nets** from the context-sensitive menu.

Using this command, you can route all the nets in the design.



Routing All Nets on the Cell Level

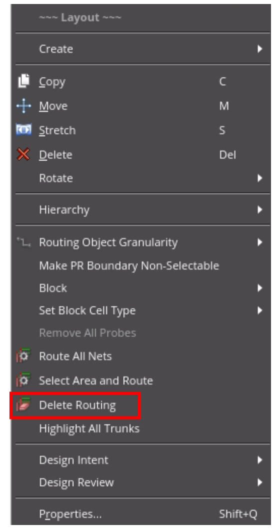
To route all the nets in a cellview, *right*-click the cell and choose the *Route All Nets* command from the context-sensitive menu.

Using this command, you can route all the nets in the design.

Deleting Routing on the Cell Level

To delete the routing at the cell level > **right-**click the cell and choose **Delete Routing** from the context-sensitive menu.

Using this command, you can delete all the routing information of the cellview.



Deleting Routing on the Cell Level

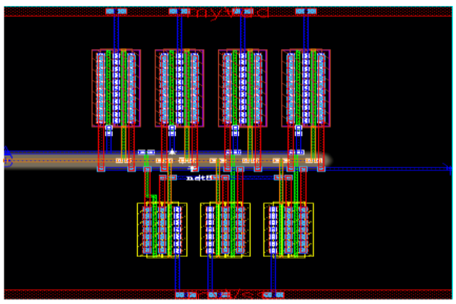
To delete the routing at the cell level, *right*-click the cell and choose the *Delete Routing* command from the context-sensitive menu.

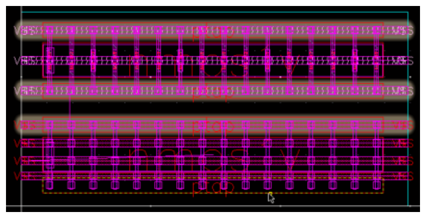
Using this command, you can delete all the routing information of the cellview.

Highlighting the Trunks

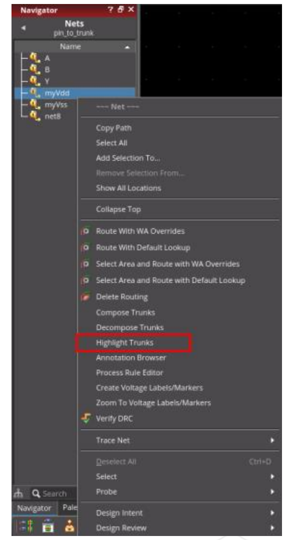
To highlight the trunks in the layout canvas > **right**-click the net in the Navigator and choose **Highlight Trunks** from the context-sensitive menu.

- Use the *Highlight Trunks* command to identify the trunks for which the twigs will be created.
- Use this option to identify the composed trunks of a selected net and to highlight the trunks and twigs in the layout canvas.





When debugging the routing, it is useful to highlight which segments have been used as trunks.



Highlighting Trunks

To highlight the trunks in the layout canvas, *right*-click the net in the Navigator assistant and choose the *Highlight Trunks* command from the context-sensitive menu.

You can use this command to identify the trunks for which the twigs will be created.

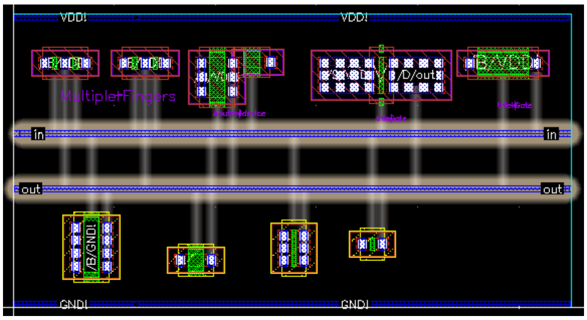
This option helps you to identify the composed trunks of a selected net and to highlight the trunks and twigs in the layout canvas.

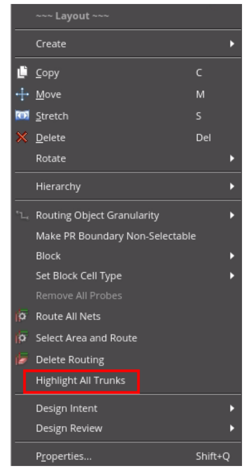
When debugging the routing, it is useful to highlight which segments have been used as trunks.

Highlighting All the Trunks

To highlight the trunks for all the nets that have a valid composed trunk in a cellview > **right**-click the cell and choose **Highlight All Trunks** from the context-sensitive menu.

Use the *Highlight All Trunks* command to highlight the trunks and twigs that are legal and can be recognized by the router.





Highlighting All Trunks

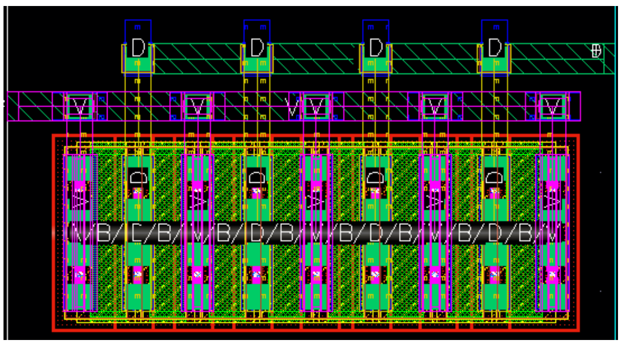
To highlight the trunks for all the nets that have a valid composed trunk in a cellview, *right*-click the cell and choose the *Highlight All Trunks* command from the context-sensitive menu.

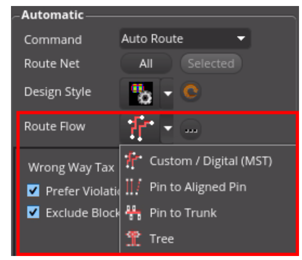
You can use this command to highlight the trunks and twigs that are legal and can be recognized by the router.

What Is the Pin-to-Trunk Routing?

The **Pin-to-Trunk** routing topology style is used to connect an individual pin to a trunk.

- Only the connections that are within the scope of a trunk are routed.
- The remaining opens are completed by the **Custom / Digital (MST)** routing style or manually.
- The **Pin-to-Trunk** routing style is usually used when a spine structure is required.





What Is Pin-to-Trunk Routing?

The *Pin-to-Trunk* routing topology style is used to connect an individual pin to a trunk.

Only the connections that are within the scope of a trunk are routed.

The remaining opens are completed by the *Custom / Digital (MST)* routing style or manually.

The *Pin-to-Trunk* routing style is usually used when a spine structure is required.

How to Do Pin-to-Trunk Routing with Default Lookup/Route With WA Overrides

The Pin to Trunk routing style connects an individual pin to a trunk. Only the connections that are within the scope of a trunk are routed. The remaining opens are completed by the Custom / Digital (MST) routing style or manually.

This page does not contain notes.

Step 1: Composing Trunks

Use the **Compose Trunks** command to convert the selected set of shapes into a trunk object.

The **Pin-to-Trunk** routing step is performed after composing trunks.

To connect the PMOS transistors to the VDD power rail (net name: **myVdd**) in the layout design:

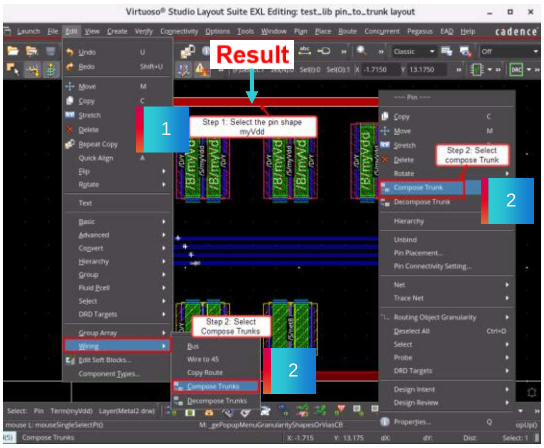
1. Define a trunk for the power net: select the pin shape **myVdd**.
2. Choose **Edit – Wiring – Compose Trunks** or **right**-click and choose **Compose Trunk**.

Result: The pin shape **myVdd** is declared as trunk.

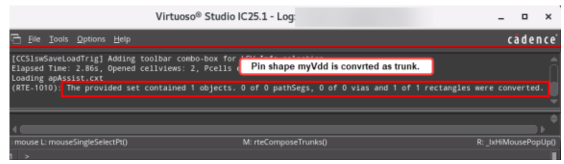
This page does not contain notes.

Result: Composing Trunks

Result: The pin shape **myVdd** declared as trunk will be used as source for all the branches.



Observe the CIW window or CDS.log file for the trunk conversion information.



Result: Starting Pin-to-Trunk Router with Route With Default Lookup / Route With WA Overrides – Composing Trunks

The pin shape *myVdd*, which is declared as trunk, will be used as the source for all the branches.

You can observe the CIW window or the CDS.log file for the trunk conversion information.

Step 2: Start Pin-to-Trunk Routing

The **Pin-to-Trunk** router is started by default when you run the Route command from the **Net** context-sensitive menu after selecting the trunk net in the **Navigator** assistant.

To route the trunk net **myVdd**, in the layout design:

1. Select the net **myVdd** in the Navigator assistant.
2. Start the **pin-to-trunk** router on the defined trunk **myVdd**: **Right**-click and choose **Route With Default Lookup**.

Result: The trunk **myVdd** is routed.

In this case, there would be no difference between *Route With Default Lookup* and *Route With WA Overrides* since you have not changed anything in the *Override Constraints* section of the Wire Assistant.

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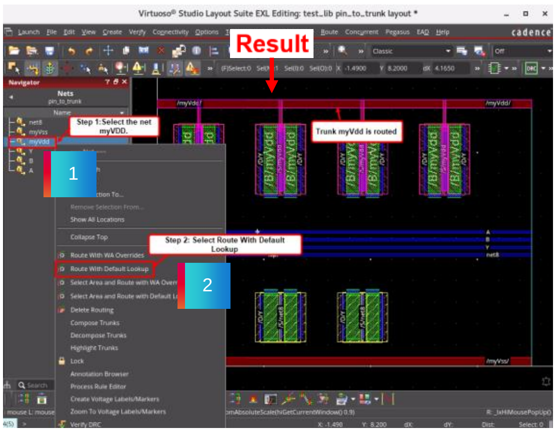
Result: Starting Pin-to-Trunk Router with Route With Default Lookup / Route With WA Overrides

Result: Routing for the net **myVdd** has been completed using the trunk approach.



Observe the CDS.log file or CIW window for the **Route Summary**.

Result: Starting Pin-to-Trunk Router with Route With Default Lookup / Route With WA Overrides

You see here that the routing for the net *myVdd* has been completed using the trunk approach.

You can observe the CDS.log file or the CIW window for the *Route Summary* information.

Step 3: Highlight the Trunks

Use the **Highlight Trunks** command to highlight the trunks of a selected net in the layout canvas.

When debugging the routing, it is useful to highlight which segments have been used as trunks.

To highlight the trunks of a selected net in the layout canvas:

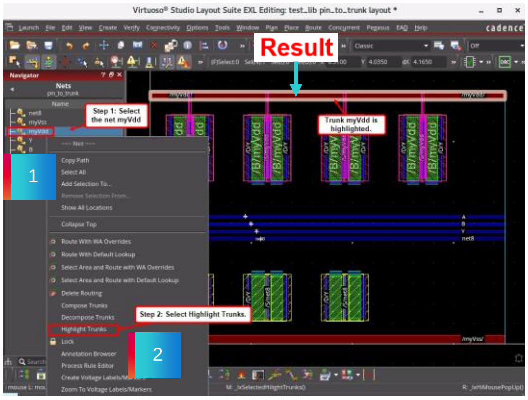
1. Select the net, say for example, **myVdd** in the Navigator assistant.
2. Now, **right**-click the net in the Navigator and choose **Highlight Trunks** from the context-sensitive menu.

Result: The trunk **myVdd** is highlighted.

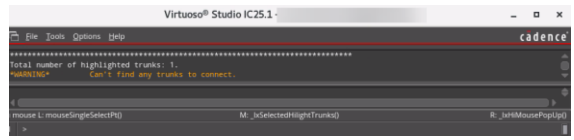
This page does not contain notes.

Result: Highlighting the Trunks

Result: The trunk **myVdd** is highlighted.



Observe the CDS.log file or CIW window for the highlighted trunks information.



Result: Highlighting the Trunks

You see here that the net *myVdd* has been routed and the corresponding trunk is highlighted.

You can observe the CDS.log file or the CIW window for the highlighted trunks information.

Starting Pin-to-Trunk Router with Wire Assistant
Composing Trunks

Use the **Compose Trunks** command to convert the selected set of shapes into a trunk object.

The **Pin-to-Trunk** routing step is performed after composing trunks.

To connect the NMOS transistors to the VSS ground rail (net name: **myVss**) in the layout design:

1. Define a trunk for the ground net: select the pin shape **myVss**.
2. Choose **Edit – Wiring – Compose Trunks** or **right**-click and choose **Compose Trunk**.

Result: The pin shape **myVss** is declared as trunk.

Starting Pin-to-Trunk Router with Wire Assistant – Composing Trunks

You can use the *Compose Trunks* command to convert the selected set of shapes into a trunk object.

The *Pin-to-Trunk* routing step is performed after composing the trunks.

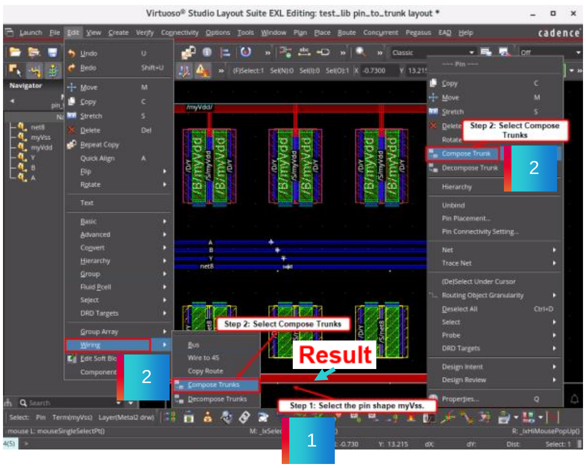
In the layout design, to connect the NMOS transistors to the VSS ground rail, for example, to connect to the net name *myVss*, first, define a trunk for the ground net: For that, select the pin shape which you want to convert as trunk.

Then, choose the *Edit – Wiring – Compose Trunks* menu command or *right*-click and choose the *Compose Trunk* command.

Now, the pin shape *myVss* is declared as trunk.

Result: Starting Pin-to-Trunk Router with Wire Assistant
Composing Trunks

Result: The pin shape **myVss** declared as trunk will be used as source for all the branches.



Observe the CDS.log file or CIW window for the trunk conversion information.

Result: Starting Pin-to-Trunk Router with Wire Assistant – Composing Trunks

The pin shape *myVss* which is declared as trunk will be used as the source for all the branches.

You can observe the CDS.log file or the CIW window for the trunk conversion information.

Starting Pin-to-Trunk Router with Wire Assistant

Use the Wire Assistant's **Pin-to-Trunk** router to route the trunk object.

To route the trunk net **myVss**, in the layout design:

1. Select the net **myVss** in the Navigator assistant.
2. Choose the **Pin to Trunk** *Route Flow* in the Wire Assistant.
3. Start the **Pin to Trunk** router on the defined trunk **myVss**: Click on the **Selected** button in the Wire Assistant.

Result: The trunk **myVss** is routed.

Starting Pin-to-Trunk Router with Wire Assistant

You can use the Wire Assistant's *Pin-to-Trunk* router to route the trunk object.

To route the trunk net *myVss*, firstly, select the net in the Navigator assistant.

Then, choose the *Pin to Trunk Route Flow* in the Wire Assistant.

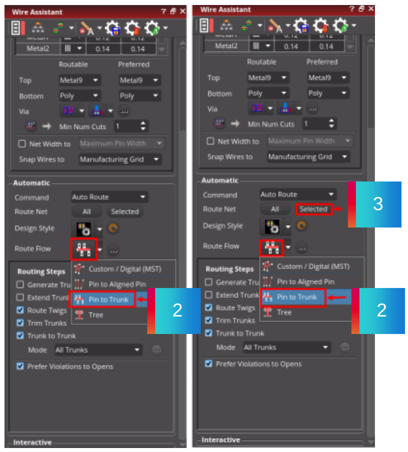
Now, start the *Pin to Trunk* router on the defined trunk *myVss*:

For that, click on the *Selected* button in the Wire Assistant.

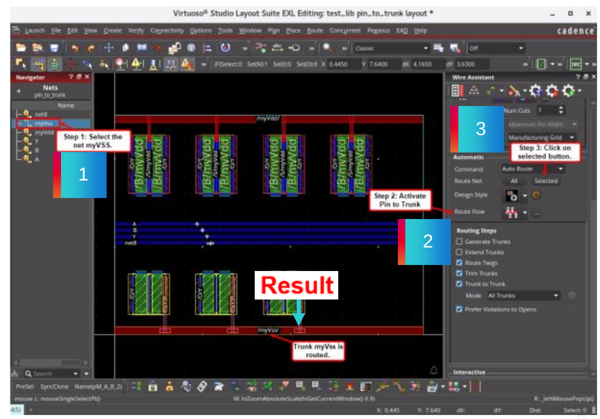
You can see that the trunk *myVss* is routed.

Result: Starting Pin-to-Trunk Router with Wire Assistant

3. Start the **Pin to Trunk** router on the defined trunk **myVss**: Click on the **Selected** button in the Wire Assistant.



Result: Routing for the net **myVss** has been completed using the trunk approach.



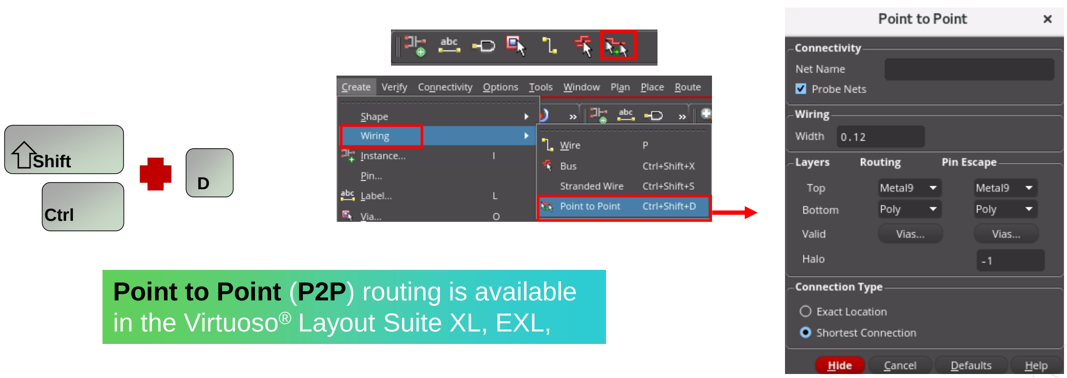
Result: Starting Pin-to-Trunk Router with Wire Assistant

When you run the *Pin-to-Trunk* router on the defined trunk net, you see here that the routing for the net *myVss* has been completed using the trunk approach.

Invoking the Point-to-Point Command

In the design canvas, choose the **Create – Wiring – Point to Point** menu command or use the **Create Point to Point** icon from the Create toolbar or use the bindkey **Ctrl+Shift+D** to invoke the Point to Point form.

Use the *Point-to-Point* command, which is an interactive routing command that automatically generates routing between two digitized points in a given cellview.



Point to Point (P2P) routing is available in the Virtuoso® Layout Suite XL, EXL,

Invoking the Point to Point Command

In the design canvas, choose the *Create – Wiring – Point to Point* menu command or use the *Create Point to Point* icon from the Create toolbar or use the bindkey *Ctrl+Shift+D* to invoke the Point to Point form.

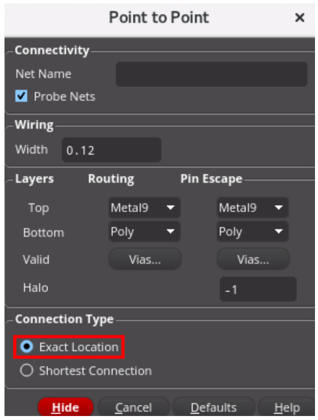
You can use this command which is an interactive routing command that automatically generates routing between two digitized points in a given cellview.

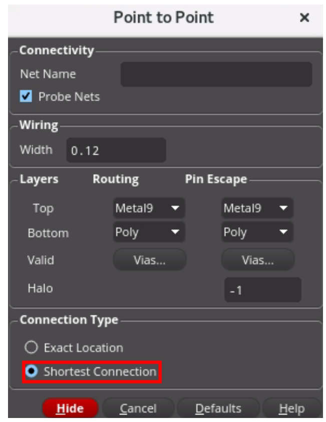
Note that *Point to Point Routing* is available in Virtuoso Layout Suite XL,EXL,MXL tiers.

Point to Point Form: Connection Type

Connection Type is of two types, namely, **Exact Location** and **Shortest Connection**.

Environment variable: **p2pSeedStyle**





Exact Location: Uses the exact digitized points to create routing between two given points.

Shortest Connection: Uses the shortest route for connecting points and may also use pre-existing wires for creating the route.

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Point to Point Form – Connection Type

In the *Point to Point* form, *Connection Type* is of two types, namely: *Exact Location* and *Shortest Connection*.

When you select the *Exact Location* option, it uses the exact digitized points to create routing between two given points.

When you select the *Shortest Connection* option, it uses the shortest route for connecting points and may also use the pre-existing wires for creating the route.

The Environment variable for this option is *p2pSeedStyle*.

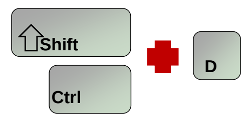
Using Point to Point Routing

Point to Point routing is an acceleration of Interactive Wire Editing when creating short local connections while also allowing you to maintain tight control of the routing.

To do point-to-point routing:

1. In the design canvas, choose the **Create – Wiring – Point to Point** menu command or use the **Create Point to Point** icon from the Create toolbar or use the bindkey **Ctrl+Shift+D** to invoke the Point to Point form.
2. Then, select the net, for example, **net1<0>** in the Navigator assistant.
3. Now, *Point at the from point of the P2P*.
4. Next, *Point at the to point of the P2P*.

Result: The routing is completed.



Using Point to Point Routing

Point to Point routing is an acceleration of Interactive Wire Editing when creating short local connections while also allowing you to maintain tight control of the routing.

To do the point-to-point routing:

In the design canvas, choose the *Create – Wiring – Point to Point* menu command or use the *Create Point to Point* icon from the Create toolbar or use the bindkey *Ctrl+Shift+D* to invoke the Point to Point form.

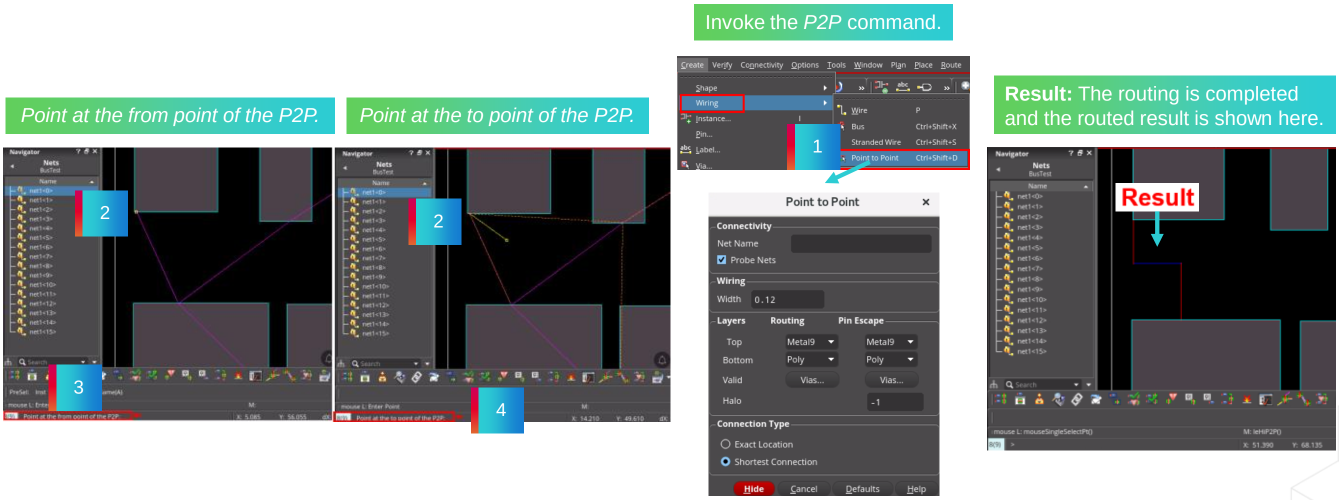
Then, select the net which you want to route in the Navigator assistant.

Now, *Point at the from point of the P2P*.

Next, *Point at the to point of the P2P*.

You can observe that the routing is completed from the selected net in the design canvas.

Result: Using Point to Point Routing



Result: Using Point to Point Routing

In this example, *net1<0>* is selected in the Navigator assistant and *Point to Point* routing is done on this net.

You see that the routing is completed on the selected net and the routed result is shown here.

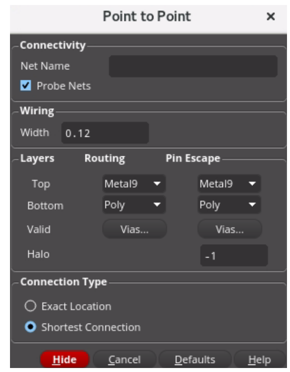
Displaying the Point-to-Point Context-Sensitive Menu

When creating ***Point to Point*** routing, you can use the options available in the **Point-to-Point** context-sensitive menu which enhances the routing functionality.

This context-sensitive menu can be accessed when the ***Point to Point*** (XL/EXL/MXL) command is active and the selected net is being routed.

1. To create point-to-point routing, in the design canvas, choose the **Create – Wiring – Point to Point** menu command or use the **Create Point to Point** icon from the Create toolbar or use the bindkey **Ctrl+Shift+D** to invoke the Point-to-Point form.
2. Now, **right**-click in the design canvas when the *Point to Point* (XL/EXL/MXL) command is running.

Result: The **Point-to-Point** context-sensitive menu is displayed in the layout canvas.



Displaying the Point to Point Context-Sensitive Menu

When creating *Point to Point* routing, you can use the options available in the *Point to Point* context-sensitive menu which enhances the routing functionality.

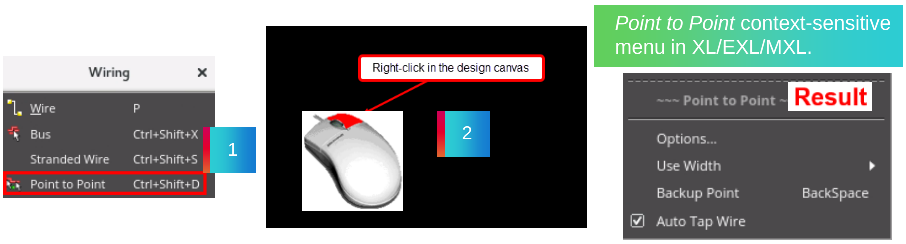
This context-sensitive menu can be accessed when the *Point to Point* command is active and the selected net is being routed.

To create point-to-point routing, in the design canvas, choose the *Create – Wiring – Point to Point* menu command or use the *Create Point to Point* icon from the Create toolbar or use the bindkey *Ctrl+Shift+D* to invoke the Point to Point form.

Now, *right*-click in the layout canvas when the *Point to Point* command is running in XL/EXL/MXL tiers of Virtuoso Layout Suite.

The *Point to Point* context-sensitive menu is displayed in the design window.

Result: Displaying the Point-to-Point Context-Sensitive Menu

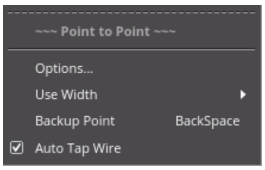


Result: Displaying the Point to Point Context-Sensitive Menu

When creating point-to-point routing, you can *right*-click in the design canvas to display the *Point to Point* context-sensitive menu.

You can use the options available in this context-sensitive menu to enhance your routing features when creating the Point to Point connection.

The Point-to-Point Context-Sensitive Menu Options



Options	Opens the **Point to Point** form. You can also use the **F3** bindkey to open the **Options** form.
Use Width	Use this option to specify the width.
Backup Point	Removes the last digitized point. If a via had been added to the last digitized point, the via is also removed.
Auto Tap Wire	Automatically sets the layer of the wire being created to be the same as that of an existing clicked wire. This option is on by default.

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The Point to Point Context-Sensitive Menu Options

When creating *Point to Point* routing, you can use the options available in the *Point to Point* context-sensitive menu which enhances the routing functionality.

Clicking the *Options* button opens the *Point to Point* form. You can also use the *F3* bindkey to open the *Options* form.

Use the *Use Width* option to specify the width.

The *Backup Point* option removes the last digitized point. If a via had been added to the last digitized point, the via is also removed.

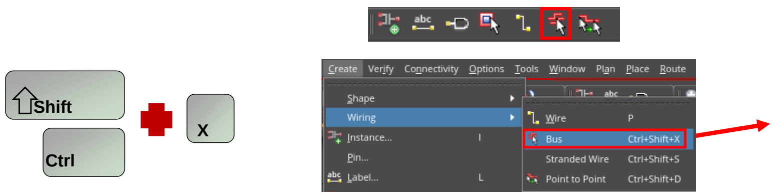
The *Auto Tap Wire* option automatically sets the layer of the wire being created to be the same as that of an existing clicked wire. This option is on by default.

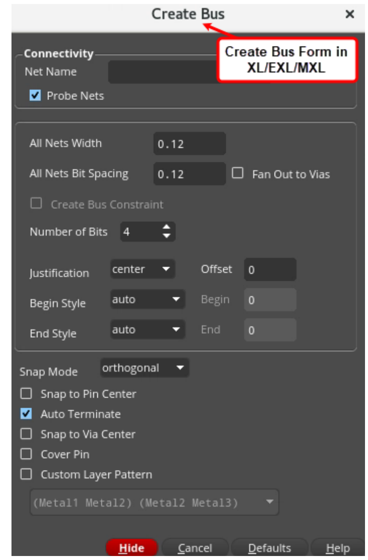
The *Constraint-aware Editing* option enforces constraints for nets grouped in a bus constraint and for shielding wires. This option is on by default and is available in Layout XL and higher tiers.

Invoking the Create Bus Command

In the design canvas, choose the **Create – Wiring –Bus** menu command or use the **Create Bus** icon from the Create toolbar or use the bindkey **Ctrl+Shift+X** to invoke the Create Bus form.

Using the *Create Bus* form, you can create bus wires in the design.





You can create a bus in XL/EXL/MXL tiers of the Virtuoso Layout Suite.

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Invoking the Create Bus Command

In the design canvas, choose the *Create – Wiring – Bus* menu command or use the *Create Bus* icon from the Create toolbar or use the bindkey *Ctrl+Shift+X* to invoke the Create Bus form.

Using this form, you can create bus wires in the design.

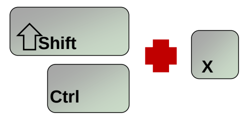
Note that you can create a bus in all the tiers of the Virtuoso Layout Suite.

Using Bus Routing

Use the **Create Bus** command to create bus wires in the design.

To create bus wires in the design:

1. In the design canvas, choose the **Create – Wiring – Bus** menu command or use the **Create Bus** icon from the Create toolbar or use the bindkey **Ctrl+Shift+X** to invoke the Create Bus form.
2. Set the values in the Create Bus form as below:
   a. *All Nets Width* – **0.14**
   b. *All Nets Bit Spacing* – **0.14**
   c. *Number of Bits* – **4**
   d. *Net Name*: enter **net1<4>** to **net1<7>** as the members.



3. Digitize and create the bus.
4. Restart the bus by starting **Create Bus** and selecting only one member of the bus.

Result: All the members of the bus will route along with the one you selected.

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Using Bus Routing

You can use the *Create Bus* command to create bus wires in the design.

To create bus wires in the layout window:

Choose the *Create – Wiring – Bus* menu command or use the *Create Bus* icon from the Create toolbar or use the bindkey *Ctrl+Shift+X* to invoke the Create Bus form.

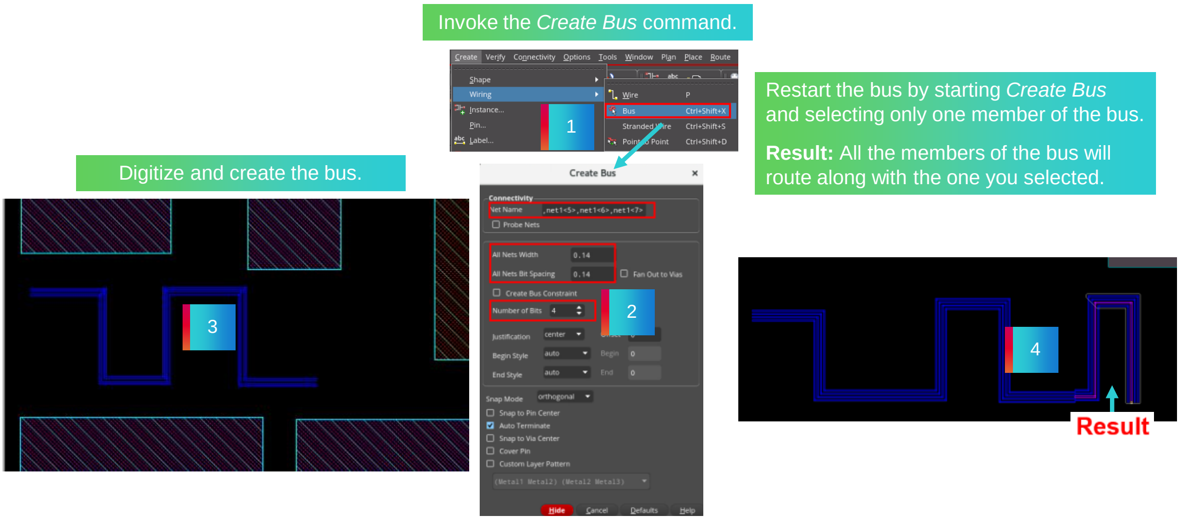
Then, set the required values in the Create Bus form.

Next, digitize and create the bus.

Restart the bus again by starting the *Create Bus* command and selecting only one member of the bus.

You can notice that all the members of the bus will route along with the one you selected.

Result: Using Bus Routing



Result: Using Bus Routing

In the *Create Bus* form, after specifying the required values for your design, you can digitize and create the bus.

When you restart the *Create Bus* command again by selecting only one member of the bus, you can see that all the members of the bus will route along with the one you selected.

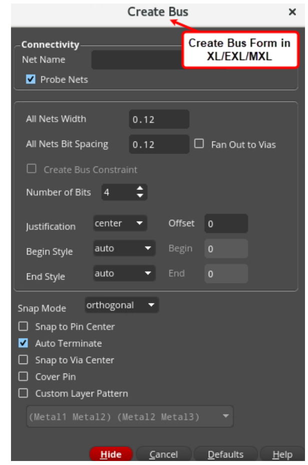
Displaying the Create Bus Context-Sensitive Menu

When creating bus wires, you can use the options available in the **Create Bus** context-sensitive menu which enhances the routing functionality.

This context-sensitive menu can be accessed when the **Create Bus** (XL/EXL/MXL) command is active and multiple wires are being routed.

- To create bus wires, in the design canvas, choose the **Create – Wiring – Bus** menu command or use the **Create Bus** icon from the Create toolbar or use the bindkey **Ctrl+Shift+X** to invoke the Create Bus form.
- Now, **right**-click in the design canvas when the *Create Bus* (XL/EXL/MXL) command is running.

Result: The **Create Bus** context-sensitive menu is displayed in the layout canvas.



Displaying the Create Bus Context-Sensitive Menu

When creating bus wires, you can use the options available in the *Create Bus* context-sensitive menu which enhances the routing functionality.

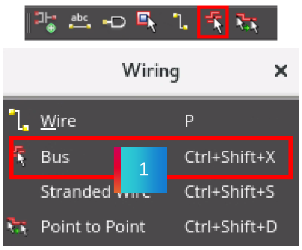
This context-sensitive menu can be accessed when the *Create Bus* command is active and multiple wires are being routed.

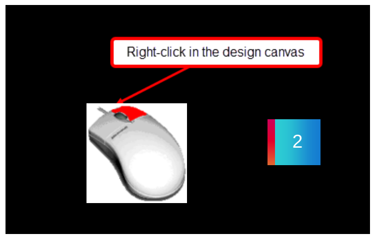
To create bus wires, in the design canvas, choose the *Create – Wiring – Bus* menu command or use the *Create Bus* icon from the Create toolbar or use the bindkey *Ctrl+Shift+X* to invoke the Create Bus form.

Now, *right*-click in the design canvas when the *Create Bus* command is running in all the tiers of the Virtuoso Layout Suite.

The *Create Bus* context-sensitive menu is displayed in the layout canvas.

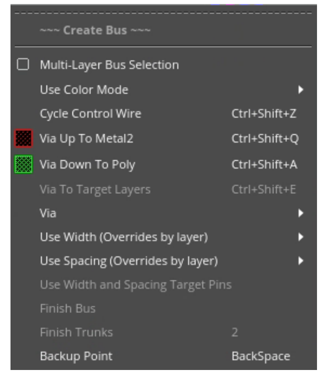
Result: Displaying the Create Bus Context-Sensitive Menu





Create Bus context-sensitive menu in XL/EXL/MXL.

Result



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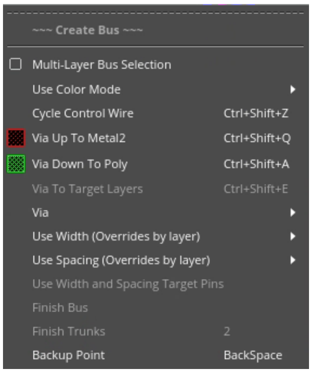
Result: Displaying the Create Bus Context-Sensitive Menu

When creating bus routing, you can *right*-click in the design canvas to display the *Create Bus* context-sensitive menu.

You can use the options available in this context-sensitive menu to enhance your routing features when creating multiple wires with the *Create Bus* command.

The Create Bus Context-Sensitive Menu Options



Multi-Layer Bus Selection	Provides support to create and digitize the bits of a bus on different metal layers.
Use Color Mode	Controls the color mode of the bus.
Cycle Control Wire	Cycles the control wire function among the extreme and middle wires of a bus.
Via Up To	Changes the layer to the next layer up and places the corresponding default via.
Via Down To	Changes the layer to the next layer down and places the corresponding default via.
Via To Target Layers	Adds via stacks to the layer of the target objects.
Via	Lets you select the via options.

The Create Bus Context-Sensitive Menu Options

When creating bus wires, you can use the options available in the *Create Bus* context-sensitive menu which enhances the routing functionality.

The *Multi-Layer Bus Selection* option provides support to create and digitize the bits of a bus on different metal layers.

The use color mode options controls the color mode of the bus

The *Cycle Control Wire* option cycles the control wire function among the extreme and middle wires of a bus.

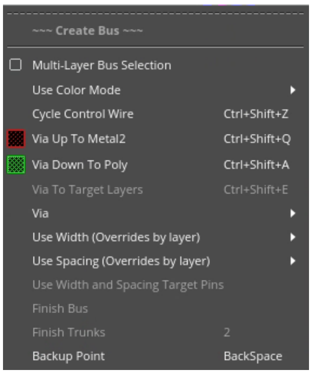
The *Via Up To* option changes the layer to the next layer up and places the corresponding default via.

The *Via Down To* option changes the layer to the next layer down and places the corresponding default via.

The *Via To Target Layers* option adds via stacks to the layer of the target objects.

The *Via* option let’s you select the via options.

The Create Bus Context-Sensitive Menu Options (continued)



Use Width	Use this option to specify the width.
Use Spacing	Use this option to specify the spacing.
Use Width and Spacing Target Pins	Sets both **Use Width**: *Target Pins* and **Use Spacing**: *Target Pins*.
Finish Bus	Automatically routes the active flightline following the layer, via and width specifications defined in the application default constraint group or any overrides defined in the **Create Bus** Form or in the **Wire Assistant**. This command is available only from XL tier onwards.
Finish Trunks	Automatically converts the pathSegs that are being created using the **Create Wire** or **Create Bus** command into trunks. The default bindkey is **2**.
Backup Point	Removes the last digitized point. If a via was added to the last digitized point, that via is also removed.

To enable the Finish Bus option in the Create Bus Context Sensitive Menu, run the ***`envSetVal("we" "enableFinishRoutingCmds" 'boolean t)`*** *command if not enabled.*

The Create Bus Context-Sensitive Menu Options

Use the *Use Width* option to specify the width.

Use the *Use Spacing* option to specify the spacing.

The *Use Width and Spacing Target Pins* option sets both the Use Width: Target Pins and Use Spacing: Target Pins.

The *Finish Bus* option automatically routes the active flightline following the layer, via and width specifications defined in the application default constraint group or any overrides defined in the Create Bus Form or in the Wire Assistant. This command is available only from XL tier onwards.

The *Finish Trunks* option automatically converts the pathSegs that are being created using the Create Wire or Create Bus command into trunks. The default bindkey is 2.

The *Backup Point* option removes the last digitized point. If via was added to the last digitized point that via is also removed.

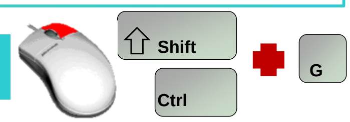
Bus Routing: Gather/Spread

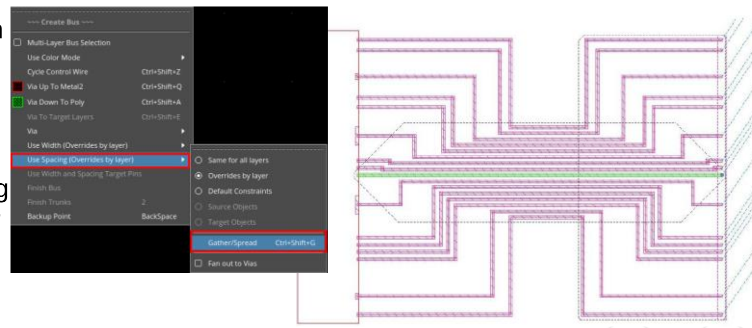
When creating bus routing, you can **right**-click in the design canvas to display the **Create Bus** context-sensitive menu.

The bus wires can be **gathered** or **spread** using the Source or Target Objects spacing options from this context-sensitive menu.

- Gathers or spreads bus wires before the next digitized point. It toggles to the Cancel Gather/Spread command.
- When spreading or gathering a bus, you can control the spacing of a bus.
- Environment variable: *gatherAndSpreadSpacing*
- The environment variable affects the spacing measured on the middle segments of gather and spread.

Create Bus Right Mouse Button





Bus Routing: Gather/Spread

When creating bus routing, you can *right*-click in the design canvas to display the *Create Bus* context-sensitive menu.

The bus wires can be *gathered* or *spread* using the Source or Target Objects spacing options.

This option gathers or spreads the bus wires before the next digitized point. It toggles to the *Cancel Gather/Spread* command.

When spreading or gathering a bus, you can control the spacing of a bus.

The environment variable for this option is *gatherAndSpreadSpacing*.

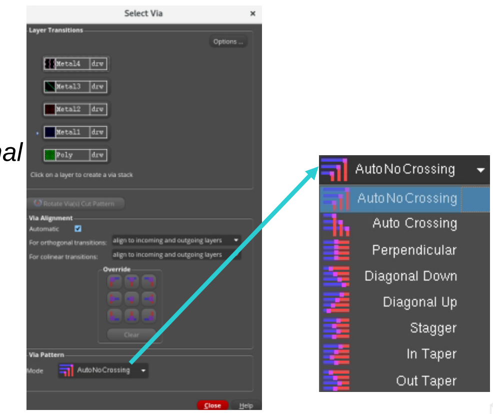
The environment variable affects the spacing measured on the middle segments of gather and spread.

Bus Routing: Place Via

When creating bus routing, you can use the **Select Via** form to change the bus layers.

- **Via Pattern** helps you specify a pre-defined manner in which multiple vias can be added simultaneously when routing multiple wires.
- You can choose from the pre-defined via patterns: *Auto No Crossing*, *Auto Crossing*, *Perpendicular*, *Diagonal Down*, *Diagonal Up*, *Stagger*, *In Taper*, and *Out Taper.*
- The display in the *Pattern* list next to the pattern name indicates the pattern in which the vias would be placed for the current routing direction.

When placing vias for buses, the “via-staggering” can be defined by the user.



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Bus Routing: Place Via

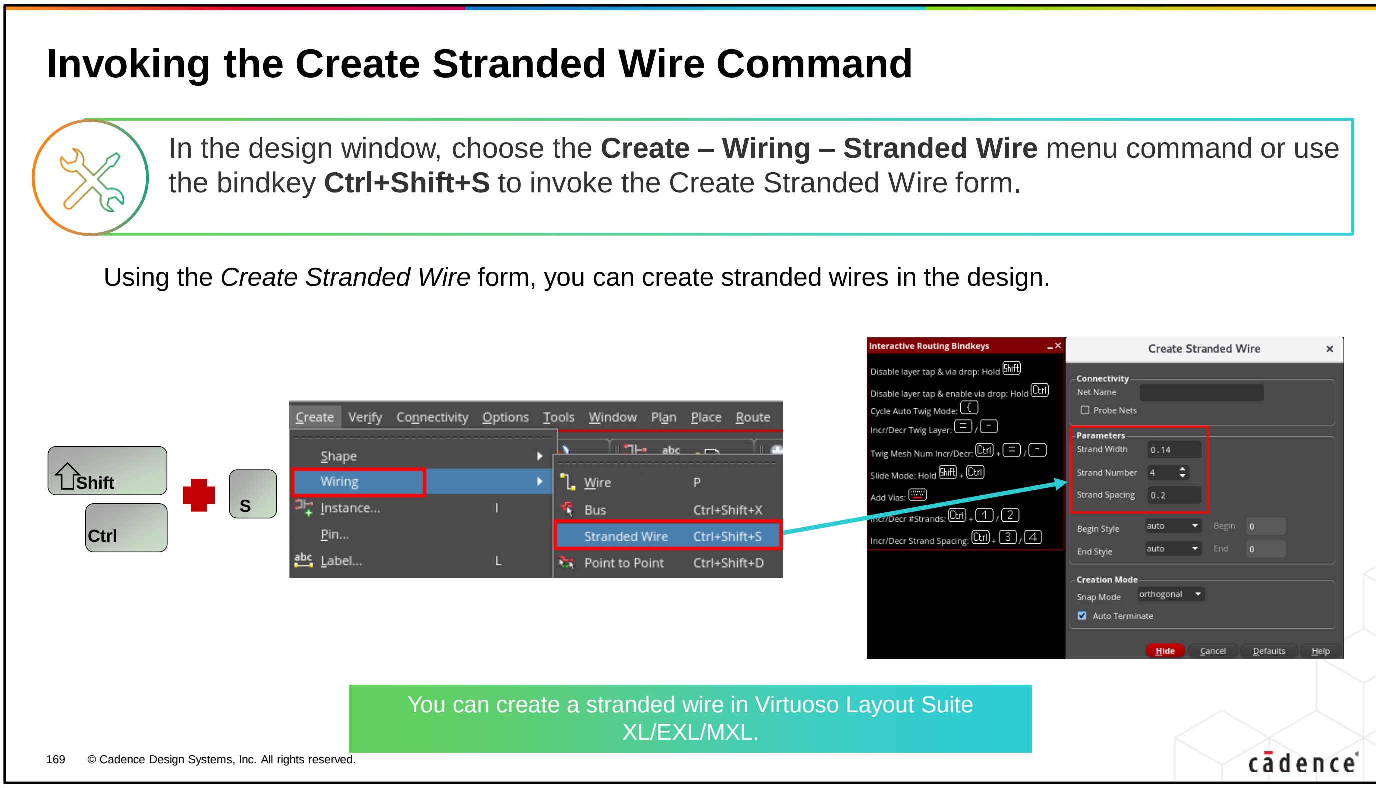
When creating bus routing, you can use the *Select Via* form to change the bus layers, rotate the via, and control the via alignment which can be invoked from the *Create Bus* context-sensitive menu or by using the bindkey.

When placing the vias for buses, the “via-staggering” can be defined by the user.

The *Via Pattern* option helps you to specify a pre-defined manner in which multiple vias can be added simultaneously when routing multiple wires.

You can choose from the pre-defined via patterns: namely, *Auto No Crossing*, *Auto Crossing*, *Perpendicular*, *Diagonal Down*, *Diagonal Up*, *Stagger*, *In Taper*, and *Out Taper*.

The display in the *Pattern* list next to the pattern name indicates the pattern in which the vias would be placed for the current routing direction.



Invoking the Create Stranded Wire Command

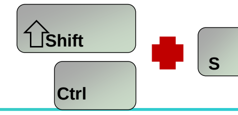
In the design window, choose the *Create – Wiring – Stranded Wire* menu command or use the bindkey *Ctrl+Shift+S* to invoke the Create Stranded Wire form.

Using this form, you can create stranded wires in the design.

You can create a stranded wire in Virtuoso Layout Suite XL,EXL,MXL tiers.

Using Stranded Wiring

Use the **Create Stranded Wire** command to create stranded wires in the design.



To create stranded wires in the design:

1. In the layout window, choose the **Create – Wiring – Stranded Wire** menu command or use the bindkey **Ctrl+Shift+S** to invoke the Create Stranded Wire form.
2. Set the values in the Create Stranded Wire form as below:
   a. *Strand Width* – **0.14**
   b. *Strand Number* – **4**
   c. *Strand Spacing* – **0.2**
   d. *Net Name* – **A**
3. Digitize and create the stranded wires.

Result: The stranded wires are created with connectivity (**A**).

Using Stranded Wiring

You can use the *Create Stranded Wire* command to create the stranded wires in the design.

To create stranded wires in the layout window:

Choose the *Create – Wiring – Stranded Wire* menu command or use the bindkey *Ctrl+Shift+S* to invoke the Create Stranded Wire form.

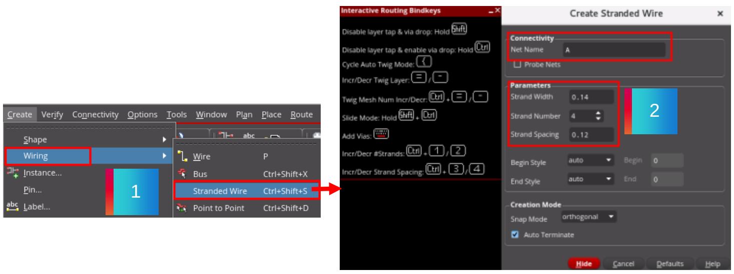
Then, set the required values in this form.

Now, digitize and create the stranded wires.

You can see that the stranded wires are created with connectivity.

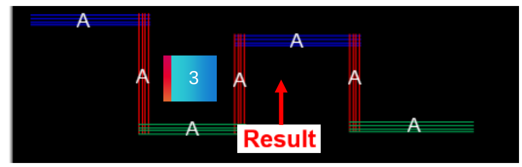
Result: Using Stranded Wiring

Invoke the *Create Stranded Wire* command.



Digitize and create the stranded wires.

Result: The stranded wires are created with connectivity (**A**).



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Result: Using Stranded Wiring

In the *Create Stranded Wire* form, after specifying the required values for your design, you can digitize and create the stranded wires.

You can observe that, in this example, the stranded wires are created with net connectivity *A* as you specified in the *Net Name* field of the *Create Stranded Wire* form.

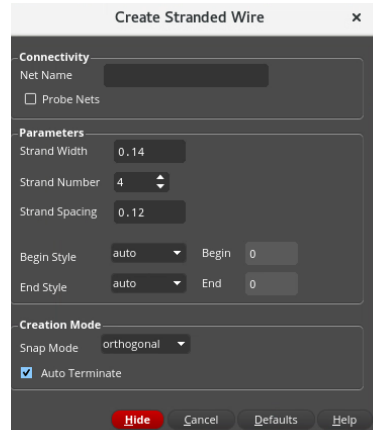
Displaying the Create Stranded Wire Context-Sensitive Menu

When creating stranded wires, you can use the options available in the **Create Stranded Wire** context-sensitive menu which enhances the routing functionality.

This context-sensitive menu can be accessed when the Create Stranded Wire (XL/EXL/MXL) command is active and the stranded wires are being routed.

1. To create stranded wires, in the design canvas, choose the **Create – Wiring – Stranded Wire** menu command or use the bindkey **Ctrl+Shift+S** to invoke the Create Stranded Wire form.
2. Now, **right**-click in the design canvas when the *Create Stranded Wire* (XL/EXL/MXL) command is running.

Result: The **Create Stranded Wire** context-sensitive menu is displayed in the layout canvas.



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Displaying the Create Stranded Wire Context-Sensitive Menu

When creating stranded wires, you can use the options available in the *Create Stranded Wire* context-sensitive menu which enhances the routing functionality.

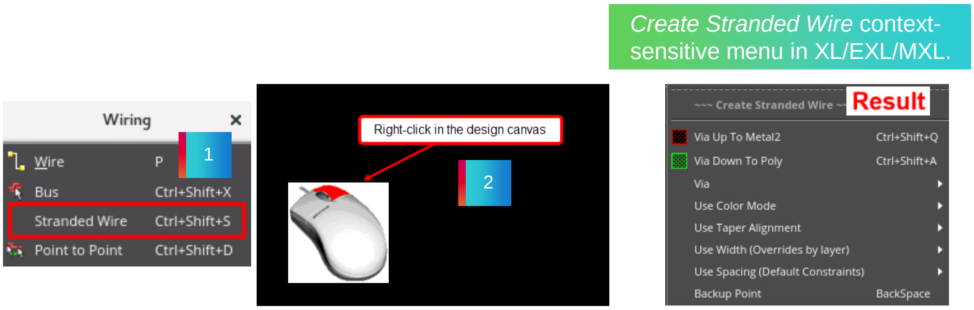
This context-sensitive menu can be accessed when the *Create Stranded Wire* command is active and the stranded wires are being routed.

To create the stranded wires, in the design canvas, choose the *Create – Wiring – Stranded Wire* menu command or use the bindkey *Ctrl+Shift+S* to invoke the Create Stranded Wire form.

Now, *right*-click in the design canvas when the *Create Stranded Wire* command is running in XL,EXL,MXL tiers of Virtuoso Layout Suite.

The *Create Stranded Wire* context-sensitive menu is displayed in the layout canvas.

Result: Displaying the Create Stranded Wire Context-Sensitive Menu

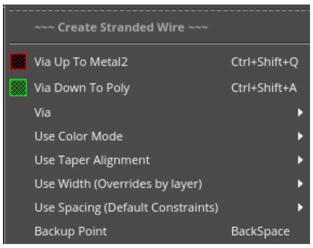


Result: Displaying the Create Stranded Wire Context-Sensitive Menu

When creating stranded wires, you can *right*-click in the design canvas to display the *Create Stranded Wire* context-sensitive menu.

You can use the options available in this context-sensitive menu to enhance your routing features when creating the stranded wires.

The Create Stranded Wire Context-Sensitive Menu Options



Via Up To	Changes the layer to the next layer up and automatically places the corresponding default via.
Via Down To	Changes the layer to the next layer down and automatically places the corresponding default via.
Via	Let's you select the via options.
Use Color Mode	Controls the color mode of the bus.
Use Taper Alignment	Let's you align the strands in a stranded wire when the number of strands are changed on each section of a wire in the same direction on the same layer or different metal layers.
Use Width	Use this option to specify the width.
Use Spacing	Use this option to specify the spacing.
Backup Point	Removes the last digitized point. If a via was added to the last digitized point, that via is also removed.

The Create Stranded Wire Context-Sensitive Menu Options

When creating stranded wiring, you can use the options available in the *Create Stranded Wire* context-sensitive menu which enhances the routing functionality.

The *Via Up To* option changes the layer to the next layer up and automatically places the corresponding default via.

The *Via Down To* option changes the layer to the next layer down, and automatically places the corresponding default via.

The *Via* option lets you select the via options.

The use color mode controls the color mode of the bus.

Use Taper Alignment option lets you align the strands in a stranded wire when the number of strands are changed on each section of a wire in the same direction on the same layer or different metal layers.

Use the *Use Width* option to specify the width.

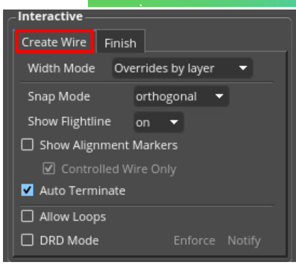
Use the *Use Spacing* option to specify the spacing.

The *Backup Point* option removes the last digitized point. If a via was added to the last digitized point, that via is also removed.

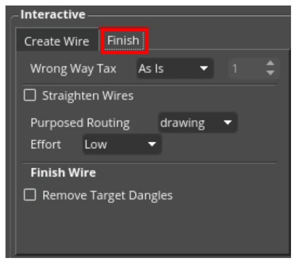
Create Wire: Finish Wire

To complete creating the wire automatically, use the **Finish Wire** command from the **Create Single Wire** context-sensitive menu.

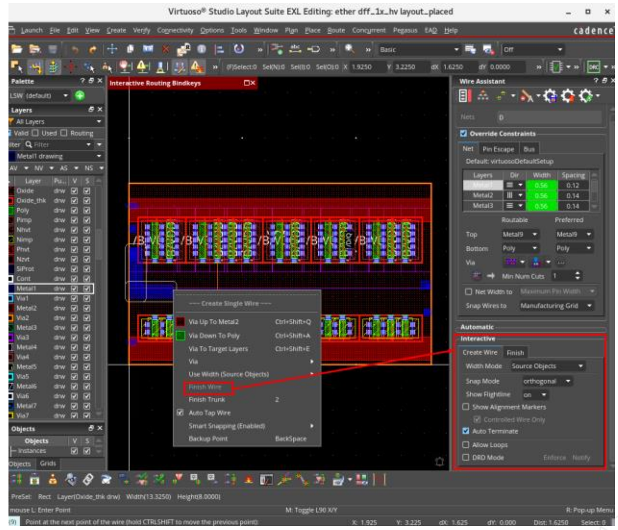
This context-sensitive menu can be displayed when the **Create Wire** (XL/EXL/MXL) command is



Create Wire option



Finish option



Create Wire: Finish Wire

To complete creating the wire automatically, use the *Finish Wire* option from the *Create Single Wire* context-sensitive menu.

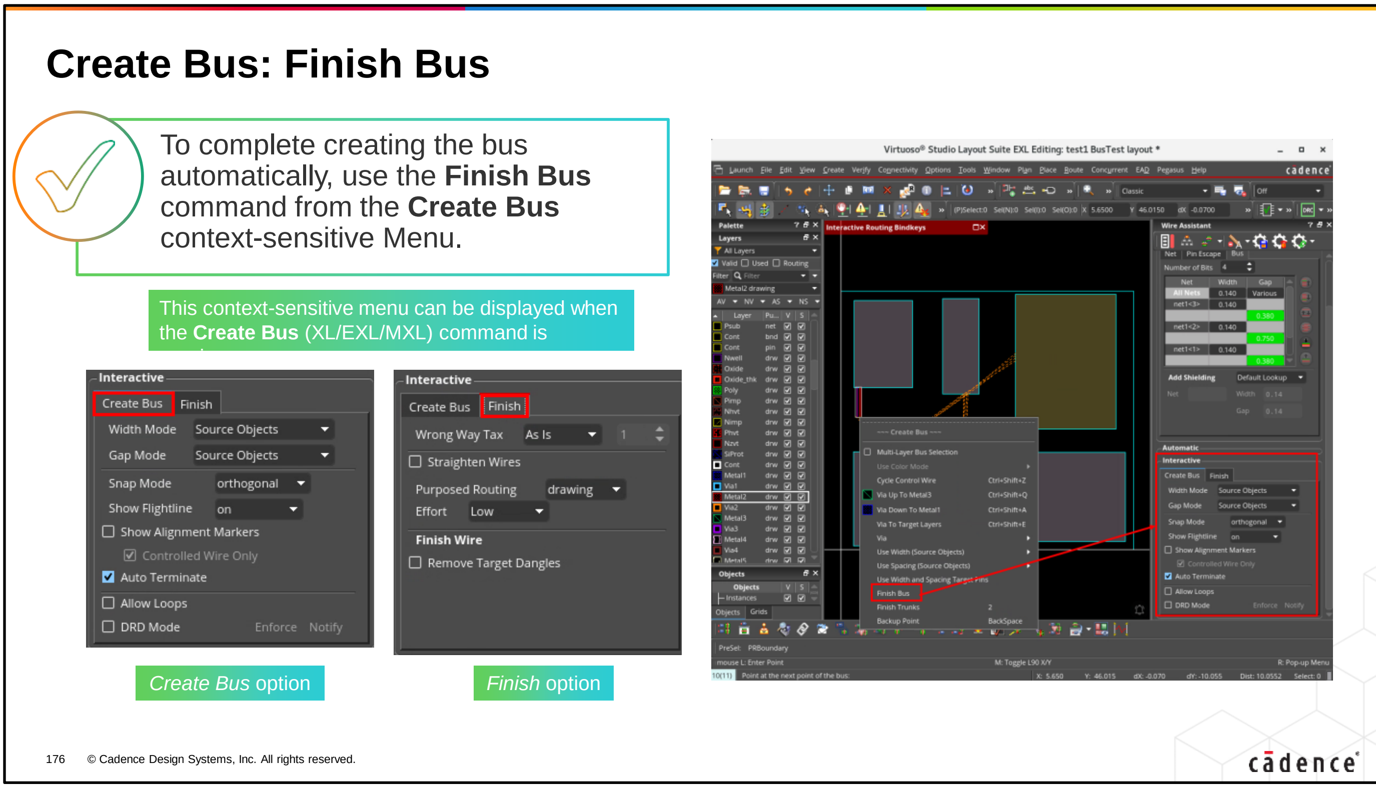
This context-sensitive menu can be displayed when the *Create Wire* command is running.

This command automatically routes the active flightline following the layer, via and width specifications defined in the application default constraint group or any overrides defined in either the *F3* Options form or the *Wire Assistant*.

The command routes only the active flightline even if the current net contains more than one open flightline.

This command is available only when the *Create Wire* command is active and there is an active flightline.

By default, it stays open if it cannot connect without creating a violation.



Create Bus: Finish Bus

To complete creating the bus automatically, use the *Finish Bus* option from the *Create Bus* context-sensitive menu.

This context-sensitive menu can be displayed when the *Create Bus* command is running.

This command automatically routes the active flightline following the layer, via and width specifications defined in the application default constraint group or any overrides defined in the *Create Bus* form or in the *Wire Assistant*.

The command is available only from XL tier onwards.

The command routes only the active flightline even if the current net contains more than one open flightline.

This command is available only when the *Create Bus* command is running and there is an active flightline, that is, dynamic flightline from the current cursor position to the closest target.

By default, the *Finish Bus* command stays open if it cannot create the routes without creating a violation.

Using Auto Router on the Selected Nets
Route With Default Lookup

Use the **Route With Default Lookup** command to route the selected net with default lookup values.

To route the selected net with default lookup values, in the layout canvas:

1. From the **Seed Attributes From a Constraint Group** drop-down list in the Wire Assistant toolbar, choose the **virtuosoDefaultSetup** constraint group.
2. From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option from the Wire Assistant toolbar.
3. Select the net **top** in the Navigator assistant.
4. Now, **right**-click and choose **Route With Default Lookup**.

Result: The router has routed with default lookup values.

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Using Auto Router on the Selected Nets – Route With Default Lookup

You can use the *Route With Default Lookup* command to route the selected net with the default lookup values.

To route the selected net with the default lookup values, in the layout canvas:

From the *Seed Attributes From a Constraint Group* drop-down list in the Wire Assistant toolbar, choose the *virtuosoDefaultSetup* constraint group.

Then, from the *Route Flow* groupbox, choose the *Custom / Digital (MST)* option from the Wire Assistant toolbar.

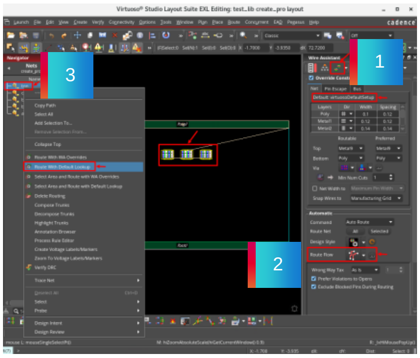
Next, select the net to route, say for example, net *top* in the Navigator assistant.

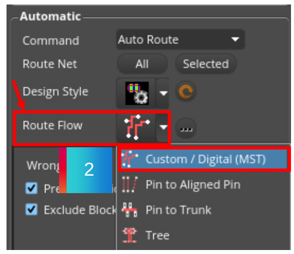
Now, *right*-click and choose the *Route With Default Lookup* command.

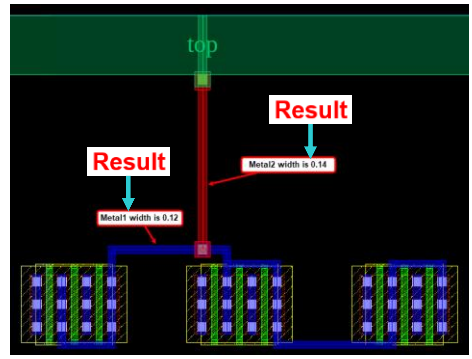
You can notice that the router has routed with the default lookup values.

Result: Using Auto Router on the Selected Nets
Route With Default Lookup

Result: The router has routed with default constraint lookup values for routing.







Result: Using Auto Router on the Selected Nets – Route With Default Lookup

In this example, with the *virtuosoDefaultSetup* constraint group and *Custom / Digital (MST)* routing topology set in the Wire Assistant, when you route the net *top* with the *Route With Default Lookup* command, you can see that, the router has routed with the default constraint lookup values for routing.

Here *Metal1 Width* is *0.12* and *Metal2 Width* is *0.14* which is taken from the default constraint group.

Using Auto Router on the Selected Nets
Route With WA Overrides

Use the **Route With WA Overrides** command to route the selected nets with the Wire Assistant Overrides.

To route the selected net with the Wire Assistant Overrides, in the layout canvas:

1. From the **Seed Attributes From a Constraint Group** drop-down list in the Wire Assistant toolbar, choose the **dsn:M1_M3_200n (Net)** constraint group.
2. From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option from the Wire Assistant toolbar.
3. Select the net **top** in the Navigator assistant.
4. Now, **right**-click and choose **Route With WA Overrides**.

Result: The router has routed with WA Overridden (PRO) values.

Using Auto Router on the Selected Nets – Route With WA Overrides

You can use the *Route With WA Overrides* command to route the selected nets with the Wire Assistant Overrides.

To route the selected net with the Wire Assistant Overrides, in the layout canvas:

From the *Seed Attributes From a Constraint Group* drop-down list in the Wire Assistant toolbar, choose the seeded constraint group, for example, *dsn:M1_M3_200n (Net)* constraint group.

Then, from the *Route Flow* groupbox, choose the *Custom / Digital (MST)* option from the Wire Assistant toolbar.

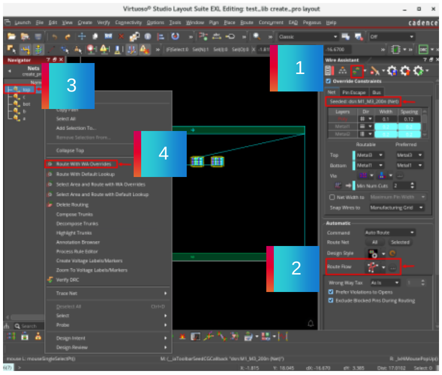
Next, select the net to route, for example, net *top* in the Navigator assistant.

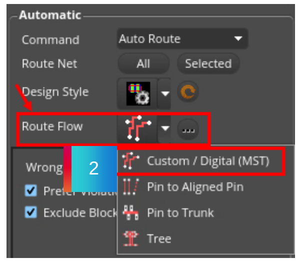
Now, *right*-click and choose the *Route With WA Overrides* command.

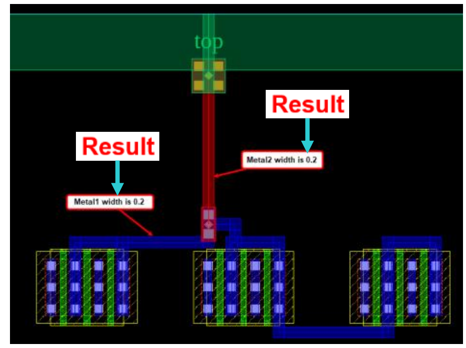
You will notice that the router has routed with the WA Overridden values.

Result: Using Auto Router on the Selected Nets
Route With WA Overrides

Result: The router has routed with the Wire Assistant Overrides.







Result: Using Auto Router on the Selected Nets – Route With WA Overrides

In this example, with the seeded constraint group and *Custom / Digital (MST)* routing topology set in the Wire Assistant, when you route the net *top* with the *Route With WA Overrides* command, you can see that the router has routed with the seeded constraint group values for routing.

Here, both *Metal1* and *Metal2 Widths* are *0.2,* which are taken from the seeded constraint group.

Using Auto Router on the Selected Nets
Route With WA Overrides with Dynamically Created PRO

Use the **Route With WA Overrides** command to route the selected nets with dynamically created PRO.

To route the selected net with the dynamically created PRO, in the layout canvas:

1. Modify the rules in the **Override Constraints** section in the Wire Assistant: for example, **Width** value of **Metal1** and **Metal2** to **0.3**.
2. From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option from the Wire Assistant toolbar.
3. Select the net **top** in the Navigator assistant.
4. Now, **right**-click and choose **Route With WA Overrides**.

Result: The router has routed with WA Overridden (PRO) values created dynamically.

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Using Auto Router on the Selected Nets – Route With WA Overrides with Dynamically Created PRO

You can use the *Route With WA Overrides* command to route the selected nets with the dynamically created Process Rule Overrides (PRO).

To route the selected net with the dynamically created Process Rule Overrides, in the layout canvas:

First, modify the rules in the *Override Constraints* section in the Wire Assistant: for example, *Width* value of *Metal1* and *Metal2* to *0.3*.

Then, from the *Route Flow* groupbox, choose the *Custom / Digital (MST)* option from the Wire Assistant toolbar.

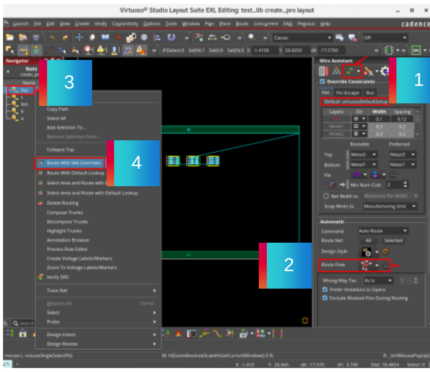
Next, select the net to route, for example, net *top* in the Navigator assistant.

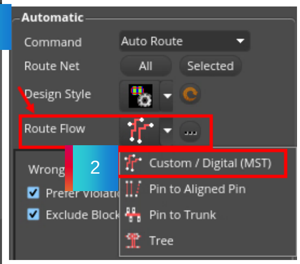
Now, *right*-click and choose the *Route With WA Overrides* command.

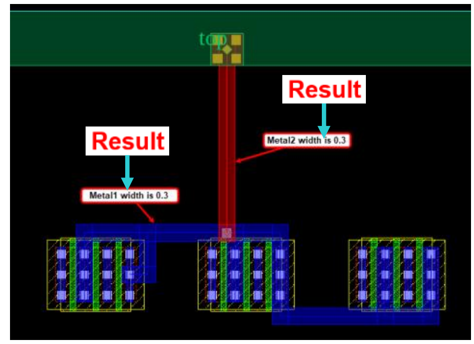
You will notice that the router has routed with the WA Overridden values created dynamically.

Result: Using Auto Router on the Selected Nets
Route With WA Overrides with Dynamically Created PRO

Result: The router has routed with Process Rule Overrides created dynamically.







Result: Using Auto Router on the Selected Nets – Route With WA Overrides with Dynamically Created PRO

In this example, with the modified rules in the *Override Constraints* section and *Custom / Digital (MST)* routing topology set in the Wire Assistant, when you route the net *top* with the *Route With WA Overrides* command, you can see that the router has routed with the Process Rule Overrides created dynamically for routing.

Here, both *Metal1* and *Metal2 Widths* are *0.3*, which are entered in the *Override Constraints* section in the Wire Assistant.

FAQ: Composing the Trunks

How do you compose the trunks?

To convert the selected set of shapes into a trunk object, in the layout canvas, choose the **Edit – Wiring – Compose Trunks** menu command.

How do you compose the trunks?

To convert the selected set of shapes into a trunk object, in the layout canvas, choose the *Edit – Wiring – Compose Trunks* menu command.

FAQ: Highlighting the Trunks

When debugging the routing, which command can you use to highlight which segments have been used as trunks?

When debugging the routing, you can use the **Highlight Trunks** command to highlight which segments have been used as trunks.

When debugging the routing, which command can you use to highlight which segments have been used as trunks?

When debugging the routing, you can use the *Highlight Trunks* command to highlight which segments have been used as trunks.

FAQ: Routing Topology

Which routing topology style can you use to connect an individual pin to a trunk?

You can use the **Pin-to-Trunk** routing topology style to connect an individual pin to a trunk.

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Which routing topology style can you use to connect an individual pin to a trunk?

You can use the *Pin-to-Trunk* routing topology style to connect an individual pin to a trunk.

FAQ: The Create Single Wire Context-Sensitive Menu

Which command on the **Create Single Wire** context-sensitive menu can you use to complete creating the wire automatically?

You can use the **Finish Wire** command on the **Create Single Wire** context-sensitive menu to complete creating the wire automatically.

Which command on the *Create Single Wire* context-sensitive menu can you use to complete creating the wire automatically?

You can use the *Finish Wire* command on the *Create Single Wire* context-sensitive menu to complete creating the wire automatically.

FAQ: The Create Stranded Wire Context-Sensitive Menu

How can you display the **Create Stranded Wire** context-sensitive menu?

You can access the **Create Stranded Wire** context-sensitive menu when the *Create Stranded Wire* (XL/EXL/MXL) command is active and the stranded wires are being routed.

How can you display the *Create Stranded Wire* context-sensitive menu?

You can access the *Create Stranded Wire* context-sensitive menu when the *Create Stranded Wire* (XL/EXL/MXL) command is active and the stranded wires are being routed.

Module Summary

In this module, you learned how to

- Use Pin-to-Trunk Routing
  - Start Pin-to-Trunk Router with Route With Default Lookup / Route With WA Overrides
  - Start Pin-to-Trunk Router with Wire Assistant
- Create Assisted Wires
  - Use Point to Point Routing
  - Use Bus Routing
  - Use Stranded Wiring
  - Use Finish Bus Command
- Launch Auto Router on the Selected Nets
  - Route With Default Lookup with default CG
  - Route With WA Overrides with seeded CG
  - Route With WA Overrides with dynamically created PRO

Module Summary

In this module, you learned how to use the Pin-to-Trunk Router with the Route With Default Lookup and with the Route With Wire Assistant Overrides commands.

You learned how to use the Pin-to-Trunk Router with the Wire Assistant toolbar.

You learned how to create the assisted wires using the Point to Point Routing, Bus Routing, Stranded Wiring, and Finish Bus commands.

You learned how to launch the auto router on the Selected Nets with the Default Lookup with the default Constraint Group, with the Wire Assistant Overrides with the seeded Constraint Group, and with the dynamically created Process Rule Overrides.

Labs

Lab 4-1 Using Pin-to-Trunk Routing

Lab 4-2 Creating Assisted Wires

Lab 4-3 Using Auto Router on Selected Nets

Labs

Now, you will do lab exercises for Module 4.

There are 3 labs in this module.

In the first lab, you will explore on how to use the Pin-to-Trunk Routing command.

In the second lab, you will explore on how to Create the Assisted Wires.

In the third lab, you will explore on how to run the Auto Router on the selected nets.

Quiz: The Bus Wires

The bus wires can be gathered or spread using the Source or Target Objects spacing options.

A. True

B. False

Answer: A

The bus wires can be gathered or spread using the Source or Target Objects spacing options.

Answer: A

The bus wires can be gathered or spread using the Source or Target Objects spacing options.

Quiz: Auto Router in Virtuoso

What are the options for using the **Auto Router** in Virtuoso?

Select all that apply.

A. Route With Default Lookup

B. Supply a TCL script to control routing

C. Route Symbolic and ignore rules

D. Route With WA Overrides

E. All of the above

Answer: A and D

The options available are *Route With Default Lookup* and *Route With WA Overrides* for using the **Auto Router** in Virtuoso.

Answer: A and D

The options available are *Route With Default Lookup* and *Route With WA Overrides* for using the *Auto Router* in Virtuoso.

Quiz: Interactive Routing Command

Which interactive routing command can you use to automatically generate routing between two digitized points in a given cellview?

Select all that apply.

A. Bus Routing
B. Pin-to-Trunk Routing
C. Point to Point Routing
D. Stranded Wiring
E. None of the above

Answer: C

You can use the **Point to Point** interactive routing command to automatically generate routing between two digitized points in a given cellview.

Answer: C

You can use the *Point to Point* interactive routing command to automatically generate routing between two digitized points in a given cellview.

Quiz: Route Flow Routing Topology Styles

What are the **Route Flow** routing topology styles available in the **Wire Assistant**?

Select all that apply.

A. Custom / Digital (MST)

B. Pin to Aligned Pin

C. Pin to Trunk

D. Bus Routing

E. All of the above

Answer: A, B, and C

The **Route Flow** routing topology styles available in the **Wire Assistant** are *Custom / Digital (MST)*, *Pin to Aligned Pin*, and *Pin to Trunk*.

Answer: A, B, and C

The *Route Flow* routing topology styles available in the *Wire Assistant* are *Custom / Digital (MST) Pin to Aligned Pin*, and *Pin to Trunk.*

Quiz: The Create Bus Context-Sensitive Menu

Which command on the **Create Bus** context-sensitive menu can you use to complete creating the bus automatically? ------ ---

Enter the answer in the blank field.

Answer: **Finish Bus**

You can use the **Finish Bus** command on the **Create Bus** context-sensitive menu to complete creating the bus automatically.

Answer: *Finish Bus*

You can use the *Finish Bus* command on the *Create Bus* context-sensitive menu to complete creating the bus automatically.

Module 5

Course Conclusions

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Summary

In this course, you learned to

- Analyze the basic concepts and settings in wire creation
- Create and modify the wires
- Use the assisted wires creation
- Use the smart auto via feature (Appendix)

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References

Refer to the links below for further details on the topics in this course

Virtuoso Layout Suite XL: Basic Editing User Guide IC25.1

This document describes using the Virtuoso Layout Suite XL Layout Editor (Layout XL) in a connectivity-based environment.

Virtuoso Layout Suite EXL Reference IC25.1

This document describes using the Virtuoso Layout Suite EXL Layout Editor (Layout EXL) in a connectivity-based environment.

Virtuoso Layout Suite MXL Reference IC25.1

This document describes using the Virtuoso Layout Suite MXL Layout Editor (Layout MXL) in a connectivity-based environment.

Virtuoso Space-based Router User Guide IC25.1

The Virtuoso Space-Based Router enables high-speed shape-based routing for physical designs.

Virtuoso Interactive and Assisted Routing User Guide IC25.1

The Virtuoso Interactive and Assisted Routing capabilities enable you to route connections interactively within the Virtuoso environment.

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Module 6

Next Steps

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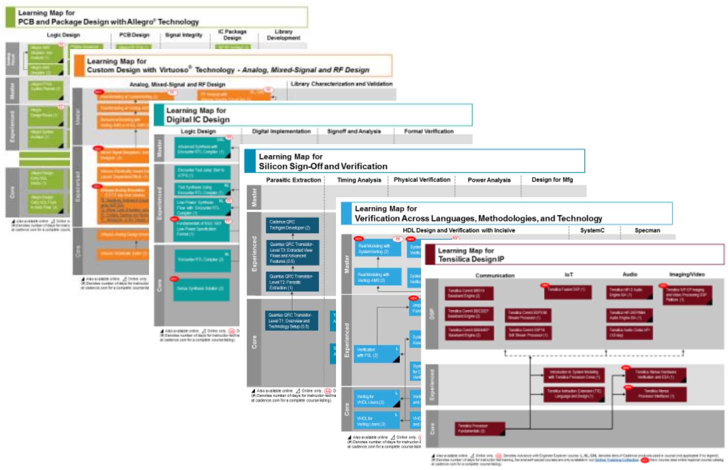
Learning Maps

Cadence® Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers.

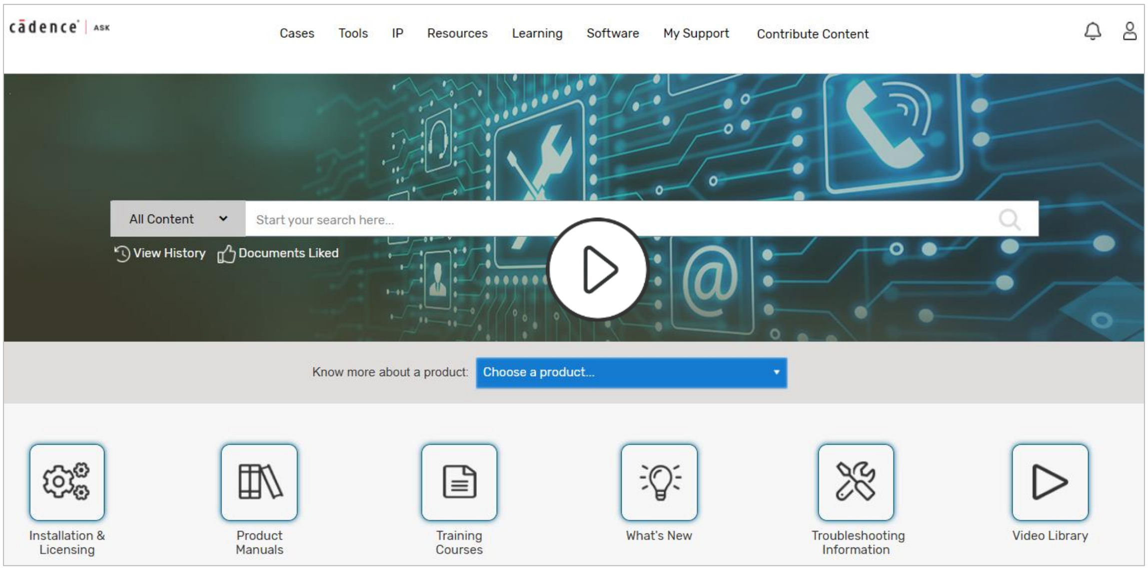
Click here to see all our courses in each technology area and the recommended order in which to take them.



Go here to view the learning maps:

http://www.cadence.com/Training/Pages/learning_maps.aspx

Cadence Learning and Support – ASK Portal



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Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion
- Complete the Course's Digital Badge exam to earn your Badge

Thank you!

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Appendix A

Smart Auto Via

In this appendix, we will discuss Smart Auto Via.

Appendix Objectives

In this appendix, you

- Use Auto Via Preview
  - Rectangular Overlap
  - Polygonal Overlap
  - Via Locked
- Use Fast Interactive Via Editing – FIVE
  - Rectangular Overlap
  - Polygonal Overlap
  - Update to Double Cut
  - Fast Edit on Selected Vias
  - Update Via After Change of Techfile

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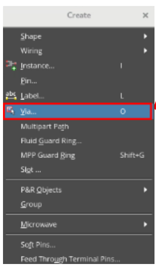
In this appendix, you will use the Auto Via Preview with Rectangular Overlap, with Polygonal Overlap, and with Via Locked.

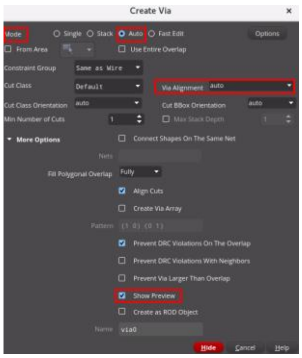
You will use the Fast Interactive Via Editing with Rectangular Overlap, with Polygonal Overlap, with Update to Double Cut, with Fast Edit on Selected Vias, and with Update Via After Change of Techfile.

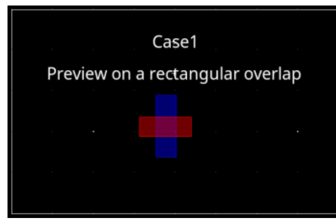
Create Via: Show Preview

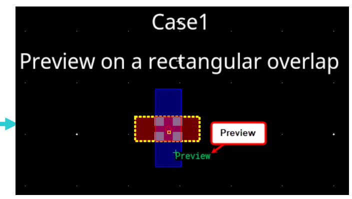
Use the **Show Preview** option in the **Create Via** form to see a preview of the via that will be created by clicking the **LMB** button in the layout window.

To create via to connect objects on two layers within a cellview, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar, or use the bindkey **O** to invoke the Create Via form.









Move the mouse over the overlap.

A label **Preview** is displayed close to mouse.

Create Via: Show Preview

You can use the *Show Preview* option in the *Create Via* form to see a preview of the via that will be created by clicking the *Left Mouse Button* in the layout window.

To create via to connect objects on two layers within a cellview, choose the *Create – Via* menu command or use the *Create Via* icon in the Create toolbar or use the bindkey *O* to invoke the Create Via form.

When you move the mouse over the overlap, a label *Preview* is displayed close to the mouse.

Using Auto Via Preview: Rectangular Overlap

Use the **Auto Via Preview** functionality on Rectangular Overlap.

Show Preview enables a preview of the via that you are going to create by clicking the **LMB** button.

Using Auto Via Preview: Rectangular Overlap

You can use the *Auto Via Preview* functionality on the Rectangular Overlap.

The *Show Preview* option in the *Create Via* form enables a preview of the via that you are going to create by clicking the *Left Mouse Button*.

On the Rectangular Overlap to use the *Auto Via Preview* functionality, perform the following:

1. Invoke and set the values in the Create Via form.
2. Move the mouse cursor over the overlap.
3. Use the Via Preview Bindkeys.
4. Do a center alignment.
5. Place the via.

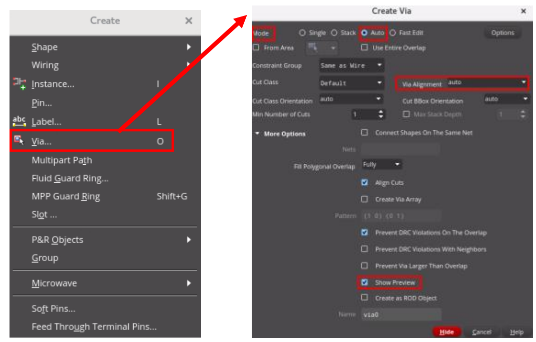
Step 1: Invoke & Set the Values in the Create Via Form

Invoke & set the values in the Create Via form → Move the mouse cursor over the overlap → Use the Via Preview Bindkeys → Do a center alignment → Place the via

To create via to connect objects on two layers within a cellview, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar, or use the bindkey **O** to invoke the Create Via form.

Set the following values in the Create Via form:

- Set **Mode** to *Auto*.
- Set **Via Alignment** as *auto*.
- Expand the **More Options** *Progressive Disclosure Widget*.
  - Select the option: **Show Preview**.
  - Leave the other fields as default values.



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Step 1: Invoke & Set the Values in the Create Via Form

To create via to connect objects on two layers within a cellview, choose the *Create – Via* menu command or use the *Create Via* icon in the Create toolbar or use the bindkey *O* to invoke the Create Via form.

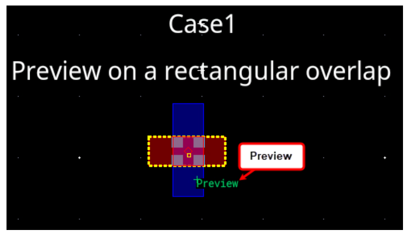
Set the required values in the *Create Via* form to create the via.

Step 2: Move the Mouse Cursor Over the Overlap

Invoke & set the values in the Create Via form → **Move the mouse cursor over the overlap** → Use the Via Preview Bindkeys → Do a center alignment → Place the via

In the layout canvas, move the mouse cursor over the overlap.

- A label **Preview** is displayed close to the mouse cursor.
- You see the via that you will drop when clicking the **LMB** button.



Step 2: Move the Mouse Cursor Over the Overlap

In the layout canvas, move the mouse cursor over the overlap.

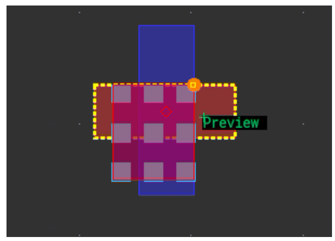
A label *Preview* is displayed close to the mouse cursor.

You see the via that you will drop when clicking the *Left Mouse Button*.

Step 3: Use the Via Preview Bindkeys

Using the **Via Preview Bindkeys** form, you can change the number of *Rows* and *Columns* with the **Ctrl+Wheel up/down** and/or **Shift+Wheel up/down**.

- Use the **Via Preview Bindkeys** form and increase the number of *Rows* to **3** and *Columns* to **3** in order to get a via, which is bigger than the overlap.



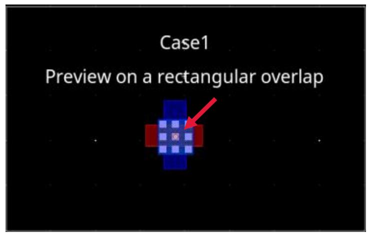
Step 3: Use the Via Preview Bindkeys

Use the **Via Preview Bindkeys** form and increase the number of *Rows* to **3** and *Columns* to **3** in order to get a via, which is bigger than the overlap.

Step 4: Do a Center Alignment

Invoke & set the values in the Create Via form → Move the mouse cursor over the overlap → Use the Via Preview Bindkeys → **Do a center alignment** → Place the via

- Do a center alignment.
- You can do that graphically by moving the mouse cursor.



An orange circle helps you to do the alignment.

Step 4: Do a Center Alignment

Do a center alignment.

You can do that graphically by moving the mouse cursor.

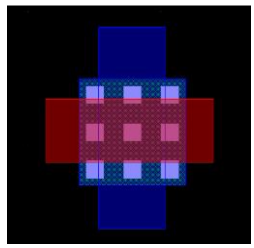
An orange circle helps you to do the alignment.

Step 5: Place the Via

Invoke & set the values in the Create Via form → Move the mouse cursor over the overlap → Use the Via Preview Bindkeys → Do a center alignment → Place the via

Click the **LMB** button in the layout canvas.

- The via that you were seeing in the **Preview** is now dropped in the design window.



Once dropped, the label **Preview** is not displayed anymore, which helps the user to know that **Preview** mode is not enabled now.

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Step 5: Place the Via

Click the *Left Mouse Button* in the layout canvas.

The via that you were seeing in the *Preview* is now dropped in the design window.

Once dropped, the label *Preview* is not displayed anymore which helps the user to know that *Preview* mode is not enabled now.

Using Auto Via Preview: Polygonal Overlap

Use the **Auto Via Preview** functionality on Polygonal Overlap.

Show Preview enables to see the via that you are going to create by clicking the **LMB** button.

To create via to connect objects on two layers within a cellview:

1. In the design canvas, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar or use the bindkey **O** to invoke the Create Via form.
2. Set the values in the Create Via form as below:
   - Set **Mode** to *Auto*.
   - Set **Via Alignment** as *auto*.
   - Expand the **More Options** *Progressive Disclosure Widget*.
     - Select the option: **Show Preview.**
     - Leave the other fields as default values.
3. In the layout canvas, move the mouse cursor over the overlap.
4. Click the **LMB** button in the layout canvas.

Result: The via is dropped in the design window.

Using Auto Via Preview: Polygonal Overlap

You can use the *Auto Via Preview* functionality on the Polygonal Overlap.

The *Show Preview* option in the *Create Via* form enables you to see the preview of the via that you are going to create by clicking the *Left Mouse Button*.

To create the via to connect the objects on two layers within a cellview:

Invoke the *Create Via* form and set the values in the form as needed.

Then, in the layout canvas, move the mouse cursor over the overlap and click the *LMB* button.

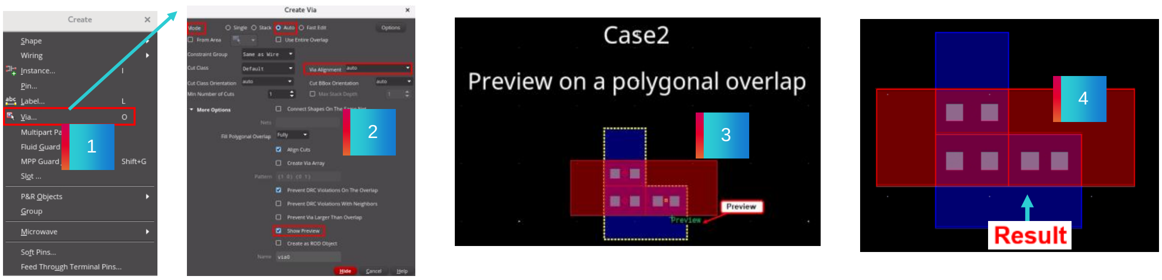
You will notice that the via is dropped in the design window.

Result: Using Auto Via Preview: Polygonal Overlap

Invoke the **Create Via** form and set the values.

Move the mouse cursor over the overlap. A label **Preview** is displayed close to the mouse cursor.

Result: Click the **LMB** button to place the via.



The number of *Rows* and *Columns* can **NOT** be changed in a polygonal overlap because for the same transition, several vias are created.

Result: Using Auto Via Preview: Polygonal Overlap

After invoking the *Create Via* form and setting the required values in the form, when you move the mouse cursor over the overlap, you can see that a label *Preview* is displayed close to the mouse cursor.

You see the via that you will drop when clicking the *Left Mouse Button*.

Click the *LMB* button in the layout canvas.

The via that you were seeing in the *Preview* is now dropped in the design window.

Once dropped, the label *Preview* is not displayed anymore which helps the user to know that *Preview* mode is not enabled now.

The number of *Rows* and *Columns* cannot be changed in a polygonal overlap because for the same transition several vias are created.

Using Auto Via Preview: Via Locked

Use the **Auto Via Preview** functionality to lock the via and drop it in another overlap.

Show Preview enables you to see the via that you are going to create by clicking the **LMB** button.

To create via to connect objects on two layers within a cellview:

1. In the design canvas, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar or use the bindkey **O** to invoke the Create Via form.
2. Set the values in the Create Via form as below:
   - Set **Mode** to *Auto*.
   - Set **Via Alignment** as *auto*.
   - Expand the **More Options** *Progressive Disclosure Widget*.
     - Select the option: **Show Preview.**
     - Leave the other fields as default values.
3. In the layout canvas, move the mouse cursor over the overlap.
4. Use the **Via Preview Bindkeys** and **Via Locking** features to drop the vias in the design.

Result: Once a customized via has been dropped, this via is locked, and you can drop it in another overlap.

Using Auto Via Preview: Via Locked

You can use the *Auto Via Preview* functionality to lock the via and drop it in another overlap.

The *Show Preview* option in the *Create Via* form enables you to see the preview of the via that you are going to create by clicking the *Left Mouse Button*.

To create the via to connect the objects on two layers within a cellview:

Invoke the *Create Via* form and set the values in the form as needed.

Then, in the layout canvas, move the mouse cursor over the overlap.

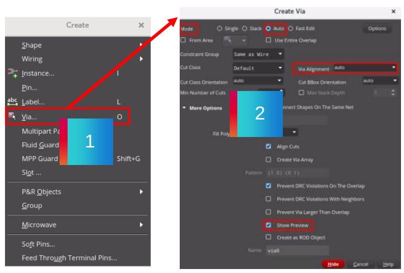
Use the *Via Preview Bindkeys* and *Via Locking* features to drop the vias in the design.

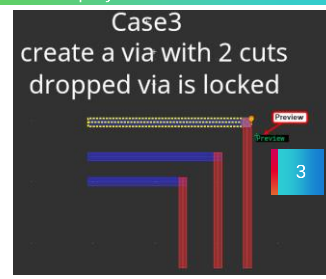
You will notice that once a customized via has been dropped, this via is locked and you can drop it in another overlap.

Result: Using Auto Via Preview: Via Locked

Invoke the **Create Via** form and set the values.

Move the mouse cursor over the overlap. A label **Preview** is displayed close to the mouse cursor.





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Result: Using Auto Via Preview: Via Locked

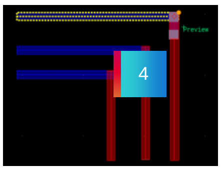
Here you see that the *Create Via* form is invoked and the values are set in the form.

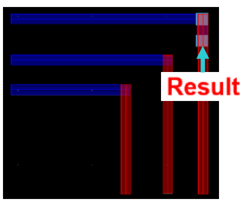
When you move the mouse cursor over the overlap, you notice that, a label *Preview* is displayed close to the mouse cursor.

Result: Using Auto Via Preview: Via Locked (continued)

Once a customized via has been dropped, this via is locked and you can drop it in another overlap.

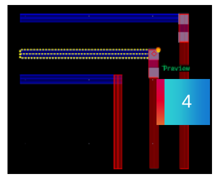
Use the **Via Preview Bindkeys** form to change the number of *Rows* to **2**.

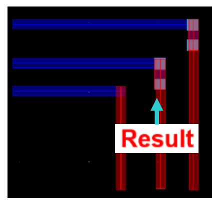




Result: Click the **LMB** button to place the via.

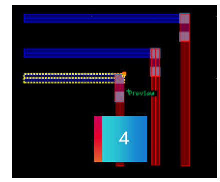
Move the mouse over the second overlap. The previous via is locked and you can drop it on the second overlap.

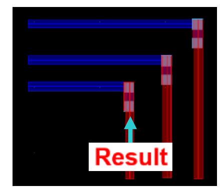




Result: Click the **LMB** button to place the via.

Move the mouse over the third overlap. The previous via is locked and you can drop it on the third overlap.





Result: Click the **LMB** button to place the via.

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Result: Using Auto Via Preview: Via Locked

Use the *Via Preview Bindkeys* form to change the number of *Rows* to *2*.

Then, click the *LMB* button to place the via.

Next, move the mouse over the second overlap. Now, the previous via is locked and you can drop it on the second overlap.

Similarly, when you move the mouse over the third overlap, you can see that, the previous via is locked and you can drop it on the third overlap.

Hence you observe that once a customized via has been dropped, this via is locked and you can drop it in another overlap.

Using Fast Interactive Via Editing (FIVE)
Rectangular Overlap

Use the **Fast Interactive Via Editing** (**FIVE**) to edit/modify the existing vias when the overlapping area increases/decreases to match the new topology on the Rectangular Overlap.

Show Preview enables you to see the via that you are going to create by clicking the **LMB** button.

To create via to connect objects on two layers within a cellview:

1. In the design canvas, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar, or use the bindkey **O** to invoke the Create Via form.
2. Set the values in the Create Via form as below:
   - Set **Mode** to *Fast Edit*.
   - Expand the **More Options** Progressive Disclosure Widget.
     - Select the two options: **Show Preview** and **Show Hints**.
   - Leave the other fields as default values.

Using Fast Interactive Via Editing – FIVE: Rectangular Overlap

You can use the *Fast Interactive Via Editing* feature to edit or modify the existing vias when the overlapping area increases or decreases to match the new topology on the Rectangular Overlap.

The *Show Preview* option in the *Create Via* form enables you to see the preview of the via that you are going to create by clicking the *Left Mouse Button*.

To create the via to connect the objects on two layers within a cellview:

Invoke the *Create Via* form and set the values in the form as required.

Then, in the layout canvas, move the mouse cursor over the overlap, which is smaller or bigger than the dropped via.

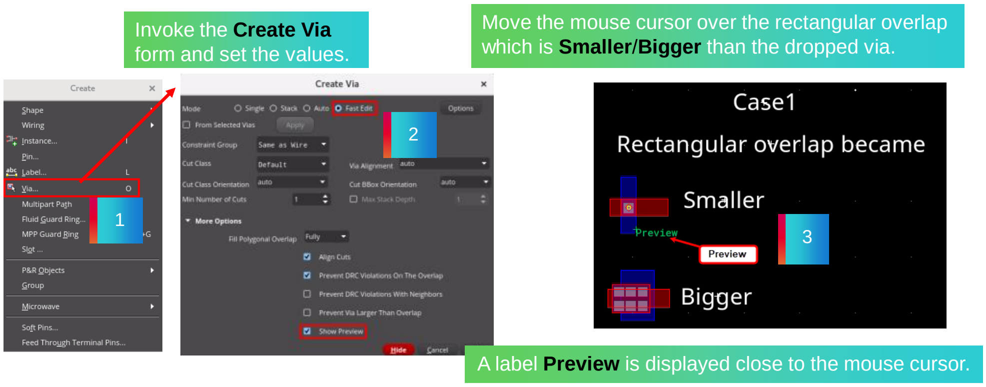
Now, modify the existing vias that was fitting a previous overlap that became smaller or bigger.

You will notice that the modified via will fit the new overlap.

Result: Using Fast Interactive Via Editing (FIVE)
Rectangular Overlap

3. In the layout canvas, move the mouse cursor over the rectangular overlap, which is **Smaller**/**Bigger** than the dropped via.
4. Modify the existing vias that were fitting a previous overlap that became **Smaller**/**Bigger**.

Result: The modified via will fit the new overlap.



Result: Using Fast Interactive Via Editing – FIVE: Rectangular Overlap

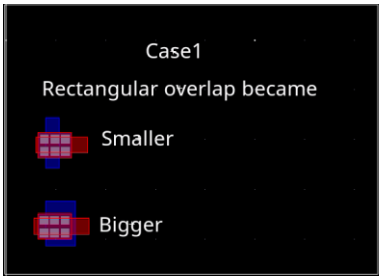
Here you see that the *Create Via* form is invoked and the values are set in the form.

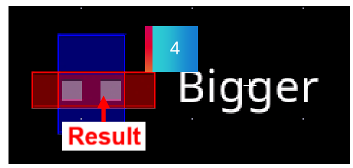
When you move the mouse cursor over the rectangular overlap, which is smaller or bigger than the dropped via, you notice that a label *Preview* is displayed close to the mouse cursor.

Result: Using Fast Interactive Via Editing (FIVE)

Rectangular Overlap (continued)

Modify the existing vias that was fitting a previous overlap that became **Smaller**/**Bigger**.





The modified via fits the new overlap.

Result: Using Fast Interactive Via Editing – FIVE: Rectangular Overlap

When you modify the existing vias that were fitting a previous overlap that became smaller or bigger, you see that the modified via fits the new overlap.

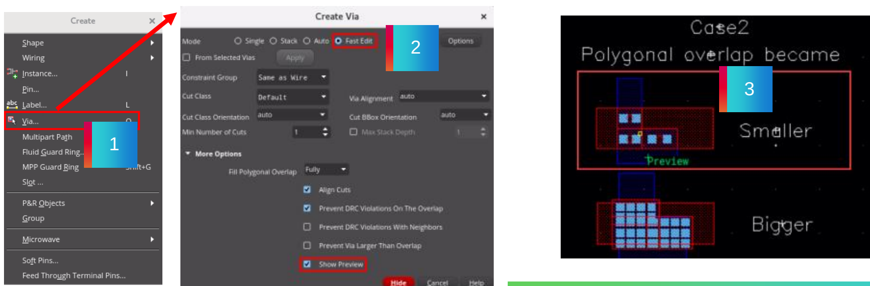
Result: Using Fast Interactive Via Editing (FIVE)
Polygonal Overlap

3. In the layout canvas, move the mouse cursor over the polygonal overlap, which is **Smaller**/**Bigger** than the dropped via.
4. Modify the existing vias that was fitting a previous overlap that became **Smaller**/**Bigger**.

Result: The modified via will fit the new overlap.

Invoke the **Create Via** form and set the values.

Move the mouse cursor over the polygonal overlap which is **Smaller**/**Bigger** than the dropped via.



A label **Preview** is displayed close to the mouse cursor.

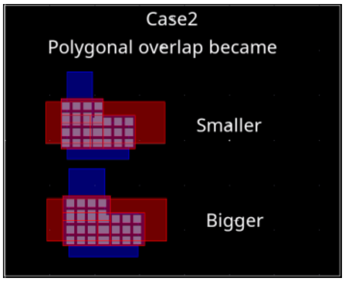
Result: Using Fast Interactive Via Editing – FIVE: Polygonal Overlap

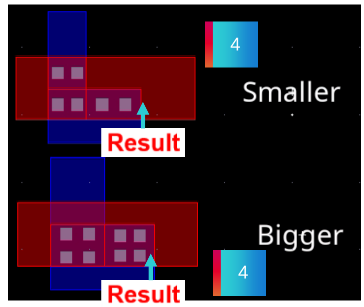
Here you see that the *Create Via* form is invoked and the values are set in the form.

When you move the mouse cursor over the polygonal overlap, which is smaller or bigger than the dropped via, you notice that a label *Preview* is displayed close to the mouse cursor.

Result: Using Fast Interactive Via Editing (FIVE)
Polygonal Overlap (continued)

Modify the existing vias that was fitting a previous overlap that became **Smaller/Bigger**.





The modified via fits the new overlap.

Result: Using Fast Interactive Via Editing – FIVE: Polygonal Overlap

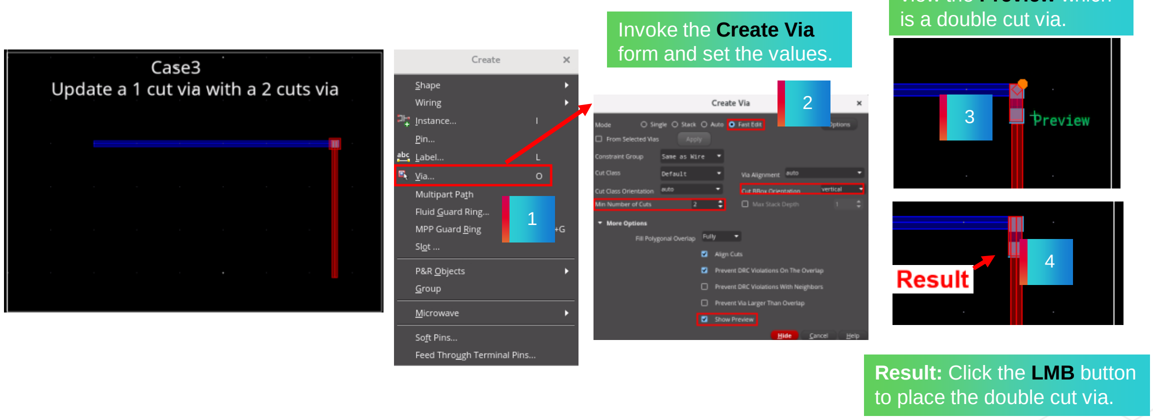
When you modify the existing vias that were fitting a previous overlap that became smaller or bigger, you see that the modified via fits the new overlap.

Result: Using Fast Interactive Via Editing (FIVE)
Update to Double Cut

3. In the layout canvas, move the mouse cursor over the existing via to view the **Preview,** which is a double cut via.
4. Click the **LMB** button in the layout canvas.

Result: The double cut via is dropped in the design window.

Move the mouse cursor over the existing via to view the **Preview** which is a double cut via.



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Result: Using Fast Interactive Via Editing – FIVE: Update to Double Cut

The *Create Via* form is invoked and the values are set in the form.

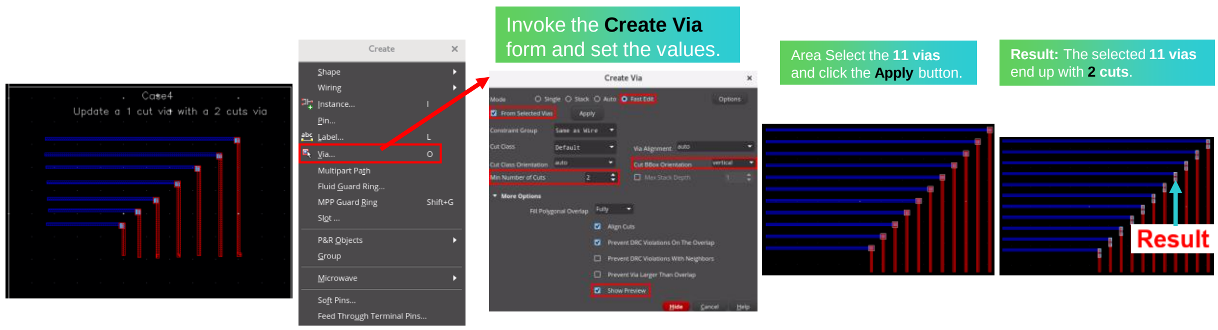
Move the mouse cursor over the existing via to view the *Preview*, which is a double cut via.

When you click the *LMB* button, the double cut via is placed in the canvas.

Result: Using Fast Interactive Via Editing – FIVE:
Fast Edit on Selected Vias

3. In the layout canvas, area select the vias; **11 via objects** should be selected.
4. Click the **Apply** button in the Create Via form.

Result: The selected **11 vias** end up with **2 cuts** in the design window.



You can have the same behavior by using the **MMB**.

Result: Using Fast Interactive Via Editing – FIVE: Fast Edit on Selected Vias

The *Create Via* form is invoked and the required values are set in the form.

Area Select the *11 vias* and click the *Apply* button.

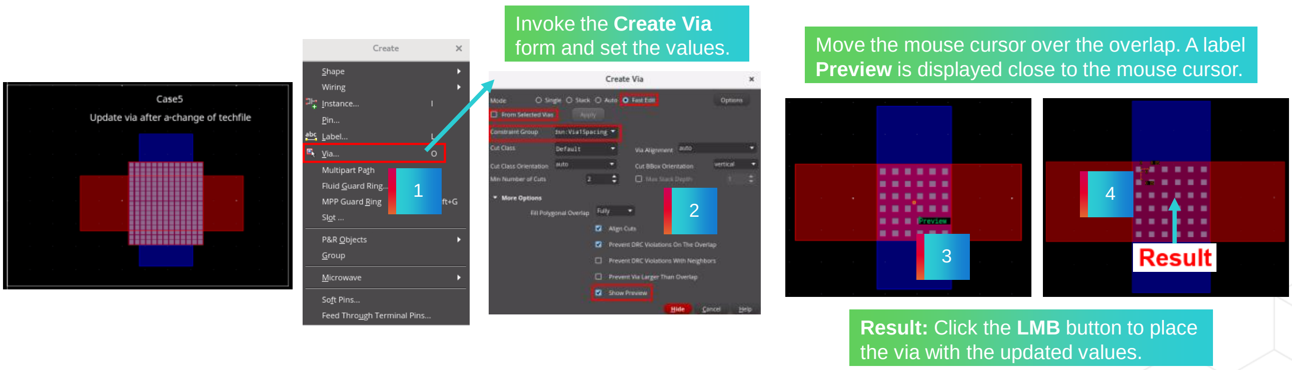
You see that the selected *11 vias* end up with *2 cuts* in the canvas.

Note that you can have the same behavior by using the *Middle Mouse Button*.

Result: Using Fast Interactive Via Editing – FIVE:
Update Via After Change of Techfile

3. In the layout canvas, move the mouse over the overlap to view the **Preview**.
4. Click the **LMB** button in the layout canvas.

Result: The via is dropped with the updated values from the Constraint Group in the design window.



Result: Using Fast Interactive Via Editing – FIVE: Update Via After Change of Techfile

The *Create Via* form is invoked and the required values are set in the form.

When you move the mouse cursor over the overlap, a label *Preview* is displayed close to the mouse cursor.

You observe that when you click the *LMB* button, the via with the updated values is placed in the design window.

Appendix Summary

In this appendix, you learned how to

- Use Auto Via Preview
  - Rectangular Overlap
  - Polygonal Overlap
  - Via Locked
- Use Fast Interactive Via Editing – FIVE
  - Rectangular Overlap
  - Polygonal Overlap
  - Update to Double Cut
  - Fast Edit on Selected Vias
  - Update Via After Change of Techfile

In this appendix, you learned how to use the Auto Via Preview with Rectangular Overlap, with Polygonal Overlap, and with Via Locked.

You learned how to use the Fast Interactive Via Editing with Rectangular Overlap, with Polygonal Overlap, with Update to Double Cut, with Fast Edit on Selected Vias, and with Update Via After Change of Techfile.

Labs

Lab A-1 Using Auto Via Preview

Lab A-2 Using Fast Interactive Via Editing – FIVE

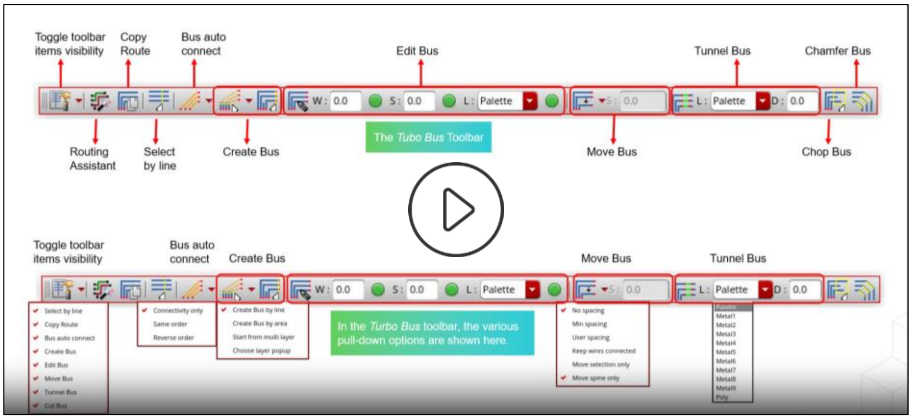
Now, you will do the lab exercises in this Appendix.

There are 2 labs in this appendix.

In the first lab, you will explore the Auto Via Preview functionality.

In the second lab, you will explore the Fast Interactive Via Editing features.

Demo: The Turbo Bus Toolbar: Pull-Down Options

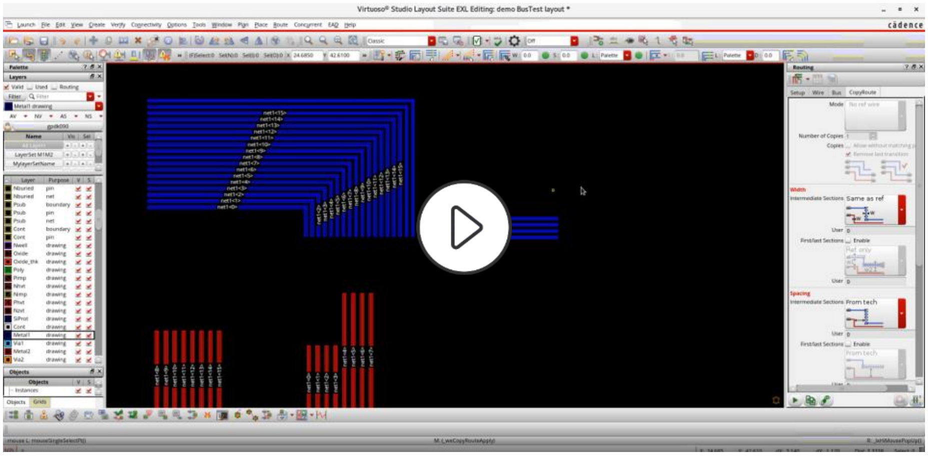


Video Play Time: **7.47** minutes

Click the Play button to start the video.

Click the link https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1OPP000001hfFR2AY&pageName=ArticleContent to play the video on support.cadence.com.

Demo: The Turbo Bus Routing



Video Play Time: **16.41** minutes

Click the Play button to start the video.

Click the link
https://support.cadence.com/apex/ArticleAttachmentPortal?id=a1OPP000001hfAb2AI&pageName=ArticleContent

to play the video on support.cadence.com.

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